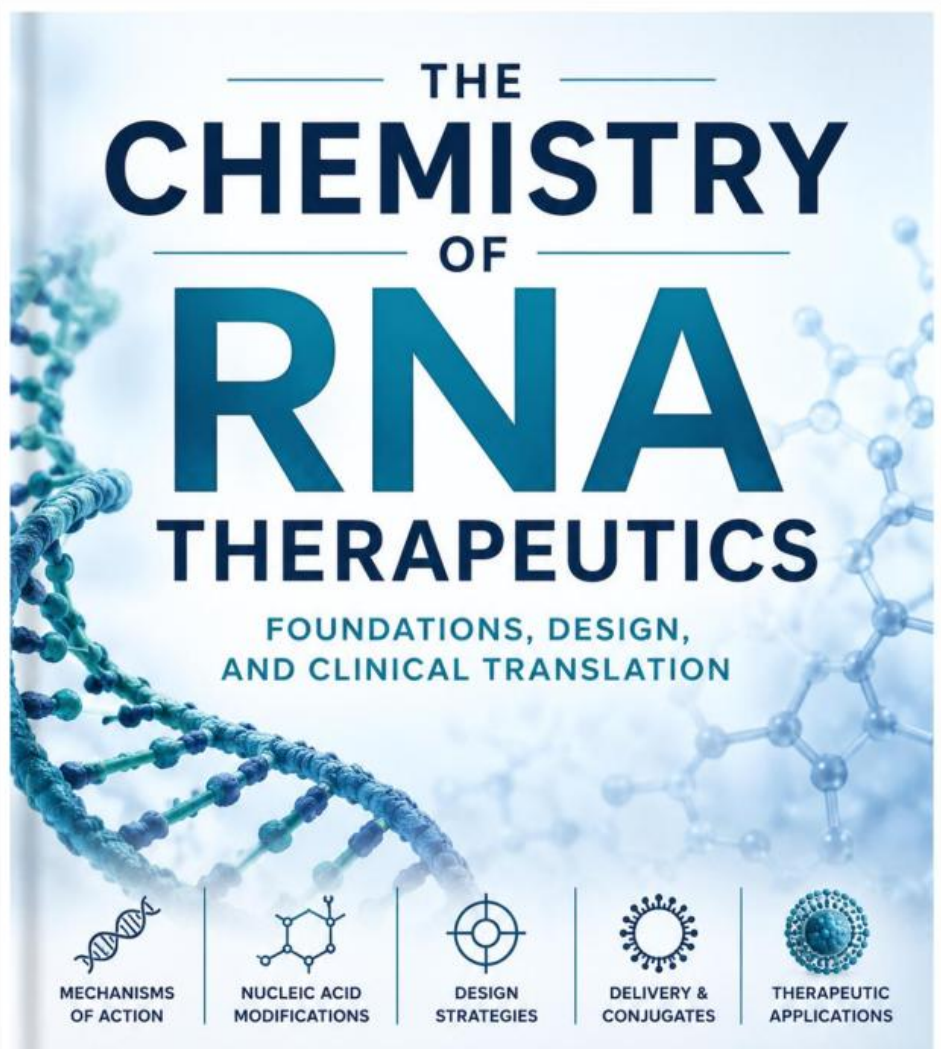




THE CHEMISTRY OF RNA THERAPEUTICS

FOUNDATIONS, DESIGN,
AND CLINICAL TRANSLATION



MECHANISMS
OF ACTION



NUCLEIC ACID
MODIFICATIONS



DESIGN
STRATEGIES



DELIVERY &
CONJUGATES



THERAPEUTIC
APPLICATIONS

From molecular insights to life-changing medicines

JEFF MILTON

Exec. Fellow
St. Jude Children's Research Hospital

It All Started with Synthetic Nucleic Acids

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The Chemistry of RNA Therapeutics Foundations, Design, and Clinical Translation

This work is dedicated to my mentor Dr. Phil Bourne



It all started with synthetic nucleic acids



Dr. Har Gobind Khorana was an Indian American biochemist and molecular biologist who shared the 1968 Nobel Prize in Physiology or Medicine with Marshall W. Nirenberg and Robert W. Holley for deciphering how nucleotides in nucleic acids control the synthesis of proteins. That discovery helped define one of the central principles of molecular biology: that information encoded in nucleic acid sequence could be translated into biological function.

For Khorana, the painstaking process of synthesizing short DNA and RNA sequences may have seemed slow, repetitive, and technically unforgiving. Yet that disciplined chemical work helped reveal something profound. Nucleic acids were not only the molecules of heredity; they could be constructed, modified, and used as experimental tools. In the mid-1950s, working in the laboratory of Sir Alexander Todd, Khorana pioneered the use of synthetic DNA and RNA to address fundamental biological questions. He showed that synthetic nucleic acids could serve as precise templates for DNA and RNA polymerases and helped establish an experimental framework that would later become central to synthetic biology, biotechnology, and therapeutic design.

RNA therapeutics begins here: once nucleic acids could be synthesized, they ceased to be mysteries of nature and became tools of design. They became molecules that could be designed, sequences defined, their chemistry altered, their stability improved, their interactions studied, and their biological effects directed toward specific genes, transcripts, proteins, cells, and tissues.

In the 1960s, Dr. Bob Letsinger, working with Dr. V. Mahadevan, developed one of the first practical methods for synthesizing DNA on an insoluble polymer: the phosphotriester method. Solid-phase synthesis transformed nucleic acid chemistry from a laborious solution-phase process into a more scalable and programmable technology. The efficiency of this

approach helped enable the creation of the first bioengineered genes. Somatostatin, human insulin, and human growth hormone became early products of recombinant DNA technology and helped establish Genentech as one of the first biotechnology companies built on the idea that biological information could be converted into medicine.

Building on this foundation, Professor Marvin Caruthers advanced the field in the 1970s by developing nucleoside phosphoramidite chemistry for the solid-phase synthesis of DNA and RNA. This chemistry allowed oligonucleotides to be produced with precision, reproducibility, and scalability. It remains a foundational method in modern nucleic acid synthesis. At nearly the same time, the importance of oligonucleotide backbone chemistry was becoming clear. Eckstein and Gindl showed that replacing a non-bridging oxygen in the phosphate backbone with sulfur, generating phosphorothioate linkages, could preserve hybridization and polymerase compatibility while improving resistance to nuclease degradation. This observation established a principle that continues to shape the field: nucleic acids can be chemically modified to improve their pharmacological properties without necessarily eliminating their biological function.

This principle is at the heart of RNA therapeutics. Native RNA is fragile, transient, highly charged, and rapidly degraded by nucleases. These properties make RNA a beautiful molecule for biology, but a difficult molecule for medicine. Therapeutic RNA technologies therefore depend on chemistry. Backbone modifications, sugar modifications, nucleobase substitutions, terminal conjugates, stereochemical control, lipid encapsulation, and ligand-directed delivery systems all exist to solve the same basic problem: how to preserve the informational power of nucleic acids while making them stable, deliverable, selective, and safe enough to function as drugs.

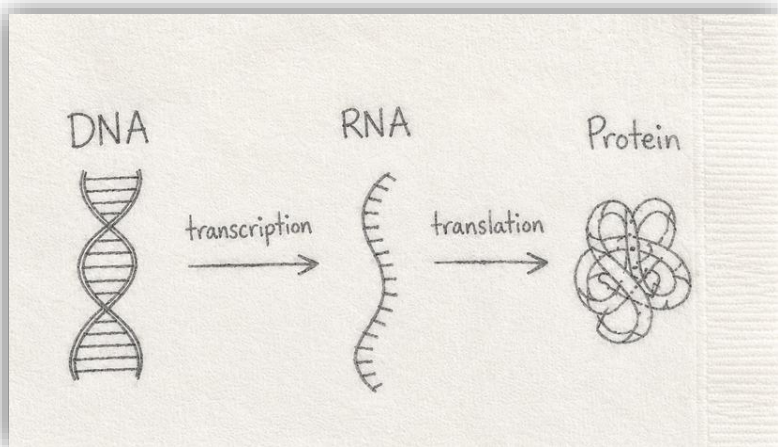
The first major therapeutic logic to emerge from synthetic oligonucleotide chemistry was antisense inhibition. Dr. Joseph Walder, founder of Integrated DNA Technologies, demonstrated that antisense oligonucleotides could inhibit gene expression either by blocking translation or by recruiting RNase H enzymes to cleave target RNA. This linked sequence-specific hybridization to enzymatic RNA degradation and helped define what is now known as the gapmer antisense oligonucleotide. In a gapmer, chemically modified nucleotides at the ends of the molecule improve stability and binding affinity, while a central DNA-like region supports RNase H recruitment. The result is a programmable drug that can bind a selected RNA transcript and direct its destruction.

A second major therapeutic logic emerged with the discovery of RNA interference. In the 1990s, Dr. Andrew Fire and Dr. Craig Mello

showed that double-stranded RNA could induce potent and sequence-specific gene silencing. This discovery revealed an endogenous cellular pathway in which double-stranded RNA is processed into small interfering RNAs, which then guide the RNA-induced silencing complex to complementary messenger RNAs. The guide strand provides specificity, while the cellular machinery provides catalytic silencing. RNA interference transformed the idea of gene knockdown from a synthetic antisense strategy into a programmable use of natural cellular regulation.

These two mechanisms, RNase H-mediated degradation and RISC-mediated degradation, became pillars of RNA-targeting therapeutics. Both use base pairing to identify a target transcript. Both can be designed from genetic sequence. Both convert molecular recognition into reduction of gene expression. Yet they differ fundamentally in chemistry, structure, intracellular compartment, protein machinery, delivery requirements, duration of action, and pharmacological behavior. Understanding these differences is essential because “RNA therapeutic” is not a single mechanism. It is a family of molecular strategies that share the use of nucleic acid information while relying on distinct biological pathways.

Alongside these degradation-based approaches, steric-blocking oligonucleotides opened a different therapeutic possibility. Rather than recruiting an enzyme to destroy RNA, these molecules bind RNA and physically block access by proteins, ribosomes, spliceosomal components, microRNAs, or other regulatory factors. Phosphorodiamidate morpholino oligomers, developed by James Summerton and colleagues, were an important advance in this area. In morpholinos, the ribose sugar is replaced by a morpholine ring and the charged phosphodiester linkage is replaced by a neutral phosphorodiamidate backbone. This architecture confers strong nuclease resistance and allows the molecule to bind RNA without activating RNase H.

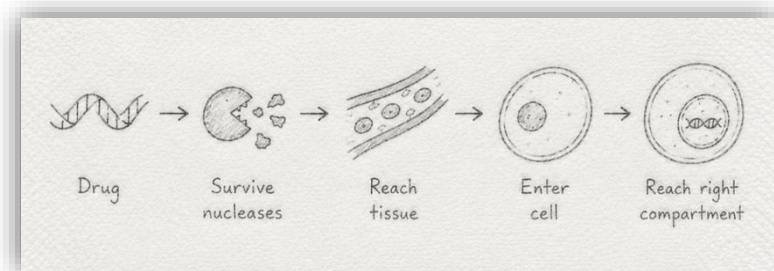


Steric-blocking chemistry made it possible to think of RNA not only as a transcript to be inhibited, but as a substrate to be remodeled. By blocking splice sites, splice enhancers, splice silencers, polyadenylation signals, upstream open reading frames, microRNA binding sites, or translational regulatory elements, oligonucleotides can alter how an RNA is processed and interpreted. This can increase productive protein output, suppress toxic isoforms, induce exon skipping, restore a reading frame, promote nonsense-mediated decay, or shift the balance between alternative transcript isoforms. In this way, RNA therapeutics move beyond simple gene silencing and enter the domain of transcript engineering.

The central dogma describes the flow of biological information from DNA to RNA to protein. RNA therapeutics intervene in the middle of that flow. They do not cause changes to DNA. Instead, they exploit the regulatory space between genome and proteome. This gives RNA therapeutics unusual flexibility. A drug can silence a toxic gain-of-function transcript, reduce a disease-associated protein, restore splicing, increase expression from a healthy allele, modulate a pathway by shifting isoform balance, or introduce a transient protein-coding message into a cell.

The clinical landscape now reflects this breadth. Antisense oligonucleotides have been approved for neurological, metabolic, and rare genetic diseases. siRNA therapeutics have validated durable gene silencing, especially in the liver. Splice-switching oligonucleotides have changed the treatment of diseases such as spinal muscular atrophy and Duchenne muscular dystrophy; in SMA, this shift has been profound. Spinal muscular atrophy is no longer a leading cause of inherited infant death in the United States, following the approval of three genetic therapies since 2016. Aptamers have shown that structured nucleic acids can bind protein targets

with antibody-like specificity. mRNA vaccines can direct transient protein expression at a global scale. Short activating RNAs, circular RNAs, self-amplifying RNAs, and microRNA-directed approaches further expand the field from gene silencing toward gene activation, protein expression, immune modulation, and programmable regulation.



mRNA therapeutics represent another conceptual approach where instead of targeting an endogenous RNA transcript, synthetic RNA introduces genetic instructions into cells. Once delivered to the cytoplasm, this RNA is translated into protein. This makes mRNA useful for vaccination, protein replacement, immunotherapy, genome editing, and cellular reprogramming. The success of COVID-19 mRNA vaccines brought this modality into public view, but the underlying technology rests on decades of work in cap biology, and sequence design, codon optimization, nucleoside modification, innate immune recognition, lipid nanoparticle delivery, and manufacturing science. Modern mRNA therapy is not simply copied from a gene; it is designed from end to end.

Across all of these modalities, delivery remains one of the defining challenges. A nucleic acid drug must survive extracellular nucleases, avoid rapid clearance, distribute to the relevant tissue, enter the appropriate cell type, escape or productively traffic through endosomal compartments, and reach the intracellular location where its mechanism operates. For siRNAs, this often means cytosolic access and RISC loading. For splice-switching ASOs, it often means nuclear delivery. For mRNA, it means cytosolic translation. For RNA editing, it may mean coordinated localization of the editing oligonucleotide, target RNA, and endogenous editing enzyme. The same sequence can be potent in a dish and ineffective in vivo if it cannot reach the right compartment.

The liver has become the most tractable organ for systemic RNA therapeutics because of efficient hepatocyte uptake mechanisms, particularly GalNAc conjugation for siRNAs and antisense oligonucleotides, as well as lipid nanoparticle delivery. Other tissues remain more difficult. The central nervous system often requires intrathecal administration. The eye can be

accessed by intravitreal injection. Muscle, lung, immune cells, and solid tumors each present different barriers of anatomy, circulation, uptake, and intracellular trafficking. For this reason, delivery is not a secondary formulation problem. It is part of the drug's mechanism.

Pharmacology is equally distinct. RNA therapeutics do not behave like small molecules or antibodies. Plasma concentration may fall rapidly while tissue activity persists for weeks or months. Total tissue concentration may not predict pharmacodynamic effect if most of the drug is trapped in nonproductive compartments. Chemical modifications may increase stability but alter protein binding, biodistribution, potency, toxicity, or clearance. A gapmer ASO, a fully modified splice-switching ASO, a GalNAc-siRNA, a lipid nanoparticle-formulated mRNA, and an aptamer each has a different relationship between dose, exposure, tissue distribution, intracellular availability, target engagement, and biological response.

Manufacturing also shapes the field. Nucleic acid therapeutics require careful control of sequence identity, purity, protecting-group removal, chain length, stereochemistry, conjugation, encapsulation, residual solvents, truncated products, double-stranded impurities, endotoxin, innate immune contaminants, and analytical characterization. For mRNA, additional considerations include capping efficiency, poly(A) tail length, nucleoside incorporation, double-stranded RNA impurities, lipid nanoparticle composition, particle size, encapsulation efficiency, and stability. These are not merely technical details. They determine whether a molecule can become a reproducible, regulatable, and scalable medicine.

RNA therapeutics sit at the intersection of chemistry, biology, medicine, and engineering. Their power comes from programmability, but programmability alone is not enough. A sequence must become a molecule. A molecule must become a drug. A drug must reach a tissue. Tissue exposure must become productive intracellular activity. Each step requires design.

The future of RNA medicine will likely be defined by increasing precision over each of these steps. Improved chemical modifications will extend stability and tune protein interactions. Better conjugates and delivery vehicles will expand tissue reach beyond the liver. Computational tools will improve target selection, structure prediction, off-target analysis, and sequence optimization. RNA editing and activation may broaden the field from suppression of disease genes to restoration, correction, and controlled enhancement of gene function. Personalized and individualized RNA medicines may make it possible to design therapies for mutations, splice defects, or rare diseases that were previously unreachable by conventional drug discovery.

GBFsen: The Speed of Urgency

Giovanni was thirty years old, a lawyer in southern Italy, when the diagnosis finally gave a name to what his family had feared for generations.

Amyotrophic lateral sclerosis. ALS.

In his family, the disease was not a rumor or a distant medical abstraction. It had passed through the years like a shadow, taking relatives, stealing movement, narrowing futures. The mutation was in a protein called TDP-43, and by the time Giovanni heard his own diagnosis, his family understood the pattern too well. They knew what progression could look like. They knew how quickly time could disappear.

And they knew there was no treatment waiting for him.

So they asked for the impossible.

Not eventually. Not someday. Now.

Their appeal reached a team of scientists who understood that rare disease does not move on institutional timelines. Biology does not pause for committee review, commercial planning, licensing negotiations, or professional ego. ALS progresses whether the world is ready or not.

In October 2025, Giovanni's patient cells arrived at La Jolla Labs.

There his disease was no longer only a family tragedy. It became a model. A system. A problem that could be studied, measured, and attacked. Illumina performed genome sequencing and phasing, mapping the pathogenic mutation onto a disease-causing haplotype. That map gave the scientists their opening: design a therapy that could selectively target the mutant copy while preserving the normal one.

At La Jolla Labs, scientists designed a series of antisense oligonucleotide candidates, or ASOs, engineered to bind RNA and knock down the mutant genetic signal. Integrated DNA Technologies, part of Danaher, synthesized the screening molecules and delivered them rapidly so the experiment could begin in Giovanni's own patient-derived cellular models.

The screen worked.

Five lead compounds showed strong selectivity in a dose-responsive manner. By late October 2025, one molecule stood out. It could alter the disease-related genetic signal and appeared relatively safe in other cell-based screens. By late November, the lead compound was ready to begin a preclinical safety path.

Giovanni and Francesca his Fiancé named the drug:

GBFsen.

GBFsen was not built as a product chasing a market. It was built as a response to urgency. It was a therapeutic candidate made for a single patient, with no obvious commercial incentive, and yet an extraordinary network of people and companies chose to help.

Illumina. La Jolla Labs. Danaher. ChemGenes. N1C. Thermo Fisher Scientific.

Each contributed expertise, resources, speed, or belief.

And still, the hardest barriers were not always scientific.

In a disease like ALS, every pause has weight.

In early 2026 GBFsen entered rodent toxicology studies. While some parts of drug development can be accelerated; others cannot. A thirteen-week safety study still takes thirteen weeks.

But Giovanni did not have thirteen weeks to spare.

So the team made a careful and thoughtful decision. Clinical dosing would be allowed to proceed after safe interim toxicology reports, while the full preclinical toxicology program continued in parallel. The path forward was conditional on both the animal safety data and Giovanni's own clinical data.

It was urgent. Giovanni was dosed.

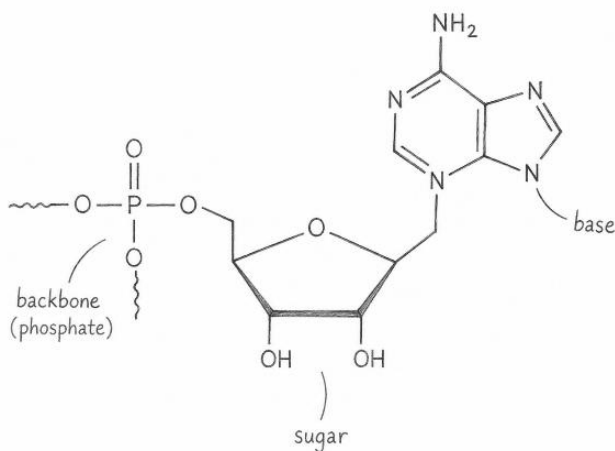
The first administration of GBFsen marked a turning point. A drug that had begun as a sequence, a mutation, a family's plea, and a set of patient cells had reached the clinic. It had crossed the distance from genetic diagnosis to human dosing in a matter of months.

For Giovanni and his family, it was not an abstract triumph. It was not a headline. It was hope.

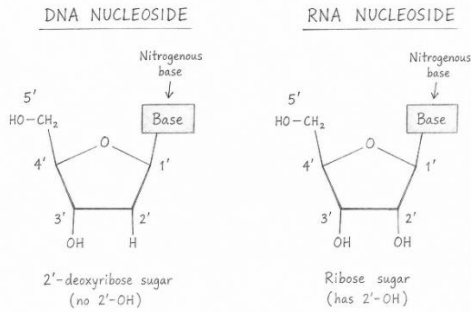


The ability to design and develop GBFsen began with Dr. Har Gobind Khorana, who helped reveal how nucleic acid sequence encodes biological meaning and showed that synthetic DNA and RNA could be used as precise tools. It continued through the work of Bob Letsinger and V. Mahadevan, who helped transform nucleic acid synthesis through solid-phase chemistry, making the construction of defined sequences more practical. It advanced further through Marvin Caruthers, whose phosphoramidite chemistry made the precise synthesis of DNA and RNA routine enough to support the scale required for modern biotechnology. It was strengthened by Fritz Eckstein and colleagues, whose work on phosphorothioate backbone chemistry showed that nucleic acids could be chemically modified to improve stability while preserving biological function. And it became therapeutically actionable through the science of Dr. Joseph Walder, who helped establish the antisense logic that sequence-specific oligonucleotides could inhibit gene expression by binding RNA and recruiting RNase H.

Chemical Modifications in Nucleic Acid Therapeutics



Nucleic acids are polymers composed of repeating nucleotide units, each consisting of a nitrogenous base, a five-carbon sugar, and a phosphate group. In DNA, the sugar is 2'-deoxyribose, whereas RNA contains ribose, which possesses an additional hydroxyl group at the 2' position. Nucleotides are connected through 3'→5' phosphodiester linkages to form a sugar-phosphate backbone, and the sequence of bases (adenine [A], guanine [G], cytosine [C], thymine [T] in DNA, and uracil [U] in RNA) encodes biological information. Understanding nucleic acid therapeutic chemistry requires familiarity with nucleotide nomenclature and numbering conventions, particularly the sugar positions 2', which is a common site for chemical modification. ASOs frequently incorporate modifications to the phosphate backbone, such as phosphorothioate (PS) linkages in which a non-bridging oxygen is replaced by sulfur, or modifications to the sugar moiety, including 2'-O-methyl (2'-OMe), 2'-O-methoxyethyl (2'-MOE), locked nucleic acids (LNAs), and constrained ethyl (cEt) nucleotides. Base modifications may also be introduced to improve stability, affinity, or specificity.



DNA is composed of repeating nucleosides that are connected by phosphate groups to form a linear polymer. Each nucleoside consists of a nitrogenous base attached to a five-carbon sugar called 2'-deoxyribose. The carbons of the sugar are numbered 1' through 5', and these numbers are essential for describing nucleic acid structure and chemistry. The 1' carbon is attached to the base, the 2' carbon contains a hydrogen atom in DNA (rather than a hydroxyl group as in RNA), the 3' carbon contains a hydroxyl (–OH) group, and the 5' carbon is attached to a phosphate group. During DNA synthesis, the phosphate group forms a phosphodiester linkage between the 3' hydroxyl of one nucleoside and the 5' position of the next nucleoside, creating the characteristic 3'→5' sugar-phosphate backbone. These numbering conventions define the directionality of a DNA strand, with one end designated the 5' end and the other the 3' end and provide the framework for understanding many nucleic acid modifications used in antisense oligonucleotides and other nucleic acid therapeutics.

Eur. J. Biochem. 13 (1970) 658–664

Polyribonucleotides Containing a Phosphorothioate Backbone

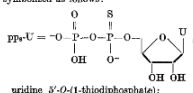
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(Received December 16, 1969/January 31, 1970)

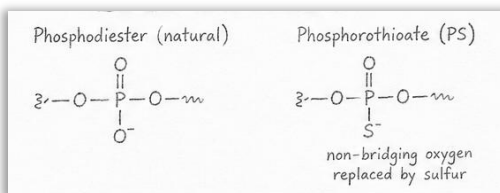
The enzymatic synthesis of two polyribonucleotides, poly r(A₃U) and poly r(A₂U) containing a sugar-phosphorothioate instead of a sugar-phosphate backbone, is described. The formation and thermal stability of double-stranded polymers is not impaired by this modification. They can also serve as templates for RNA-polymerase. Single stranded polymers can serve as messengers. Both double and single-stranded polymer exhibit some degree of resistance to nucleases.

Unusual Abbreviations. Following recommendations by Dr. W. E. Cohn (Director of the National Academy of Sciences—National Research Council, Office of Biochemical Nomenclature, Oak Ridge National Laboratory, Oak Ridge, Tennessee, U.S.A.) the phosphorothioate-containing compounds are symbolized as follows:



Recently we described [2] the synthesis and some properties of a polyribonucleotide, poly r(A₃U) in which every other phosphate was replaced by a phosphorothioate group. This modified polynucleotide was degraded by nucleases at a considerably slower rate than the unmodified polynucleotide, poly r(A₃U). The present investigation was undertaken to determine the effects of replacement of all phosphate groups in a polynucleotide by phosphorothioate. Parts of this work have been published in a preliminary report [1].

The importance of backbone chemistry in oligonucleotides dates to foundational work by Eckstein and Gindl (1970), who demonstrated that replacing a non-bridging oxygen in the phosphate backbone with sulfur (phosphorothioate) produced polyribonucleotide that retained the ability to form stable duplexes and serve as templates for RNA polymerase, while exhibiting enhanced resistance to nuclease degradation. This early work established the principle that backbone modification could preserve biological function while improving stability.



Today, backbone engineering remains a central parameter in oligonucleotide design, enabling control over stability, protein interactions, and pharmacokinetics. Phosphorothioate (PS) remains universal across ASO and many siRNA designs to improve stability, delivery, and pharmacological activity. The natural phosphodiester (PO) backbone is retained in some RNA-based modalities, including siRNA and mRNA (Egli and Manoharan, 2023; FDA, 2016; Rodrigues and Yokota, 2023).

The first antisense in the clinic

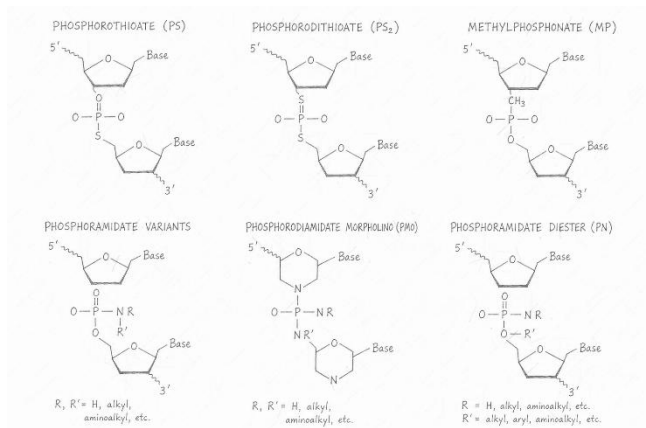


FOMIVIRSEN (BRAND NAME VITRAVENE)

Fomivirsen is a simple sequence specific antisense DNA molecule with a modified phosphorothioate backbone (PS); this backbone is the only thing protecting it from nuclease activity.

Intravitreal administration of Fomivirsen significantly prolonged the median time to disease progression in patients with CMV retinitis compared to delayed treatment, supporting its therapeutic efficacy (Geary, Henry and Grillone, 2002). Fomivirsen demonstrated activity against CMV strains resistant to conventional antivirals and exhibits additive effects when used in combination with agents such as ganciclovir or foscarnet (Azad et al., 1995).

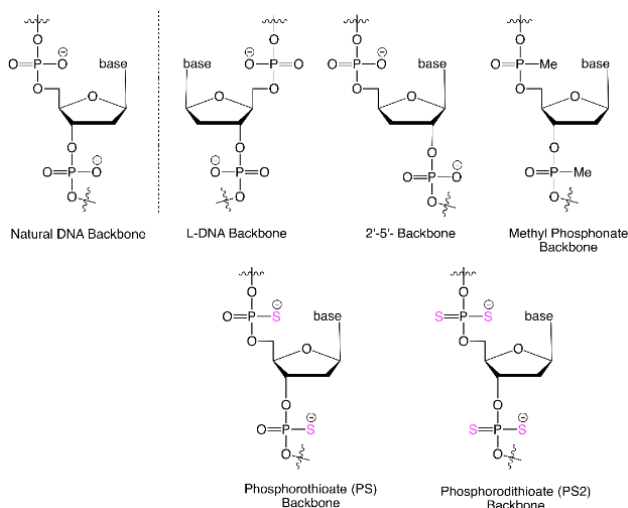
Charge, Stability and Function



Chemical modification of the oligonucleotide backbone underpins the success of modern nucleic acid therapeutics. While native phosphodiester (PO) linkages provide optimal compatibility with biological machinery, they are intrinsically unstable and pharmacologically limited. A diverse set of backbone chemistries including phosphorothioate (PS), phosphorodithioate (PS₂), methylphosphonate (MP), phosphoramidate variants, phosphorodiamidate morpholino (PMO), and emerging phosphoramidate diester (PN) chemistries has emerged to address these constraints. These modifications operate within a constrained design space

defined by three interdependent variables: charge, nuclease stability, and biological mechanism compatibility.

First, electrostatic charge governs interactions with proteins, membranes, and biological barriers. The phosphodiester backbone (PO) is fully anionic, enabling strong protein binding and efficient recognition by endogenous enzymes, but simultaneously limiting membrane permeability and promoting nonspecific interactions. Phosphorothioate (PS) linkages retain this negative charge while altering its distribution, contributing to increased protein binding and improved pharmacokinetics. In contrast, several alternative chemistries including methylphosphonates, phosphoramidates, phosphorodiamidate morpholino (PMO), and PN linkages reduce or eliminate backbone charge. PMO and PN backbones are largely charge-neutral, fundamentally altering electrostatic interactions, often reducing plasma protein binding and accelerating clearance while potentially improving intracellular distribution. However, reduced charge can also diminish hybridization affinity and weaken protein-mediated uptake pathways that are critical for productive delivery (Crooke, 2017; Juliano, 2016).



Second, nuclease stability is primarily governed by chemical substitutions at the phosphorus center and by steric protection from sugar modifications. Substitution of non-bridging oxygen atoms with sulfur (e.g., PS linkages) reduces nuclease-mediated hydrolysis and increases resistance to degradation in biological matrices (Lovric et al., 2025) (Wan et al., 2016). Importantly, stability is not solely dependent on the presence of stabilizing chemistries but also on their distribution and combination. Mixed backbone designs (e.g., PO/PS or PS/PN chimeras) and gapmer architectures allow selective tuning of degradation kinetics and pharmacokinetics. For example,

incorporation of PO linkages within a PS backbone reduces stability in a position-dependent manner, while sugar modifications such as MOE can compensate for reduced backbone protection and enable fine-tuning of tissue distribution (Lovrić et al., 2025; Geary, 2001).

The following table highlights typical rules associated with the placement of PO linkages in a gapmer ASOs:

Phosphodiester (PO) placement

At very 5' end	Often increases nuclease vulnerability, especially to exonucleases. May reduce protein binding and alter uptake.
At very 3' end	Often risky because many nucleases attack from the 3' end. A terminal or near-terminal PO can create a cleavage-sensitive site.
Near ends but not terminal	Can reduce stability, but effect depends on neighboring bases and sugar modifications. Sometimes tolerated if protected by 2'-modified sugars such as MOE, cEt, LNA, or 2'-OMe.
Central DNA gap of RNase H gapmer	Usually the most dangerous place for PO if nuclease exposure is high, because the DNA gap is already less sugar-protected than MOE wings. PO in the gap can reduce metabolic stability. It may also affect RNase H activity depending on placement.
In MOE / 2'-modified wings	More tolerated than in an unmodified DNA gap, because sugar modifications add nuclease resistance. PO here may be used to tune protein binding, clearance, or toxicity.
Multiple PO linkages clustered together	Usually lowers stability more strongly. Lovrić et al. reported that ASOs with one PO were more stable than those with two or three PO links in nuclease-containing systems.
PO next to MeC	Lovrić et al. found that the local sequence context matters; their study reports that PO placement relative to 5-methylcytidine (see modified nucleobases section) affected nuclease resistance, meaning "PO before" and "PO after" the same base are not necessarily equivalent.

ASOs in the clinic have two main backbone strategies: fully phosphorothioated designs and newer mixed PS/PO gapmers. Fomivirsen is a first-generation 21-mer DNA ASO with a fully phosphorothioate backbone, meaning all 20 internucleotide linkages are PS and there are no PO linkages (Anderson et al., 1998). Several second-generation 5-10-5 MOE/DNA gapmers follow the same all-PS backbone logic: mipomersen,

inotersen and volanesorsen are each 20-mer ASOs with all PS and no PO linkages, with five 2' MOE residues at each end and a central ten-nucleotide DNA gap (Geary et al., 2015; Ionis Pharmaceuticals, 2013; Ionis Pharmaceuticals, 2019; Witztum et al., 2019). Nusinersen is different mechanistically because it is a splice-modulating ASO rather than an RNase H gapmer, but it is also fully phosphorothioated: as an 18-mer 2'-MOE ASO, it contains 17 PS linkages and 0 PO linkages (Bennett et al., 2017). By contrast, tofersen uses an explicitly mixed PS/PO 5-10-5 MOE gapmer backbone: it is a 20-mer with 19 total internucleotide linkages, of which 15 are PS and 4 are PO; in 5'→3' notation its linkage pattern is PS–PO–PS–PO–PS–PS–PS–PS–PS–PS–PS–PS–PS–PS–PO–PS–PO–PS, placing the four PO linkages in the MOE wing regions rather than across the central DNA gap (Biogen, 2023; KEGG, 2026). This pattern preserves an all-PS DNA gap for RNase H1-competent activity while reducing total PS burden at selected wing positions.

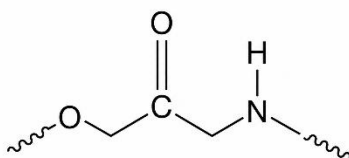
Chemistry	Pattern	Charge	Used by	In approved drugs?
PO	soooooooooooooos	full negative	Alnylam	Onpattro and others
PS	ssssssssssssssss	full negative	Ionis	Yes, many including Spinraza
PS + PO mix	sooooo:sssssssssooss	mixed	n-Lorem	Yes, neuro – gapmer
PN	ssssnssssssssnsss	reduced	Wave	
Stereopure PS		full negative	Wave	

New backbone chemistries are being developed to retain the useful pharmacology of PS ASOs while reducing liabilities associated with extensive PS content, including nonspecific protein binding, pro-inflammatory effects and sequence-dependent toxicity. One prominent example is mesyl phosphoramidate chemistry, also referred to as MsPA or μ chemistry, in which a mesyl phosphoramidate group replaces the natural phosphodiester linkage at selected positions. Unlike many neutral backbone analogues, MsPA remains negatively charged near physiological pH and acts as a close phosphodiester mimic, while providing nuclease resistance that can exceed PS and preserving compatibility with RNase H recruitment in appropriate DNA-like ASO designs (Stetsenko, 2023). This makes MsPA particularly attractive for next-generation gapmers, where mixed PS/MsPA backbones can be used to preserve RNase H1 activity and productive cellular uptake while reducing the total PS burden. In Ionis’ next-generation ASO work, site-specific replacement of PS linkages with MsPA in gapmer ASOs improved nuclease stability, reduced nonspecific protein binding, lowered cytotoxic and pro-inflammatory effects, and extended pharmacodynamic duration in vivo; importantly, replacement of selected PS linkages, including several within the DNA gap, was tolerated without loss of RNase H1-mediated activity, whereas complete replacement of all PS

linkages could reduce activity depending on design context (Migawa et al., 2021). The main advantage of this chemistry is therefore not simply greater nuclease resistance, but a broader therapeutic-index gain: MsPA enables medicinal chemists to modulate the balance between stability, RNase H activity, protein binding, tissue exposure and tolerability more precisely than with uniformly PS-modified ASOs.

Backbone chemistries in combination with sugar modifications such as 2'-O-methoxyethyl (MOE) and locked nucleic acid (LNA), provide tunable levers for ASO pharmacology. Rather than simply replacing earlier chemistries, newer modifications expand the combinatorial design space available for therapeutic optimization. Modern oligonucleotide therapeutics increasingly use hybrid architectures that integrate multiple backbone and sugar modifications within a single molecule to balance nuclease stability, pharmacokinetics, potency, safety, and mechanism of action.

Morpholino Phosphorodiamidate Linkages

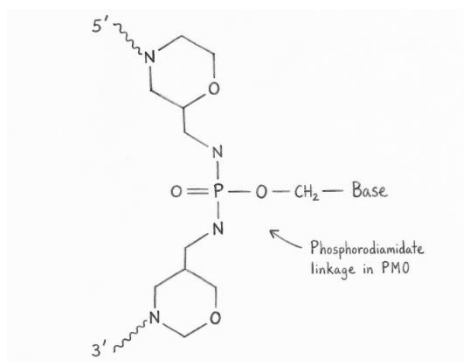


Carbamates are versatile amide–ester hybrid functional groups widely used in medicinal chemistry because they combine chemical stability, tunable conformational properties, and favorable pharmacokinetics. They serve as peptide bond surrogates, prodrug linkers, and key structural motifs in many approved drugs, enabling improved membrane permeability, metabolic stability, and controlled drug release. Their ability to modulate hydrogen bonding and resist enzymatic degradation has made them especially valuable in designing enzyme inhibitors and prodrugs across diverse therapeutic areas. In antisense technologies, carbamates are used to enhance stability and modulating molecular properties for delivery.

Carbamate-based therapeutics exhibit pharmacokinetic and pharmacodynamic profiles that are strongly influenced by their chemical stability and susceptibility to enzymatic cleavage. The carbamate-containing HIV protease inhibitor Darunavir is well absorbed orally with rapid attainment of peak plasma concentrations, and undergoes carbamate hydrolysis-dominated metabolism. PK can be significantly modulated by

co-administered agents; for example, ritonavir inhibits CYP3A metabolism, leading to markedly increased systemic exposure and prolonged half-life of darunavir. Excretion occurs via the feces, with minimal renal clearance, reflecting hepatic metabolism and biliary elimination. From a PD perspective, carbamates can either act as stable structural motifs that enhance target binding or as cleavable groups that enable controlled release of active species, thereby linking metabolic processing directly to therapeutic activity. Overall, carbamate functionalities provide a tunable balance between stability and bioactivation, which is central to optimizing both drug exposure and efficacy.

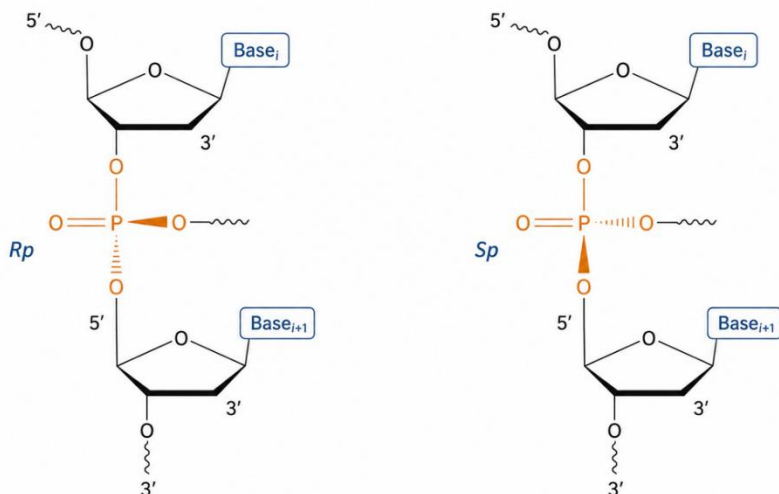
The development of phosphorodiamidate linkages (carbamate-like linkages) for morpholino antisense was a natural extension of the broader medicinal chemistry principles. Early antisense oligonucleotides were limited by the instability and charged nature of the phosphodiester backbone, prompting efforts to introduce chemically modified linkages that could enhance durability and biological performance. Building on small-molecule success of this carbamate chemistry, Dr James Summerton and colleagues introduced the phosphorodiamidate linkage as part of the morpholino platform. This modification replaced the negatively charged phosphodiester with a neutral backbone incorporating P–N bonds, conceptually analogous to carbamate functionality, yielding molecules that were highly resistant to nucleases while maintaining strong and specific hybridisation to RNA.



As this chemistry matured, phosphorodiamidate morpholino oligomers (PMOs) emerged as a distinct class of antisense agents that operate through steric blockade rather than enzymatic degradation, marking a shift toward safer and more controllable mechanisms of action (Summerton and Weller, 1997; Summerton, 2017). The introduction of phosphorodiamidates was a key translational step in antisense design.

Backbone modification is no longer used simply to increase nuclease resistance, but to fine-tune the broader pharmacological behavior of oligonucleotide drugs. When combined with sugar modifications such as MOE, LNA, or other affinity-enhancing groups, these backbone strategies provide a flexible design framework for balancing potency, stability, delivery, tissue exposure, safety, and mechanism of action. Next-generation antisense therapeutics are increasingly defined by rational hybrid architectures, in which each chemical modification is selected not as an isolated improvement, but as part of an integrated strategy to expand therapeutic index and clinical utility.

Chirality



Substitution at the phosphorus center introduces chirality into the backbone, generating Rp and Sp stereoisomers. In phosphorothioate backbones, each linkage can adopt either configuration, resulting in large mixtures of diastereomers in conventionally synthesized oligonucleotides. Chirality can influence multiple aspects of oligonucleotide behavior. Rp linkages are often associated with improved RNase H recognition and RNA binding, whereas Sp linkages confer greater resistance to nuclease degradation. These opposing effects suggest that stereochemical control could, in principle, enable fine-tuning of backbone properties beyond chemistry alone. Recent advances in stereocontrolled synthesis have enabled the generation of stereodefined oligonucleotides, allowing systematic exploration of these effects.

ASO	Stereo pattern	Tm (°C)	Percent knockdown / remaining at day 5 (%)
Mipomersen	Stereorandom	80.2	67
WW-1	Rp all	84.7	34
WW-2	Sp all	74.7	97
WW-3	Sp-rich mixed pattern	78.8	92
WW-4	Rp-rich mixed pattern	80.0	55
WW-5	Mixed Rp/Sp pattern	81.6	90
WW-6	Mixed Rp/Sp pattern	78.3	55
WW-7	Mixed Rp/Sp pattern	68.2	

Iwamoto, N., Butler, D.C.D., Svrtkapa, N., Mohapatra, S., Zlatev, I., Sah, D.W.Y., Meena, Standley, S.M., Lu, G., Apponi, L.H., Frank-Kamenetsky, M., Zhang, J.J., Vargeese, C. and Verdine, G.L., 2017. *Control of phosphorothioate stereochemistry substantially increases the efficacy of antisense oligonucleotides*. Nature Biotechnology.

A team at Wave Life Sciences demonstrated the importance of backbone stereochemistry in a 2017 study where they showed that a lack of control masked significant differences in biological activity, as individual Rp and Sp configurations at phosphorus profoundly influenced stability, protein interactions, and RNase H-mediated cleavage (Iwamoto et al., 2017). By developing methods to produce stereopure oligonucleotides with defined chirality patterns, the team demonstrated improved, in vivo, potency, and greater durability compared to conventional stereorandom ASOs (Iwamoto et al., 2017). This helped establish a key platform technology for Wave Therapeutics and laid the foundation for compounds such as their DMD compound WVE-N531.

Stereo-pure ASO WVE-N531



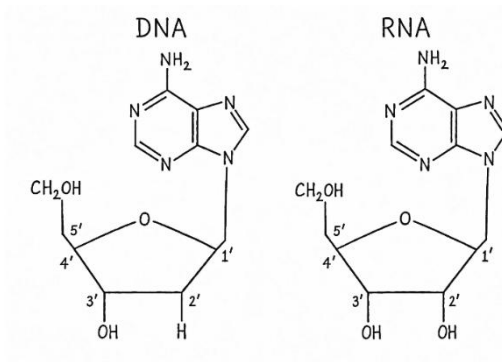
WVE-N531 is a stereopure antisense oligonucleotide (ASO) designed to modulate splicing of the dystrophin pre-mRNA for the treatment of Duchenne muscular dystrophy (DMD). It is a 20-mer single-stranded RNA-like molecule built from a precisely controlled pattern of 2'-fluoro (2'-F) and 2'-O-methyl (2'-OMe) sugar modifications, combined with a chirally defined backbone containing mostly “Sp” phosphorothioate linkages and a smaller number of Rp phosphoramidate linkages, along with a phosphodiester. This stereochemical control is a key feature of Wave Life Sciences’ design approach, as it can influence binding affinity, stability, and protein interactions. WVE-N531 binds to dystrophin pre-mRNA and promotes exon 53 skipping, enabling restoration of a shorter but functional dystrophin protein in patients with certain mutations.

Despite its conceptual appeal, experimental evidence indicates that the benefits of stereochemical control is less obvious than it may seem at first. Systematic studies by a group at Ionis pharmaceuticals demonstrate that while PS chirality modulates RNase H1 cleavage patterns in a predictable manner, these changes do not translate into improved potency or therapeutic index. Stereodefined oligonucleotides exhibit target knockdown comparable to stereorandom mixtures, with no configuration consistently outperforming the parent compounds. Indeed, fully Sp-modified backbones can exhibit substantial losses in activity.

At the mechanistic level, introduction of defined Rp and Sp linkages can bias cleavage site selection and, in some cases, produce dominant cleavage positions. Notably, such effects can be achieved with minimal stereochemical intervention, such as incorporation of a single RpSp step

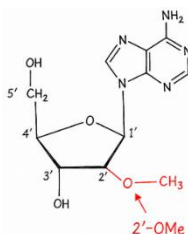
within the gap region. However, these alterations in enzymatic processing do not result in enhanced cellular or in vivo efficacy, underscoring a disconnect between biochemical precision and pharmacological outcome (Wan WB, Seth PP., 2016).

Furanose Sugar Modifications



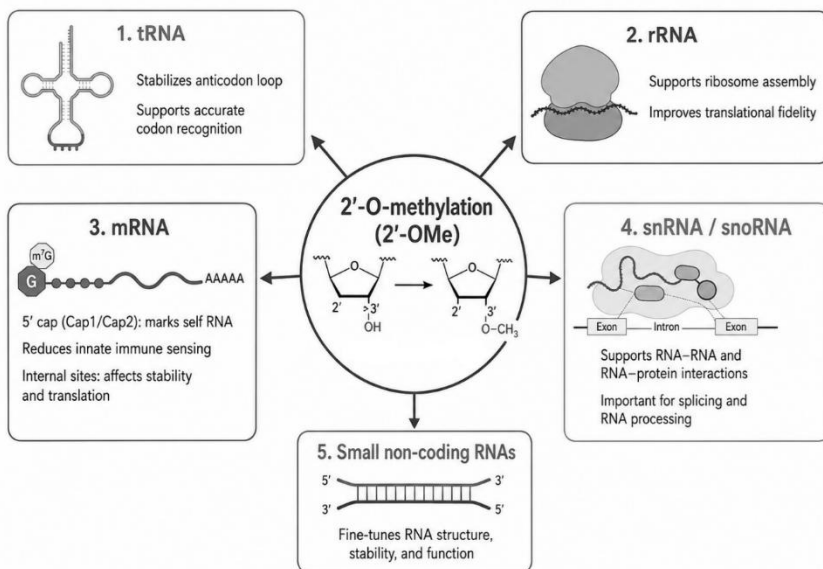
Sugar modifications are among the most important tools for converting fragile nucleic acids into durable, selective, and therapeutically useful molecules. By altering ribose or sugar-like scaffold, chemists can increase nuclease resistance, tune hybridization strength, reduce innate immune activation, improve strand selection, and shape how an oligonucleotide interacts with cellular machinery. Modifications such as 2'-O-methyl, 2'-fluoro, 2'-MOE, LNA, and GNA illustrate how small changes at or around the sugar can produce major effects on potency, safety, pharmacokinetics, and clinical performance. These chemistries form a central design layer in modern RNA therapeutics, especially for siRNAs, antisense oligonucleotides, splice-switching drugs.

2'-O-Methyl (2'-OMe)



Chemically, 2'-O-methylation replaces the ribose 2'-hydroxyl group with a methoxy substituent, increasing RNA conformational stability, nuclease resistance, and compatibility with endogenous RNA-recognition pathways while reducing innate immune stimulation. Endogenously, 2'-O-methylation is one of the most abundant and conserved RNA modifications,

occurring across multiple RNA classes including tRNA, rRNA, mRNA, snRNA, snoRNA, and small non-coding RNAs, where it fine-tunes RNA folding, stability, intermolecular recognition, and translational control (Dimitrova et al., 2019; Motorin and Helm, 2019). In tRNAs, 2'-OMe modifications help stabilize anticodon-loop structure and support accurate codon recognition, particularly at positions involved in decoding and wobble interactions. In rRNAs, 2'-OMe marks are installed largely through snoRNA-guided pathways and cluster around functionally important ribosomal regions, where they contribute to ribosome assembly, translational fidelity, and ribosome activity. In spliceosome snRNAs and snoRNAs, 2'-OMe supports RNA–RNA and RNA–protein interactions required for proper pre-mRNA splicing and ribonucleoprotein function.



In mRNAs, 2'-OMe occurs both at the 5' cap and internally: 2'-OMe generates cap1 and cap2 structures that mark host transcripts as “self” and suppress inappropriate innate immune sensing, while internal 2'-OMe can modulate translation elongation by influencing codon–anticodon decoding dynamics (Choi et al., 2018; Picard-Jean et al., 2018; Sendinc et al., 2023). Therapeutic RNA design adapts this same chemical logic for different purposes: in siRNAs, 2' ribose modifications can increase nuclease resistance, reduce immune activation, preserve RISC compatibility, and tune strand behavior.

Alnylam developed the first siRNA architecture for therapeutics; these made extensive use of the 2'OMe modifications where designs combined other modifications helped balance stability, potency, RISC loading, tissue exposure, and safety. First-generation Standard Template Chemistry (STC) used a fully modified 2'-OMe/2'-F duplex with limited

terminal phosphorothioate stabilization. Enhanced Stabilization Chemistry then increased terminal phosphorothioate linkages, raised the relative 2'-OMe content, and reduced total 2'-F content to improve metabolic stability and durability. Later advanced ESC and ESC+ designs further refined this architecture, including seed-region destabilizing modifications such as GNA to reduce off-target effects. Thus, Song et al. illustrate the mechanistic principle that modification position can influence strand selection and specificity, while Alnylam's STC-to-ESC progression shows how full-duplex chemical architectures were optimized for clinical GalNAc-siRNA performance.

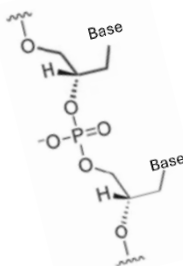
At Alnylam Pharmaceuticals, Janas and colleagues worked with Martin A. Maier and Jing-Tao Wu to evaluate the safety of 2'-F modifications in GalNAc-siRNA conjugates and to examine the transition from first-generation STC to ESC. Because some nucleoside analog drugs can cause mitochondrial toxicity, they tested whether 2'-F monomers released during siRNA metabolism posed similar risks. They compared revusiran, an STC GalNAc-siRNA, with ALN-TTRSC02/vutrisiran, an ESC GalNAc-siRNA with the same sequence, and measured 2'-F monomer release, tissue exposure, metabolic stability, and safety. Revusiran contained 22 2'-F modifications, reflecting the roughly balanced 2'OMe:2'-F composition of STC, approximately 22:22 or 1:1. In contrast, ALN-TTRSC02 contained only nine 2'-F modifications and a higher 2'-OMe content, giving an approximate 2'-OMe:2'-F ratio of 35:9, or about 4:1, assuming the same fully modified duplex length. This shift toward more 2'-OMe and less 2'-F, together with four additional terminal phosphorothioate linkages, reduced 2'-F metabolite exposure, improved metabolic stability, increased potency and duration of activity, and supported lower and less frequent dosing. Thus, ESC minimizes 2'-F metabolite exposure while preserving durable GalNAc-siRNA activity (Janas et al., 2019).

This broader principle now underlies clinically validated GalNAc-siRNA design: therapeutic siRNAs use combinations of 2'-OMe, 2'-F, phosphorothioate linkages, and ligand conjugation to protect the duplex, maintain RISC activity, reduce off-target and immune liabilities, and enable durable gene silencing. In approved and late-stage GalNAc-siRNA therapies such as givosiran, lumasiran, and inclisiran, these modification architectures support subcutaneous dosing, hepatocyte delivery, improved pharmacokinetics, and long-lasting target knockdown (Nair et al., 2014; Fitzgerald et al., 2017; Adams et al., 2020; Garrelfs et al., 2021; Raal et al., 2020).

This principle is central to Alnylam's enhanced stabilization chemistry, where 2'-OMe combined with 2'-fluoro modifications improves pharmacokinetics, reduces off-target effects, and enables durable gene

silencing in approved therapies such as givosiran, lumasiran, and inclisiran (Nair et al., 2014; Fitzgerald et al., 2017; Adams et al., 2020; Garrelfs et al., 2021; Raal et al., 2020).

Glycol Nucleic Acid (GNA)



Unlike the preceding modifications, which are derived from naturally occurring RNA nucleosides, glycol nucleic acid (GNA) represents a synthetic nucleic acid analogue. GNA contains an acyclic three-carbon glycol backbone rather than the ribose sugar found in RNA and DNA, yet it retains the ability to form Watson–Crick base pairs and stable duplex structures (Zhang et al., 2005).

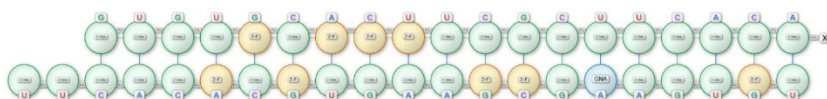
In RNA therapeutics, GNA has emerged as a valuable modification for optimizing siRNA performance. Strategic incorporation of (S)-GNA residues, particularly at position 7 within the antisense strand seed region, introduces controlled thermodynamic destabilization that reduces unintended seed-mediated off-target interactions while maintaining potent on-target gene silencing activity (Schlegel et al., 2022). GNA modifications also enhance resistance to nuclease degradation and can improve strand selection and loading into the RNA-induced silencing complex (RISC), contributing to overall therapeutic efficacy (Schlegel et al., 2017).

By enabling fine control over duplex stability and RNA structure, GNA improves specificity, reduces toxicity, and supports favorable pharmacological performance, making it an increasingly important component of next-generation siRNA therapeutics.

The success of RNA therapeutics has been driven in large part by advances in RNA chemistry. Naturally occurring modifications such as m⁵C, pseudouridine, and inosine provide mechanisms for enhancing RNA stability, regulating translation, reducing immunogenicity, and enabling programmable RNA editing, while synthetic analogues such as GNA offer

additional control over specificity and pharmacological behavior. Together, these modifications illustrate how insights from endogenous RNA biology can be translated into therapeutic design, transforming inherently unstable RNA molecules into clinically viable medicines. As RNA-based technologies continue to expand beyond vaccines into gene silencing, protein replacement, and RNA editing applications, the strategic incorporation of chemical modifications will remain central to improving efficacy, safety, and durability of therapeutic responses.

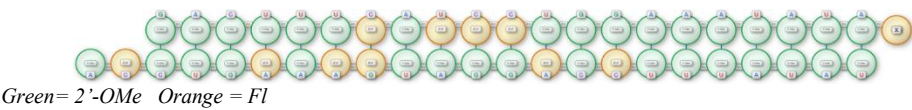
Elebsiran (ALN-HBV02)



Elebsiran (ALN-HBV02), a GalNAc-conjugated siRNA developed for chronic hepatitis B uses the “enhanced stabilization chemistry” (ESC+), within which a GNA substitution contributes to improved strand specificity and reduced off-target activity.

In Phase 1/2 studies, Elebsiran demonstrated knockdown of HBsAg, with reductions often exceeding 1.5–2.0 log₁₀ IU/mL following subcutaneous dosing at 300 mg every 4–8 weeks, and evidence of sustained activity beyond the dosing interval (Yuen et al., 2022; Gane et al., 2023). Importantly, the safety profile was favorable, with no dose-limiting toxicities and a low incidence of ALT elevations, supporting the hypothesis that chemical refinements such as GNA substitutions mitigate hepatocellular stress linked to off-target effects. Across cohorts, repeated dosing maintained efficacy without evidence of tachyphylaxis, and pharmacodynamic responses were consistent with prolonged hepatic RISC loading. (Yuen et al., 2022; Gane et al., 2023).

Lumasiran (Oxlumo)



Lumasiran (Oxlumo) is administered via subcutaneous and is an *ESC GalNAc-conjugated* siRNA, it is designed for targeted delivery to hepatocytes following systemic administration, enabling RNA interference of the HAO1 gene in the liver. Efficacy was successfully achieved in the phase 3 ILLUMINATE-A trial, where Lumasiran caused a 65% mean reduction in 24-hour urinary oxalate compared to a 12% reduction in the placebo group; this is a statistically significant difference; 54% (p<0.0001) (FDA, 2020). Approximately half of treated patients reached normalization of urinary oxalate levels by Month 6, indicating a clinically meaningful biochemical response (FDA, 2020). Consistent results were observed in younger patients in the ILLUMINATE-B study, where a 71% reduction in urinary oxalate (measured as oxalate-to-creatinine ratio) was reported, supporting efficacy across age groups (FDA, 2020). Across studies, reductions in urinary oxalate were rapid, occurring within the first few months of treatment and sustained over the study duration (FDA, 2020). The data demonstrates that Lumasiran provides substantial and consistent reductions in urinary oxalate (surrogate endpoint), supporting its efficacy in the treatment of primary hyperoxaluria type (FDA, 2020).

	siRNA	Target / indication
STC	Revusiran / ALN-TTRsc	TTR; ATTR amyloidosis
STC	ALN-AT3 / fitusiran	Antithrombin; hemophilia
ESC	ALN-TTRSC02 / vutrisiran	TTR; ATTR amyloidosis
ESC	Givosiran / GIVLAARI / ALN-AS1	ALAS1; acute hepatic porphyria
ESC	Cemdisiran / ALN-CC5	Complement C5; complement-mediated diseases
ESC	Lumasiran / ALN-GO1	HAO1; primary hyperoxaluria type 1
ESC	Inclisiran / ALN-PCSSc	PCSK9; hypercholesterolemia
ESC+	ALN-AAT02	AAT; alpha-1 liver disease
ESC+	ALN-HBV02 / VIR-2218	HBV
ESC+	ALN-AGT	Angiotensinogen; hypertension

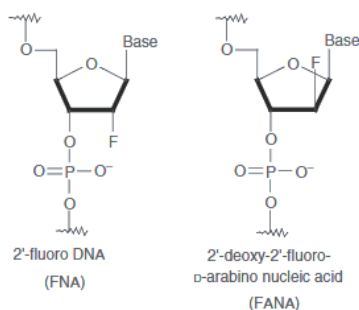
Revusiran highlights an important disconnect between preclinical toxicology, early clinical tolerability, and late-stage clinical outcome. Revusiran was a first-generation GalNAc–siRNA in which the GalNAc ligand was attached to the 3' end of the sense strand; the duplex contained 22 2'-OMe and 22 2'-F residues, with two phosphorothioate linkages at the 3' end of the antisense strand to increase nuclease resistance (Sutherland et al., 2020). This high 2'-F content raised a reasonable safety question because some fluorinated nucleoside analogues have caused mitochondrial toxicity in humans. The classic example is FIAU, a fluorinated antiviral nucleoside analogue developed for hepatitis B, which caused severe multisystem toxicity, liver failure, lactic acidosis, and mitochondrial injury after repeat dosing. Mechanistically, FIAU toxicity was linked to mitochondrial dysfunction, including abnormal swollen hepatic mitochondria with disrupted cristae. Because revusiran metabolism could theoretically release 2'-F nucleoside or nucleotide monomers, Sutherland et al. specifically examined whether chronic exposure produced a similar mitochondrial injury pattern (Sutherland et al., 2020).

The nonclinical data did not show a FIAU-like toxicity profile. In monkeys, revusiran produced no cardiovascular or respiratory effects at doses up to 100 mg/kg, no neurological abnormalities in repeat-dose studies, and no dose-limiting toxicity. In chronic monkey studies, the NOAEL was 200 mg/kg, the highest dose tested in the 39-week study, despite strong pharmacologic reductions in TTR, vitamin A, and thyroxine (Sutherland et al., 2020). Rats were more sensitive: the NOAEL was 30 mg/kg in both the 6- and 26-week repeat-dose studies, based mainly on reversible liver findings and associated clinical chemistry changes at higher doses, including hepatocellular vacuolation and modest elevations in ALT, AST, ALP, and GGT. These findings suggested species-specific sensitivity and liver adaptation or accumulation, but they were largely reversible and did not progress to overt organ failure (Sutherland et al., 2020). Revusiran also showed no genotoxicity and no carcinogenicity in a two-year rat study at doses up to 100 mg/kg. Importantly, liver ultrastructural assessments in the 26-week rat and 39-week monkey studies did not reveal mitochondrial damage. In the two-year rat carcinogenicity study, investigators did observe enlarged, elongated, ring-shaped, or cup-shaped mitochondria in liver and skeletal muscle, but these changes lacked the swollen morphology and cristae disruption typical of FIAU-associated mitochondrial toxicity and were not accompanied by functional evidence such as elevated plasma lactate (Sutherland et al., 2020).

Janas et al. further addressed the 2'-F concern by testing whether 2'-F metabolites from GalNAc-siRNAs behave like toxic nucleoside analogues. They found that 2'-F monomer exposure from revusiran and ALN-TTRSC02 was low and transient in rats and humans, and that 2'-F triphosphates were neither preferred substrates nor strong inhibitors of human polymerases. They also found no obligate or non-obligate chain termination, which argues against the kind of polymerase-mediated mitochondrial toxicity associated with some nucleoside analog drugs (Janas et al., 2019). Some in vitro assays did show modest reductions in cell viability and mitochondrial DNA in selected cell types at high 2'-F nucleoside concentrations, so the data do not prove that 2'-F monomers are biologically inert under all conditions. However, those effects generally occurred at concentrations above those achieved in vivo, and long-term rat studies did not show significant 2'-F monomer accumulation or clear mitochondrial functional injury (Janas et al., 2019; Sutherland et al., 2020).

Clinically, the early human data was also reassuring. In phase 1 healthy volunteers, Zimmermann et al., reported that single and multiple subcutaneous doses of revusiran were generally well tolerated, with injection-site reactions as the most common adverse events and no serious adverse events, treatment discontinuations, or clinically meaningful renal, hematologic, thyroid, cytokine, CRP, or urinalysis abnormalities. Multiple dosing produced approximately 90% mean serum TTR reduction, confirming potent GalNAc-mediated delivery and RNAi activity in humans (Zimmermann et al., 2017). The discrepancy emerged later in the phase 3 ENDEAVOUR trial in patients with cardiomyopathic hATTR amyloidosis, where development was stopped after an imbalance in deaths in the revusiran arm. Sutherland et al. note that no conclusive causal relationship was established, but a drug-related effect could not be excluded (Sutherland et al., 2020). Overall, the available evidence suggests that revusiran's phase 3 mortality signal was not clearly explained by intrinsic 2'-F mitochondrial toxicity. Instead, the mismatch between preclinical/phase 1 safety and ENDEAVOUR may reflect a combination of species differences, the use of healthy animals and healthy phase 1 volunteers, the older and medically fragile cardiomyopathy population in phase 3, advanced heart failure, disease-related vulnerability, and the relatively high-burden revusiran dosing regimen.

2'-Fluoro & 2'-Fluoro-arabino



2'-Fluoro modifications, including 2'F-ANA (fluoro-arabino nucleic acid) and related FNA chemistries are highly potent nucleic acid modifications in which a fluorine atom replaces the 2'-hydroxyl group of the sugar. This change enhances nuclease stability and strengthens target binding, while also supporting steric-blocking mechanisms; in some cases, these modifications can enable gymnotic cellular uptake without delivery systems.

Inclisiran (Leqvio)



Inclisiran (Leqvio) is a chemically modified, double-stranded small interfering RNA (siRNA) conjugated to GalNAc to tissue-selectively knock down PCSK9 mRNA, leading to reduced hepatic PCSK9 synthesis and increased LDL receptor recycling, thereby lowering circulating low-density lipoprotein cholesterol (LDL-C) (FDA, 2021; EMA, 2020). It is indicated as an adjunct to diet and lipid-lowering therapy, including statins, for patients with primary hypercholesterolemia or atherosclerotic cardiovascular disease who require additional LDL-C reduction, and is administered subcutaneously with an extended dosing interval of every 6 months; illustrating a robust and long hepatic half-life (FDA, 2021; EMA, 2020).

Nonclinical and clinical studies demonstrate dose-dependent and sustained reductions in PCSK9 and LDL-C, with distribution primarily to the liver, metabolism via nuclease degradation to shorter RNA fragments, and elimination mainly through urine and feces (FDA, 2021). Toxicological findings were largely non-adverse and associated with tissue accumulation,

with no evidence of genotoxicity, carcinogenicity, or reproductive toxicity, supporting a favorable nonclinical safety profile (FDA, 2021).

Givosiran (GIVLAARI)

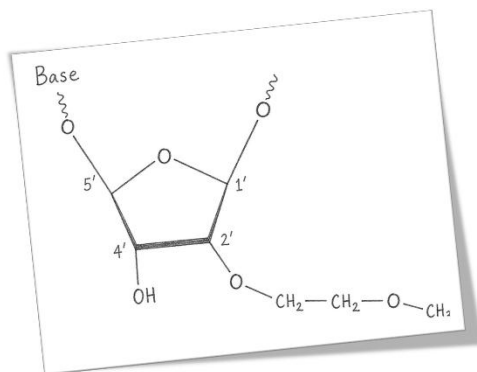


Givosiran (GIVLAARI) is a small interfering RNA (siRNA) therapeutic indicated for the treatment of adults with acute hepatic porphyria (AHP). It acts by targeting Aminolevulinate Synthase 1 (ALAS1) mRNA in hepatocytes, thereby reducing the accumulation of neurotoxic intermediates such as aminolaevulinic acid (ALA) and porphobilinogen (PBG). The drug is administered once a month through subcutaneous injection at a dose of 2.5 mg/kg (Alnylam Pharmaceuticals, Inc., 2019) or about 180mg per dose for an average person.

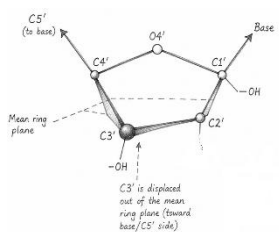
Clinical studies have demonstrated that Givosiran significantly reduces the frequency of porphyria attacks, with patients experiencing approximately 70% fewer attacks compared to placebo in the ENVISION trial (*page 10 on the drug label/US Prescribing Information (USPI)*). Common adverse effects include nausea and injection-site reactions, while more serious risks include hepatic toxicity, renal impairment, and rare anaphylactic reactions (Alnylam Pharmaceuticals, Inc., 2019).

2'F-ANA (2'-fluoro-arabino nucleic acid) is a chemically modified nucleotide in which substitution at the 2' position of the sugar enhances nuclease resistance and increases binding affinity to complementary RNA, while maintaining compatibility with RNase H activity in gapmer designs (Souleimanian et al., 2012). These properties have made 2'F-ANA a useful modification in preclinical studies, where it has demonstrated potent and sustained gene silencing. However, despite its favorable biochemical profile, this chemistry has not yet been translated into clinically approved oligonucleotide therapeutics and remains primarily confined to experimental applications.

2'-O-Methoxyethyl (2'-MOE) / MOE chemistry



In 1995, Pierre Martin, a chemist at Ciba-Geigy in Basel, Switzerland, reported the synthesis and properties of 2'-O-alkylated ribonucleosides, including the 2'-O-(2-methoxyethyl) modification, in *Liberation Serif Chimica Acta* (Martin, 1995; Hill & Hall, 2023). Martin's work was motivated by the need for ribose modifications that were compatible with solid-phase oligonucleotide synthesis and that could protect oligonucleotides from nuclease degradation without compromising hybridization to RNA. His synthetic approach provided access to 2'-O-alkylribonucleosides in which the ribose 2'-hydroxyl group was replaced with ethylene glycol-derived substituents, including what is now known as the MOE group. Oligonucleotides containing this modification displayed what Martin described as “surprising properties”: increased affinity and specificity for RNA targets, as well as enhanced resistance to nuclease digestion.



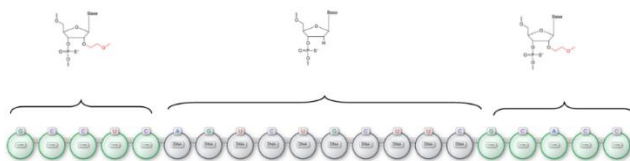
The beneficial properties of MOE-modified oligonucleotides are closely linked to their effects on duplex structure and stability. Martin's original studies showed that incorporation of MOE residues increased melting temperature against complementary RNA by approximately 0.9–1.6 °C per modification, while also improving discrimination against

mismatched RNA sequences. Subsequent structural studies helped explain these observations by showing that the MOE substituent favors a C3'-endo ribose sugar pucker, a conformation associated with A-form RNA duplexes. This preorganization enhances RNA binding while maintaining sequence specificity. Importantly, MOE modification also improves nuclease resistance; in serum stability experiments, MOE-containing oligonucleotides persisted far longer than unmodified or 2'-O-methyl-modified comparators. Together, these features made MOE chemistry particularly attractive for therapeutic ASO design.

As antisense technology evolved, MOE chemistry became a defining feature ASOs, especially when combined with phosphorothioate (PS) backbone modification. Early PS ASOs improved nuclease stability and tissue distribution but were limited by reduced RNA-binding affinity and pro-inflammatory liabilities. Incorporation of MOE-modified nucleotides helped address these shortcomings by increasing potency, extending tissue residence time, and improving tolerability. In the commonly used “gapmer” architecture, MOE-modified nucleotides are placed in the flanking “wings” of an ASO to enhance affinity and stability, while a central DNA gap supports RNase H-mediated cleavage of the target RNA. MOE chemistry is also used in fully modified splice-switching oligonucleotides, where the goal is not RNA degradation but modulation of pre-mRNA splicing.

These combined chemical innovations enabled the clinical translation of MOE-modified ASOs and helped establish oligonucleotides as a major therapeutic modality. MOE chemistry is present in multiple approved drugs, including mipomersen, nusinersen, inotersen, and volanesorsen, and it remains a foundation for numerous candidates in clinical development. Mipomersen provided an early proof of concept for RNase H-recruiting MOE gapmers capable of reducing disease-associated protein expression through mRNA targeting, while nusinersen demonstrated the therapeutic power of fully modified MOE ASOs for splice correction in spinal muscular atrophy. More broadly, the success of MOE-modified ASOs illustrates how rational chemical modification can transform fragile nucleic acids into drug-like molecules with tunable activity, durability, and tissue distribution.

Inotersen



Inotersen is administered as a once-weekly subcutaneous injection and has demonstrated clinical efficacy in phase II/III NEURO-TTR randomized controlled trial. The study showed that inotersen significantly slowed the progression of neuropathy and improved quality of life compared with placebo, as measured by the modified Neuropathy Impairment Score and the Norfolk Quality of Life–Diabetic Neuropathy questionnaire (CADTH, 2020). However, despite achieving statistical significance, the clinical meaningfulness of these improvements remains uncertain due to the absence of clearly defined thresholds for clinically important differences (CADTH, 2020).

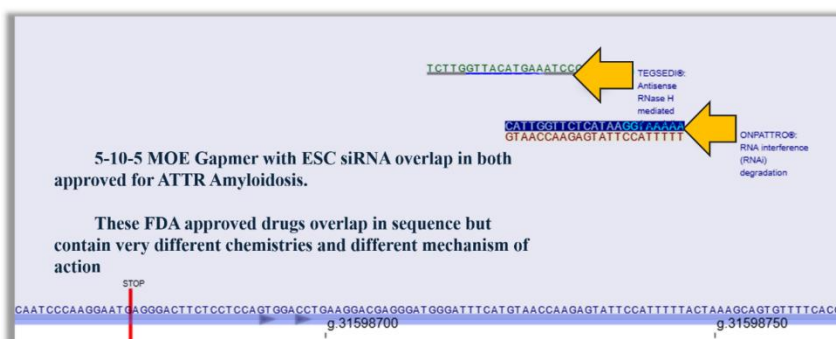
Safety remains a key concern, as treatment is associated with a high incidence of adverse events, including thrombocytopenia, renal impairment, and injection-site reactions. Serious adverse events and deaths have been reported in clinical trials, necessitating regular monitoring of platelet counts and renal function (CADTH, 2020). Clinical guidance further emphasizes the importance of ongoing monitoring through regular blood and urine tests to assess both efficacy and toxicity (Mayo Clinic, n.d.).

Inotersen can reduce platelet counts, increasing the risk of bleeding and infection, and patients are advised to take precautions to minimize injury and exposure to infectious agents (Mayo Clinic, n.d.). It has also been associated with serious organ-related complications, including kidney and liver toxicity, with symptoms such as reduced urine output, swelling, jaundice, or abdominal pain requiring immediate medical attention (Mayo Clinic, n.d.). In addition, there is an elevated risk of inflammatory or immune-mediated conditions, stroke, and arterial dissection, highlighting the need for vigilance regarding neurological symptoms (Mayo Clinic, n.d.).

Hypersensitivity reactions, including life-threatening allergic responses, have also been reported, alongside reductions in vitamin A levels that may lead to visual disturbances, necessitating ophthalmologic monitoring in some patients (Mayo Clinic, n.d.). Patients are further advised to consult their healthcare provider before taking any concomitant

medications, including over-the-counter or herbal products, due to potential interactions (Mayo Clinic, n.d.).

Overall, Inotersen represents an important advancement in RNA-targeting therapeutics and offers a more convenient, self-administered alternative to intravenous treatments. However, its use is limited by safety concerns and ongoing uncertainties regarding long-term efficacy and broader clinical benefits (CADTH, 2020; Rüger et al., 2020).



MOE-modified ASOs are widely used in two RNA-targeting strategies: RNase H-mediated RNA degradation and splice modulation via steric blocking. These two approaches enable treatment of a broad range of genetic diseases (Sang et al., 2024). For example, MOE-containing gapmer ASOs such as Inotersen and Eplontersen are used in the treatment of hereditary transthyretin amyloidosis (hATTR), where they bind transthyretin (TTR) mRNA and promote its degradation, thereby reducing pathogenic protein accumulation (Sang et al., 2024). Similarly, Tofersen, a 5-10-5 MOE-modified gapmer ASO approved for amyotrophic lateral sclerosis (ALS), targets mutant SOD1 mRNA, demonstrating the utility of MOE chemistry in neurodegenerative disease through selective reduction of toxic protein expression (Sang et al., 2024). In the context of splice modulation, Nusinersen (Spinraza) discovered at Cold Spring Harbor uses a fully MOE-modified ASO to promoting inclusion of exon 7 in SMN2 mRNA and thereby restoring production of functional SMN protein (Sang et al., 2024). Beyond these established therapies, MOE chemistry has also enabled highly personalized approaches, exemplified by Milasen, a patient-specific ASO designed to correct aberrant splicing in Batten disease, highlighting the adaptability of MOE-modified oligonucleotides for precision medicine (Sang et al., 2024). This chemistry is heavily used by nLorem for their nano-rare therapeutics programs.

Incorporation of MOE at the 2'-position of the ribose increases conformational rigidity and enhances hybridization while reducing susceptibility to nuclease-mediated cleavage (Zanardi et al., 2018). From a pharmacokinetic and pharmacodynamic (PK/PD) perspective, MOE-modified ASOs exhibit slow plasma clearance, high plasma protein binding, and extensive tissue distribution—particularly to the liver, kidney, and adipose tissue—resulting in long tissue half-lives that can extend from weeks to months (Takakusa et al., 2023). For example, 2'-MOE ASOs such as Mipomersen demonstrate terminal half-lives of 30 days in plasma and similar persistence in liver tissue, reflecting equilibrium between plasma and tissue compartments and supporting infrequent dosing regimens (Takakusa et al., 2023). These prolonged kinetics arise from strong protein binding and intracellular retention following endocytosis and endosomal trafficking, which enable sustained pharmacological activity even after plasma levels decline (Takakusa et al., 2023). Importantly, pharmacodynamic effects are more closely correlated with tissue concentrations than plasma levels, highlighting the importance of tissue PK in dose optimization and therapeutic efficacy (Christensen et al., 2025). Furthermore, advances in chemical stabilization and conjugation strategies have enabled improved delivery beyond the liver, expanding the therapeutic reach of oligonucleotides to extrahepatic tissues such as muscle, brain, and lung (Anand et al., 2025).

Fomivirsen (Vitraven[®])	Phosphorothioate backbone (PS)	Cytomegalovirus (CMV) retinitis	1999
Mipomersen (Kynamro[®])	PS & 2'-MOE	Homozygous familial hypercholesterolemia	2013
Nusinersen (Spinraza[®])	PS & 2'-MOE	Spinal muscular atrophy	2016
Inotersen (Tegsedi[®])	PS & 2'-MOE	Hereditary transthyretin-mediated amyloidosis	2018
Milasen	PS & 2'-MOE	CLN7 gene mutation associated with Batten disease	2018
Volanesorsen (Waylivra[®])	2'-MOE	Familial chylomicronemia syndrome	EMA approved 2019
Tofersen (Qalsody[®])	PS & 2'-MOE	Amyotrophic lateral sclerosis with SOD1 mutation	2023
Eplontersen (Wainua[®])	PS, 2'-MOE & GalNAc	Polyneuropathy of hereditary transthyretin-mediated amyloidosis	2023

Volanesorsen (Waylivra)

Volanesorsen (Waylivra): ApoC-III targeting ASO for familial chylomicronemia syndrome (FCS)

Base:	A G mC mU mU mC T T G T mC mC A G mC mU mU mU A mU
Backbone	Full phosphorothioate (PS)
Sugar:	E E E E E d d d d d d d d d d E E E E E

legend: E = 2'-MOE d = DNA (2'-deoxy)

Volanesorsen demonstrated substantial triglyceride-lowering efficacy but is associated with several safety considerations. The most clinically significant risk is thrombocytopenia, including cases of severe platelet reduction. In addition, injection-site reactions are very common, occurring in a substantial proportion of administrations, although these are generally mild to moderate and primarily affect tolerability rather than safety. Rare but serious immune-mediated events, such as serum sickness, have also been reported, indicating potential for hypersensitivity reactions. (Witztum et al., 2019; Tardif et al., 2016).

Olezarsen, a GalNAc-conjugated version of the Volanesorsen ASO was developed to solve its relatively poor safety profile. Thrombocytopenia with Volanesorsen was attributed to high systemic exposure and dosing burden. By incorporating N-acetylgalactosamine (GalNAc) for hepatocyte-targeted delivery, Olezarsen matches efficacy with less frequent doses, thereby reducing systemic exposure. Clinical data demonstrate that while some adverse events persist, including mild thrombocytopenia and liver enzyme elevations, these effects are dose-dependent and more pronounced at higher doses, with overall safety outcomes approaching those of placebo. This supports the concept that dose and exposure are central determinants of antisense oligonucleotide toxicity, and that GalNAc conjugation can effectively improve the therapeutic index by decoupling efficacy from systemic toxicity (Marston et al., 2026; Witztum et al., 2019).

	Volanesorsen	Olezarsen (GalNAc)
Dosing	High dose (e.g., 300 mg weekly → biweekly)	Lower dose (50–80 mg monthly)
Delivery	Systemic (no targeting)	Hepatocyte-targeted (GalNAc)
Thrombocytopenia	Clinically significant; severe cases reported	Less frequent; dose-dependent, higher at 80 mg
Injection-site reactions	Very common (~24% of injections)	Less frequent (~10–16%)
Liver enzyme elevations	Not dominant signal	Increased, especially at higher dose (80 mg)
Hepatic fat	No consistent increase	Dose-dependent increase observed
Hypersensitivity / immune	Rare (e.g., serum sickness)	Mild hypersensitivity events slightly increased
AE rate	Higher discontinuation vs placebo	Similar to placebo overall
Key safety driver	Systemic exposure / high dose	Residual dose-dependent toxicity, reduced by targeting

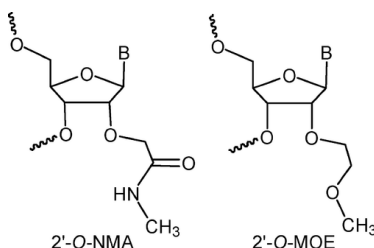
IONIS-HTTRx (Tominersen, Roche/Ionis)



In the Phase III GENERATION HD1 trial, patients received Tominersen via intrathecal injection at doses of 120 mg every 8 weeks or every 16 weeks. Although the drug successfully lowered huntingtin protein levels, it failed to improve clinical outcomes and in some cases worsened disease progression, particularly with more frequent dosing. This lack of efficacy is thought to result from non-selective reduction of both mutant and normal huntingtin along with possible overdosing and intervention occurring too late in the disease course.

Nusinersen (Spinraza®)

Nusinersen (Spinraza®) is a fully modified 2'-O-methoxyethyl (MOE) antisense oligonucleotide with a phosphorothioate backbone, designed to modulate SMN2 splicing by promoting inclusion of exon 7. In the ENDEAR (NCT02193074) and CHERISH (NCT02292537) clinical trials, this sterically blocking MOE chemistry demonstrated substantial therapeutic benefit in patients with spinal muscular atrophy (SMA), leading to significant improvements in motor function and survival compared with untreated cohorts (Finkel et al., 2017; Mercuri et al., 2018). The clinical efficacy of nusinersen reflects key properties of fully modified MOE ASOs, including high binding affinity, strong resistance to nuclease degradation and prolonged tissue residence within the central nervous system following intrathecal administration (Geary et al., 2015). This durability supported maintenance dosing at four-month intervals and established MOE-modified splice-switching oligonucleotides as a clinically validated platform for CNS splicing modulation.

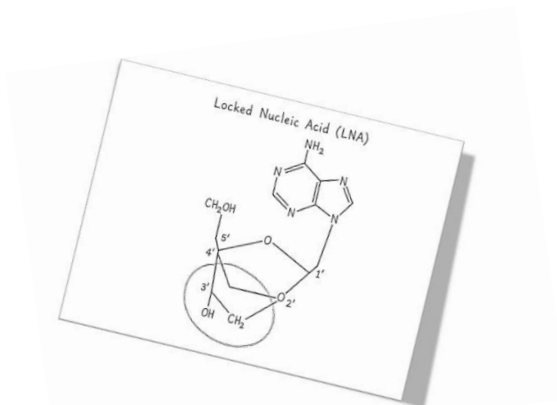


However, because CNS-directed SSOs must be administered intrathecally, further improving potency and durability remains an important therapeutic goal. Longer-lasting molecules could reduce the frequency of

lumbar punctures, lessen treatment burden for patients and caregivers and potentially improve clinical outcomes by sustaining target engagement over extended periods. These considerations have driven the development of next-generation ribose modifications that retain the favourable properties of MOE while improving pharmacological efficiency. One such chemistry is 2'-O-[2-(methylamino)-2-oxoethyl] (NMA), a 2'-O-substituted sugar modification structurally related to MOE but distinguished by an amide-containing side chain. NMA originated from efforts to improve second-generation ASO chemistries by enhancing RNA binding, nuclease resistance and metabolic stability while maintaining tolerability. Early studies showed that NMA-modified oligonucleotides could be efficiently synthesized and incorporated into ASOs, preserving strong hybridization to complementary RNA and showing pharmacological activity comparable to MOE-modified analogues (Prakash et al., 2003; Pattanayek et al., 2004; Prakash et al., 2008).

More recent work has extended NMA chemistry to splice-switching oligonucleotides, where it appears to offer advantages over MOE in CNS-relevant applications. In preclinical models of SMA, NMA-modified SSOs targeting the same SMN2 splicing regulatory region as nusinersen produced greater exon 7 inclusion at lower doses, with optimized NMA candidates showing approximately three- to four-fold higher potency than MOE comparators while maintaining prolonged activity in CNS tissues (Ling et al., 2026). This improved activity was not explained by increased tissue accumulation, suggesting that NMA enhances intracellular pharmacology rather than simply improving delivery or exposure. Potential mechanisms include altered intracellular trafficking, improved nuclear availability or modified interactions with RNA–protein complexes involved in splicing (Ling et al., 2026). Collectively, these findings position NMA as a next-generation chemistry for steric-blocking ASOs, with the potential to improve potency, extend duration of action and reduce dosing frequency for intrathecally administered CNS splicing therapies.

Locked Nucleic Acids (LNA)



Locked nucleic acids were introduced in the late 1990s:

Obika, Satoshi; Nanbu, Daishu; Hari, Yoshiyuki; Morio, Ken-ichiro; In, Yasuko; Ishida, Toshimasa; Imanishi, Takeshi (1997-12-15). "Synthesis of 2'-O,4'-C-methyleneuridine and -cytidine. Novel bicyclic nucleosides having a fixed C3', -endo sugar puckering". *Tetrahedron Letters*. 38 (50): 8735–8738. Doi:10.1016/S0040-4039(97)10322-7. ISSN 0040-4039.

These nucleosides incorporate a 2'-O,4'-C methylene bridge that locks the ribose ring into a C3'-endo conformation characteristic of RNA duplexes (Koshkin et al., 1998; Obika et al., 1998).

Unlike earlier modifications, LNA technology was a shift toward structure-driven nucleic acid engineering. Constrained ethyl (cEt) chemistry is a type of LNA modification that incorporates a 2'-O-ethyl substituent within a constrained bicyclic framework; used primarily by Ionis Pharmaceuticals. Structural analyses show that cEt and LNA adopt highly similar sugar conformations, while cEt positions its substituents closer to the phosphate backbone, contributing to enhanced resistance to enzymatic degradation (Pallan et al., 2012).

	ΔT_m per modification ($^{\circ}\text{C}$)	Relative affinity vs DNA
DNA (unmodified)	Baseline	
Phosphorothioate (PS DNA)	0	0.8–1 $\times$
2'-O-methyl (2'-OMe)	+0.5 to +1.0	1.2–1.5 $\times$
2'-fluoro (2'-F)	+1.0 to +2.0	1.5–2 $\times$
2'-O-methoxyethyl (2'-MOE)	+0.9 to +1.7	1.5–2 $\times$
Locked nucleic acid (LNA)	+3.5 to +8.0	3–10 $\times$

LNA as a biological tool

Locked nucleic acid (LNA) chemistry has become an important component of modern PCR-based technologies, particularly in applications requiring high specificity, sensitivity, and reproducibility. The enhanced hybridization affinity conferred by LNA modifications enables the design of short oligonucleotide primers and probes with elevated and tunable melting temperatures (T_m), which improves assay performance across a wide range of targets (Braasch and Corey, 2001; Grunweller and Hartmann, 2007).

Incorporation of LNA residues into PCR primers and fluorescent probes:

- Improved binding to difficult or GC-rich templates
- Shorter primer design without loss of affinity
- Enhanced mismatch discrimination
- Enhanced fluorescence signal upon hybridization
- Enable stable hybridization to short targets

Strategic placement of LNA nucleotides often near the 3'-end can significantly improve allele-specific amplification, making LNA-modified primers particularly useful for single nucleotide polymorphism (SNP) detection and genotyping assays (Di Giusto and King, 2004; Mouritzen et al., 2003). The increased thermodynamic stability also reduces nonspecific amplification by favoring correct primer–template pairing. (Morandi et al., 2007; Valoczi et al., 2004; Castoldi et al., 2006).

Miravirsen is a first-in-class LNA-modified antisense oligonucleotide that targets miR-122, a liver microRNA essential for HCV replication, leading to durable, dose-dependent viral suppression with a strong safety profile and no observed resistance. Its success validated LNA chemistry as a viable and safe platform for RNA-targeting therapeutics.

Miravirsen (Santaris/Roche) – anti-miR-122 (HCV)

L d L d d L L d d L d L d L L L = LNA ; d = DNA

Miravirsen is a 15-nucleotide LNA-modified phosphorothioate antisense oligonucleotide designed to bind and sequester miR-122, a liver-specific microRNA that is essential for hepatitis C virus (HCV) replication. miR-122 normally binds to two conserved sites in the 5' untranslated region of the HCV genome, enhancing viral RNA stability and replication. By forming a highly stable heteroduplex with miR-122, miravirsen blocks this interaction, functionally inhibiting the microRNA, destabilizing HCV RNA, and suppressing viral replication. In a Phase 2a clinical study, miravirsen was administered as five weekly subcutaneous injections at doses of 3, 5, or 7 mg/kg and demonstrated dose-dependent reductions in HCV RNA, with a 1.2 log₁₀ IU/mL reduction at 3 mg/kg and approximately 2.9–3.0 log₁₀ IU/mL reductions at the 5–7 mg/kg doses. Viral suppression persisted beyond the dosing period, and undetectable HCV RNA was observed in multiple patients receiving higher doses. Key differentiators of miravirsen include the absence of observed viral resistance or escape mutations, activity across HCV genotypes due to conserved miR-122 binding sites, favorable pharmacokinetics with a prolonged tissue half-life, and a lack of cytochrome P450 interactions, which reduces the risk of drug–drug interactions. The safety profile was also favorable, with no dose-limiting toxicities, mostly mild to moderate adverse events such as headache and fatigue, no significant hepatotoxicity or clinically relevant laboratory abnormalities, and reductions in cholesterol levels consistent with the known biology of miR-122.

Cobomarsen is an LNA-based anti-miR-155 therapy for cutaneous T-cell lymphoma that showed encouraging early clinical results, including a favorable safety profile and disease improvement in most patients, supporting its potential as a novel microRNA-targeted cancer treatment.

L d L d d L L d L d L L L L = LNA ; d = DNA

Phase 1 clinical trial evaluated Cobomarsen in patients with mycosis fungoides (MF), the most common subtype of CTCL, assessed safety, tolerability, pharmacokinetics, and preliminary efficacy across multiple routes of administration, including intralesional, subcutaneous (SC), and intravenous (IV) delivery at doses up to 900 mg.

Cobomarsen demonstrated a favorable safety profile, with no serious adverse events attributed to the drug. The most reported AE was fatigue, neutropenia, injection-site pain, nausea, pruritus, and headache. Dose-limiting toxicities were limited to Grade 3 pruritus and tumor flare, and the maximum tolerated dose was not reached.

91% showed improvement in disease burden as measured by the modified Severity Weighted Assessment Tool mSWAT -- the percentage of Body Surface Area affected by patches, plaques, and tumors.

Importantly, improvements in clinical endpoints were accompanied by enhanced quality of life and meaningful patient benefits beyond tumor reduction. These results highlight the potential therapeutic potential of LNA-based anti-miR strategies in oncology. The strategy is not without controversy, however. Targeting microRNA can have unintended consequences as the intended target typically regulates hundreds of other genes. Therefore, it is essential to understand the biological consequences of a potential “on-target” tox effect.

RG-101 is a GalNAc-conjugated anti-miR-122 therapy



RG-101 is a GalNAc-conjugated anti-miR-122 therapy that showed strong antiviral activity and good initial tolerability but later revealed safety concerns due to transient hyperbilirubinemia from off-target effects. This led to the development of RG6650, an optimized follow-up compound designed to maintain efficacy while improving safety by reducing interference with bilirubin transport.

RG-101	LNA/MOE/DNA	Clinically strong antiviral, but safety issue
RG6650	cEt/DNA	selected for improved safety

In early clinical studies (phase 1B study), RG-101 was well tolerated after single-dose administration, with no serious adverse events and mostly mild, self-limiting side effects such as fatigue and injection site reactions (van der Ree et al., 2017). Transient increases in alkaline phosphatase and a single case of cholestasis were observed, but there was no consistent evidence of hepatocellular injury or significant bilirubin elevation across the

cohort (van der Ree et al., 2017). Later studies of RG-101 demonstrated strong antiviral efficacy but showed transient hyperbilirubinemia that was attributed to off-target inhibition of the MRP2 transporter (van der Ree et al., 2017; Drygin, 2022). Following this finding, Regulus conducted a focused optimization effort to identify next-generation compounds that retained potent miR-122 inhibition while minimizing interference with bilirubin transport (Drygin, 2022). This work involved a screening and structure–activity relationship (SAR) campaign that ultimately leading to the identification of RG6650 (Drygin, 2022).

	RG-101	RG6650 / RG7443
Base	A m5C A m5C m5C A m5U T G U C A C A C T C C A	U C A C A C T C C
Backbone	Full phosphorothioate	Full phosphorothioate
Sugar	E E E E E E E d d L L d L d L d L L d	K K d K d K d K K
Legend	E = 2'-MOE; d = DNA; L = LNA; A = adenine; T = thymine; G = guanine; C = cytosine; U = uracil; m ⁵ C = 5-methylcytosine; m ⁵ U = 5-methyluracil.	
Conjugation	3' GalNAc-conjugated	3' GalNAc-conjugated

EZN-4176



In early clinical development, the LNA gapmer EZN-4176 exemplified the balance between potency and safety that characterizes high-affinity antisense chemistries. In a first-in-human phase I study in patients with castration-resistant prostate cancer, EZN-4176 demonstrated limited clinical activity despite promising preclinical rationale. Treatment was generally tolerated at lower dose levels; however, dose escalation was constrained by hepatotoxicity, with reversible elevations in alanine aminotransferase (ALT) and aspartate aminotransferase (AST) observed as dose-limiting toxicities at 10 mg/kg (Bianchini et al., 2013). These transaminase elevations occurred in the absence of concomitant bilirubin increases, suggesting hepatocellular injury rather than cholestatic dysfunction (Bianchini et al., 2013). Despite achieving systemic exposures consistent with pharmacological activity, no meaningful reduction in androgen receptor (AR) expression or sustained antitumor responses were observed, and the study was discontinued (Bianchini et al., 2013).

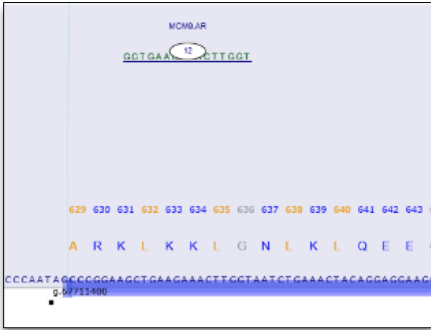
Mechanistically, these findings are consistent with the known properties of locked nucleic acids. The increased binding affinity conferred by LNA chemistry enhances target hybridization but also increases the likelihood of hybridization-dependent off-target effects and nonspecific interactions with cellular macromolecules. Both preclinical and clinical evidence suggest that such interactions may drive hepatic accumulation and toxicity, a phenomenon observed across multiple LNA-based oligonucleotides (Swayze et al., 2007; Bianchini et al., 2013). In the case of EZN-4176, its high-affinity binding profile likely contributed not only to hepatotoxicity but also to reduced specificity, limiting its ability to achieve selective and effective AR suppression in patients.

Importantly, although EZN-4176 was designed to target exon 4 of AR mRNA and suppress both full-length and splice variant receptors, clinical biopsy data failed to demonstrate meaningful AR downregulation following treatment (Bianchini et al., 2013). This disconnect between target engagement in preclinical models and clinical outcomes underscores a central challenge in antisense drug development: increasing binding affinity can improve in vitro potency but may also exacerbate off-target effects, narrow the therapeutic window, and ultimately hinder clinical translation. However, it should be noted that this compound also hits MCM9, a DNA repair gene linked to cancer. This seems to have been a bioinformatics oversight likely because the genome was not fully annotated at the time.

These limitations parallel observations from other antisense programs targeting the androgen receptor. For example, EZN-4176 can be contrasted with later-generation compounds such as ISIS-AR Rx (AZD5312), which incorporates some constrained ethyl (cEt) modifications designed to retain potency while improving tolerability and specificity. While cEt chemistry reduces some of the toxicity risks associated with LNA, clinical outcomes have shown that improved safety alone does not guarantee efficacy, particularly if sufficient target engagement or downstream biological effect is not achieved.

Taken together, these findings highlight a fundamental tension in antisense therapeutics: while increasing binding affinity (as seen with LNA chemistry) can enhance potency, it often comes at the cost of increased off-target interactions and toxicity. Conversely, newer chemistries such as cEt may improve safety margins but do not inherently ensure clinical success. Achieving an optimal balance between affinity, specificity, and tolerability remains a key determinant of successful antisense drug development.

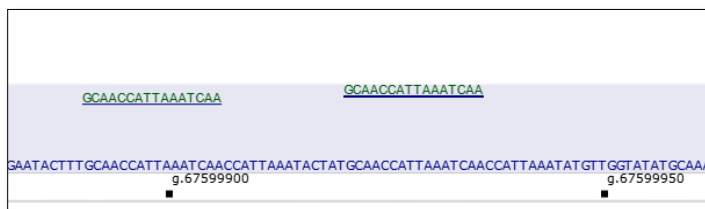
	EZN-4176	ISIS-AR Rx (AZD5312)
Developer	Enzon	Ionis / AstraZeneca
Chemistry	LNA gapmer	cEt gapmer
Target	Androgen receptor (AR)	Androgen receptor (AR)
Backbone	Phosphorothioate	Phosphorothioate
Sugar mods	LNA + DNA	cEt + DNA
Clinical outcome	ALT elevations (toxicity signal)	Developed to improve potency/safety



The Enzon–Santaris LNA gapmer program progressed into the clinic by targeting oncogenic drivers like AR and PI3K/Akt, but the same compound also unintentionally suppressed MCM9, a DNA repair gene linked to cancer predisposition. --a critical failure in bioinformatics off-target analysis that made it all the way into the clinic!

Feature	ISIS-AR Rx / AZD5312 (AKA ISIS 560131)
Base	TTGATTTAATGGTTGC
Backbone	Full phosphorothioate
Sugar	k k k d d d d d d d d k k E

Legend k = (S)-cEt; d = DNA; E = 2'-MOE; C = 5-methylcytosine

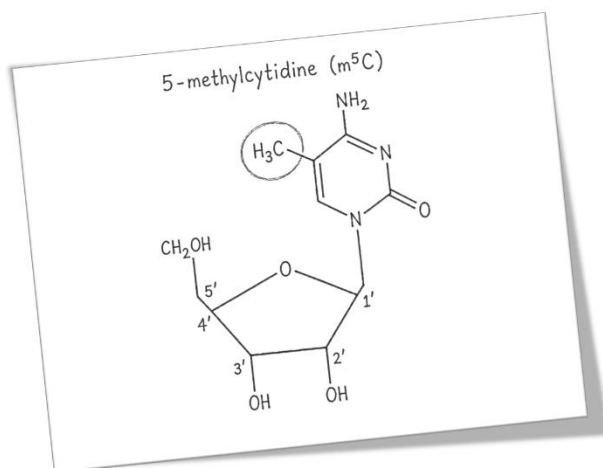


ISIS AR RX HITS ANDROGEN RECEPTOR TWICE.

Recent data (2024–early 2026) indicate that AZD5312 shows promising efficacy in prostate cancer with observed PSA declines in some heavily pre-treated patients and enhanced activity in combination with AKT inhibitors (e.g., Capivasertib). Its safety profile is consistent with antisense oligonucleotides, with common adverse events including fatigue, injection site reactions, and flu-like symptoms, while more serious but manageable risks include thrombocytopenia, hepatotoxicity, and renal effects requiring routine monitoring (AstraZeneca, 2024; ClinicalTrials.gov, 2025; de Bono et al., 2023).

Locked nucleic acid (LNA) chemistry is one of the most potent antisense modification strategies for RNA targeting due to its conformationally “locked” sugar, which markedly increases binding affinity and duplex stability relative to earlier chemistries such as 2'-O-methoxyethyl (MOE) and 2'-O-methyl modifications (Qassem et al., 2024; Crooke et al., 2018). This enhanced affinity enables shorter oligonucleotides, improved potency, and flexible designs (e.g., LNA/DNA gapmers) that retain RNase H activity while achieving stronger target engagement than MOE-based ASOs (Qassem et al., 2024). However, these advantages come with important trade-offs: compared with MOE chemistries, which generally exhibit lower toxicity and reduced immunogenicity, LNA-modified ASOs show a higher incidence of hybridization-dependent off-target effects due to their increased binding strength and shorter sequence length, as well as a greater risk of hepatotoxicity observed in preclinical models (Qassem et al., 2024; Crooke et al., 2018). Thus, while MOE modifications offer a more balanced safety profile with moderate potency, LNA chemistry shifts the balance toward maximal efficacy at the expense of a narrower therapeutic window, making careful sequence design and dose optimization critical for clinical application.

Nucleobase modifications



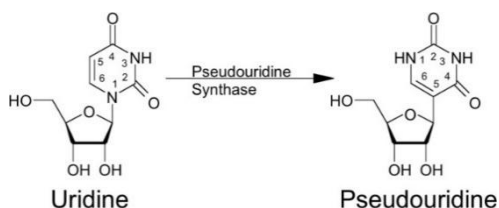
Chemical modification of RNA is essential for modern RNA therapeutics, enabling stability, reduced immunogenicity, enhanced target engagement, and greater pharmacological efficacy. Many of these modifications are inspired by naturally occurring post-transcriptional RNA modifications that regulate RNA structure and function in cells. By incorporating modified nucleosides or alternative backbone chemistries into therapeutic oligonucleotides, researchers can overcome many of the limitations associated with unmodified RNA, including rapid degradation, poor cellular persistence, and activation of innate immune responses. Among the most widely utilized modifications are 5-methylcytidine (m⁵C), pseudouridine (Ψ), inosine, and glycol nucleic acid (GNA), each contributing distinct structural and functional advantages.

5-Methylcytidine (m⁵C)

5-Methylcytosine (m⁵C; 5-methylcytidine, 5-Me-C) is a naturally occurring methylated RNA base found in tRNA and numerous other cellular RNAs, where it functions as a post-transcriptional modification that contributes to RNA stability, processing, and translation (Motorin and Helm, 2011; Squires et al., 2012). Its best-characterized role is in tRNA, where m⁵C protects transcripts from endonucleolytic cleavage and helps preserve structural integrity under cellular stress (Blanco et al., 2014).

These endogenous stabilizing properties have been leveraged in therapeutic RNA design. Incorporation of 5-methylcytidine into synthetic RNA oligonucleotides enhances base pairing and duplex stability through increased hydrophobicity at the C5 position, typically increasing melting temperature by approximately 1–1.5 °C per modification. Improved duplex stability strengthens target hybridization, facilitates disruption of target secondary structures, and can enable the use of shorter antisense oligonucleotides while maintaining binding affinity. In addition, m5C reduces susceptibility to nuclease degradation and attenuates activation of innate immune receptors that preferentially recognize unmodified RNA, thereby improving both stability and tolerability of RNA therapeutics (Karikó et al., 2005; Anderson et al., 2010).

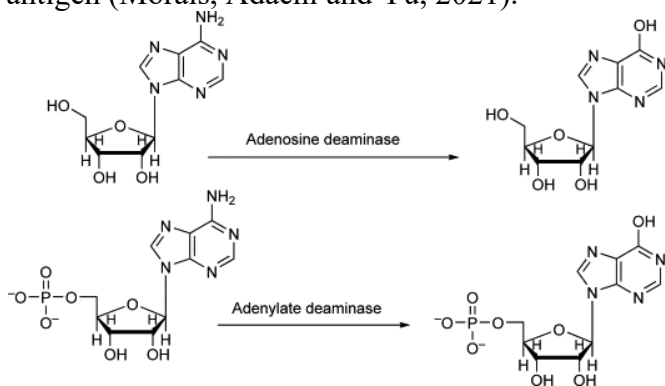
Pseudouridine (Ψ)



Pseudouridine (Ψ), the C5-glycoside isomer of uridine, is the most abundant naturally occurring RNA modification and is widely distributed across tRNA, rRNA, mRNA, and non-coding RNAs (Lin et al., 2021). Conversion of uridine to pseudouridine introduces a carbon–carbon glycosidic bond and an additional hydrogen-bond donor, enhancing base stacking interactions, strengthening RNA–RNA and RNA–protein interactions, and increasing structural rigidity. As a result, Ψ contributes to the stability of tRNA architecture, supports ribosomal function through rRNA modification, and regulates mRNA stability and translational efficiency (Zhao and He, 2015; Lin et al., 2021).

These properties have made pseudouridine one of the most important modifications in RNA therapeutics. Incorporation of Ψ into synthetic RNAs enhances duplex stability, improves resistance to nuclease degradation, and increases translational output, often resulting in substantially higher protein expression compared with unmodified RNA (Lin et al., 2021). Equally important, pseudouridine reduces activation of innate immune sensors and downstream cytokine responses, thereby enabling sustained translation and improved tolerability of exogenous RNA (Karikó et al., 2008; Lin et al., 2021).

The clinical significance of pseudouridine is exemplified by mRNA COVID-19 vaccines. These vaccines consist of a nucleoside-modified mRNA encoding the SARS-CoV-2 spike protein, encapsulated within a lipid nanoparticle (LNP) delivery system. The mRNA contains regulatory elements including 5' and 3' untranslated regions, a coding sequence encoding a stabilized pre-fusion spike protein, and multiple stop codons to ensure accurate translation (Morais, Adachi and Yu, 2021). A defining feature of these vaccines is the substitution of uridine with N1-methylpseudouridine, which enhances mRNA stability, suppresses innate immune activation, and improves protein translation efficiency (Morais, Adachi and Yu, 2021). The surrounding LNP protects the mRNA from degradation, facilitates cellular uptake, and promotes delivery to antigen-presenting cells such as dendritic cells, ultimately enabling robust immune responses against the encoded antigen (Morais, Adachi and Yu, 2021).



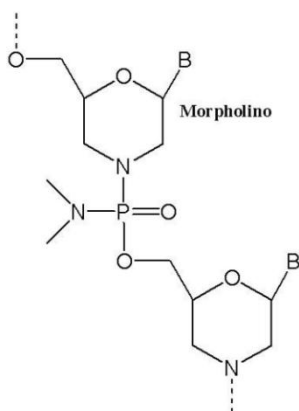
Inosine

Inosine is a functionally unique RNA modification generated through adenosine-to-inosine (A-to-I) editing by adenosine deaminase acting on RNA (ADAR) enzymes. Because inosine is interpreted by cellular machinery as guanosine, A-to-I editing enables dynamic modification of RNA information without altering the underlying genome. This process can recode proteins, alter splice-site selection, and modify RNA secondary structure, thereby influencing transcript stability, localization, and protein expression (Eisenberg and Levanon, 2018; Srinivasan et al., 2021).

In naturally occurring tRNAs, inosine located at the wobble position expands codon recognition, allowing a single tRNA to decode multiple synonymous codons and thereby enhancing translational efficiency and proteome diversity (Srinivasan et al., 2021). In mRNAs, site-specific editing can produce amino acid substitutions or regulate RNA processing, with important roles in neurological function and development.

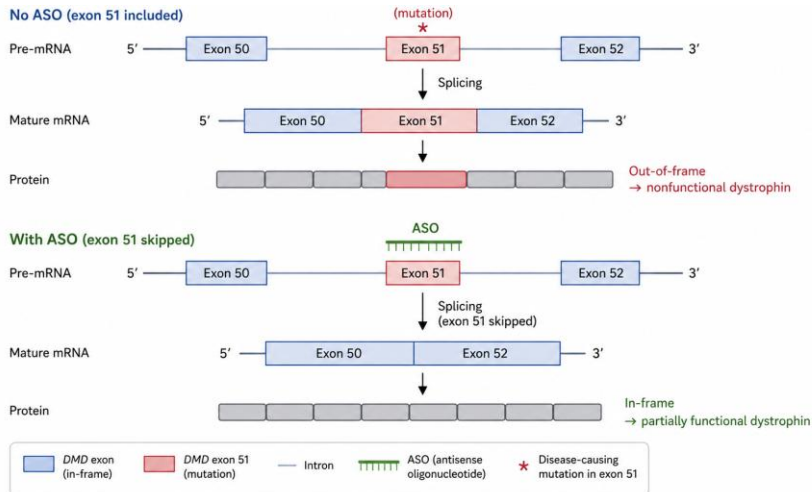
These biological functions have provided the foundation for RNA editing therapeutics. Programmable A-to-I editing exploits endogenous ADAR enzymes to correct disease-causing mutations at the RNA level, offering a reversible and potentially safer alternative to permanent genome editing. In addition, inosine-containing RNAs exhibit reduced activation of innate immune sensors such as MDA5, improving the safety profile of therapeutic RNAs. Its flexible base-pairing properties also provide opportunities to modulate RNA structure and translation, making inosine relevant to the design of antisense oligonucleotides, siRNAs, and mRNA therapeutics. Despite its promise, challenges including off-target editing, variable editing efficiency, and delivery limitations continue to constrain broader clinical application.

Morpholinos



The development of morpholino oligonucleotides arose from the need to overcome fundamental limitations of early antisense chemistries, particularly the poor stability of natural oligonucleotides in biological systems due to nuclease degradation and the nonspecific interactions caused by their negatively charged phosphodiester backbone (Stein and Cheng, 1993; Summerton, 1999). To address these issues, James Summerton in 1985 working at Antivirals which later became Santarus Inc replaced the ribose sugar with a chemically robust morpholine ring and substituted the charged phosphodiester linkage with a neutral phosphorodiamidate backbone. (Summerton and Weller, 1997; Li, 2017, p. 40). This elegant design preserved the Watson–Crick base pairing and allowed for effective sequence-specific binding to target RNA and subsequent steric blocking for transcript manipulation (Moulton and Moulton, 2017, pp. v–vi, 11).

In the early 1990s, Zbigniew Dominski and Ryszard Kole at the University of North Carolina demonstrated that targeting specific splice elements in β -globin pre-mRNA could redirect the splicing machinery and restore correct mRNA production *in vitro*. This seminal work laid the foundation for the modern splice-switching therapeutic strategies we have today, showing that RNA splicing is not only mechanistically tractable but also chemically programmable. The favorable safety of morpholinos subsequently enabled the first clinical applications of splicing therapeutics.



Researchers at Kobe University first demonstrated that ASOs could redirect splicing of the dystrophin transcript to induce exon skipping, laying the foundation for modern RNA therapeutics for Duchenne muscular dystrophy (DMD).

Takeshima Y, et al. (1995). "Modulation of in vitro splicing of the upstream intron by modifying an intra-exon sequence in the dystrophin gene." *Journal of Clinical Investigation* 95(2): 515–520.
DOI: 10.1172/JCI117693

AVI-4658, an exon 51–targeting PMO, showed in a proof-of-concept clinical study showed that direct intramuscular administration could induce sequence-specific exon skipping and restore dystrophin expression locally in patients, without significant safety concerns (Kinali et al., 2009). This study provided the first clinical validation of PMOs as a therapeutic platform, establishing proof-of-mechanism in humans and laying the foundation for subsequent systemic PMO-based therapies.

Eteplirsen



Eteplirsen, a phosphorodiamidate morpholino oligomer, was evaluated in a small randomized, placebo-controlled trial in boys with Duchenne muscular dystrophy amenable to exon 51 skipping, demonstrating that sustained treatment restores production of truncated but functional dystrophin (Mendell et al., 2013). While no meaningful dystrophin increase was observed at 12 weeks, significant increases in dystrophin-positive muscle fibers were seen by 24 weeks and reached 40-50% by 48 weeks,

with corresponding restoration of associated proteins such as nNOS and sarcoglycans (Mendell et al., 2013). Functionally, Eteplirsen-treated patients showed stabilization of walking ability compared with the expected decline in placebo patients, with an approximate 67 m benefit in the 6-minute walk test among evaluable patients (Mendell et al., 2013). The study suggests that treatment duration, rather than dose, is the key driver of efficacy, and that Eteplirsen is generally well tolerated, though conclusions are limited by the small sample size and variability in disease progression (Mendell et al., 2013).



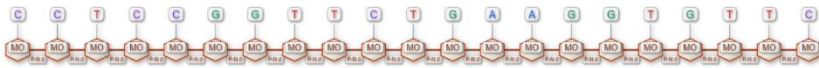
Golodirsen (Vyondys 53)

Golodirsen (Vyondys 53) is a phosphorodiamidate morpholino oligomer (PMO) antisense oligonucleotide indicated for Duchenne muscular dystrophy (DMD) patients with mutations amenable to exon 53 skipping. It binds to exon 53 of dystrophin pre-mRNA, promoting exon skipping and enabling production of a truncated but partially functional dystrophin protein (FDA, 2019). It received accelerated approval in the U.S. based on increased dystrophin production rather than confirmed clinical outcomes (FDA, 2019). The recommended dosing is 30 mg/kg IV weekly (FDA, 2019).

Exon skipping	Increased in all patients; 28.9-fold increase at 48 weeks (FDA, 2019; Servais et al., 2022)
Dystrophin	0.10% → 1.02% of normal (mean +0.92%)
Dose	30 mg/kg IV weekly
PK linearity	Dose-proportional exposure; minimal accumulation
Cmax variability	38–72% CV
AUC variability	34–44% CV
Vd (steady-state)	668 mL/kg
Protein binding	33–39%
Half-life	3.4 hours
Clearance	346 mL/hr/kg

Metabolism	Minimal; no metabolites detected
Excretion	Primarily unchanged in urine
Renal impairment	AUC ↑ 1.2× (Stage 2 CKD), 1.9× (Stage 3 CKD) (FDA, 2019)
DDI potential	Low (FDA, 2019)

Viltolarsen (Viltepso)

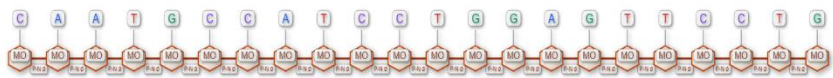


Viltolarsen (Viltepso) is a phosphorodiamidate morpholino oligomer that induces exon 53 skipping in dystrophin pre-mRNA, restoring production of a truncated but functional dystrophin protein (FDA, 2020). Treatment results in measurable increases in dystrophin expression (~5–6% of normal) and consistent exon skipping at the mRNA level, confirming target engagement (FDA, 2020; Clemens et al., 2020). Pharmacokinetically, viltolarsen demonstrates dose-proportional exposure, minimal accumulation, and rapid renal elimination of unchanged drug, with low metabolic and drug–drug interaction potential (FDA, 2020). Clinically, it is generally well tolerated, with mostly mild adverse events (e.g., respiratory infections, injection-site reactions), no serious drug-related adverse events in trials, and low immunogenicity; however, renal toxicity remains a theoretical class risk, requiring ongoing monitoring (FDA, 2020; Clemens et al., 2020).

Viltolarsen is generally well tolerated, with most adverse events being mild and including upper respiratory tract infections, cough, pyrexia, and injection-site reactions (FDA, 2020). Clinical trial data show no serious drug-related adverse events or treatment discontinuations, supporting an overall favorable safety profile (Clemens et al., 2020). However, as with other antisense oligonucleotides, there is a potential risk of renal toxicity, necessitating routine monitoring during treatment (FDA, 2020). Immunogenicity appears minimal, with no significant immune responses reported (FDA, 2020).

Administration	IV infusion (60 min)
Bioavailability	100%
Tmax	1 hour (end of infusion)
Exposure	Dose-proportional (1.25–80 mg/kg/week)
Accumulation	Minimal with weekly dosing
Volume of distribution	300 mL/kg
Plasma protein binding	39–40%
Metabolism	Metabolically stable; no metabolites detected
Elimination	Primarily renal (unchanged drug)
Half-life (t _{1/2})	2.5 hours
Clearance	217 mL/hr/kg
Drug interactions	Low potential (no major CYP/transporter effects)

Casimersen (AMONDYS 45)



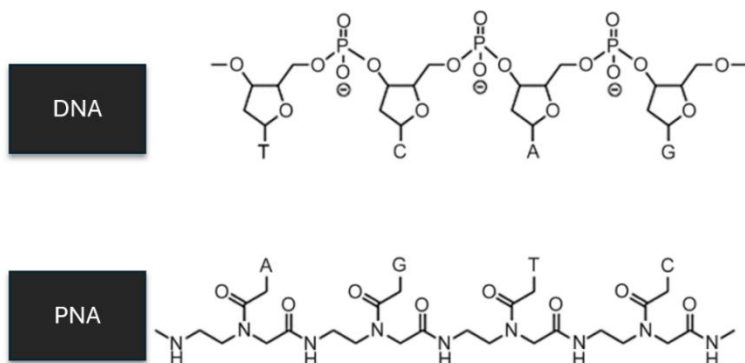
Casimersen (AMONDYS 45) is a phosphorodiamidate morpholino oligomer (PMO) antisense oligonucleotide approved for the treatment of Duchenne muscular dystrophy (DMD) in patients with mutations amenable to exon 45 skipping (FDA, 2021). DMD is an X-linked recessive disorder characterized by absence of functional dystrophin, leading to progressive degeneration and premature mortality (Assefa et al., 2024). Casimersen received accelerated FDA approval based on its ability to increase dystrophin production, a surrogate biomarker, with confirmatory trials required to establish clinical benefit (FDA, 2021).

Casimersen acts at the level of pre-mRNA splicing. It binds specifically to exon 45 of dystrophin pre-mRNA, preventing its inclusion during mRNA processing. This restores the translational reading frame and allows synthesis of a truncated but partially functional dystrophin protein (FDA, 2021; Assefa et al., 2024). In a placebo-controlled study dystrophin increased from 0.93% to 1.74% of normal after 48 weeks

Casimersen is generally well tolerated, with most adverse events being mild to moderate and consistent with common intercurrent illnesses. The most frequently reported adverse reactions include upper respiratory tract infection (65%), cough (33%), pyrexia (33%), headache (32%), arthralgia (21%), and oropharyngeal pain (21%) (FDA, 2021). Despite this favorable tolerability profile, an important safety concern is the potential for renal toxicity, which has been observed in animal studies and is considered a class effect of antisense oligonucleotides. Consequently, careful renal monitoring is recommended during treatment, including assessment of serum cystatin C, urine protein, and urine protein-to-creatinine ratio to detect early signs of nephrotoxicity (FDA, 2021) (Assefa et al., 2024).

Key PD outcome	+0.81% dy'strophin vs +0.22% placebo
Dose	30 mg/kg IV weekly
Cmax	End of infusion
Half-life	3.5 h
Clearance	180 mL/hr/kg
Vd	367 mL/kg
Protein binding	8.4–31.6%
Metabolism	Minimal (non-hepatic)
Excretion	>90% renal (unchanged)
Accumulation	None
Key safety risk	Renal toxicity
Common AEs	URTI, cough, pyrexia, headache

Peptide Nucleic Acids (PNAs)



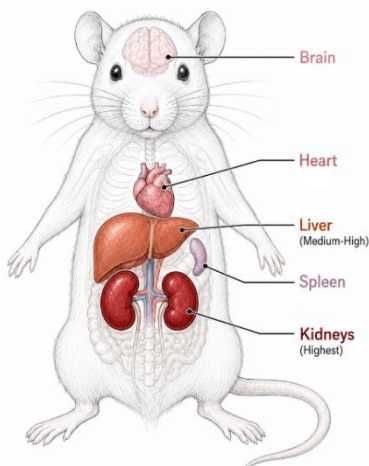
Peptide nucleic acids (PNAs) are synthetic DNA/RNA analogues in which the negatively charged sugar-phosphate backbone of natural nucleic acids is replaced by a neutral pseudopeptide backbone composed of N-(2-aminoethyl)-glycine units. The nucleobases adenine, guanine, cytosine and thymine are attached to this backbone through methylene carbonyl linkers, allowing PNAs to hybridize with complementary DNA or RNA through Watson-Crick base pairing. This distinctive backbone chemistry gives PNAs unusually strong binding affinity, high thermal stability and resistance to enzymatic degradation, making them attractive tools for gene targeting, antisense inhibition and sequence-specific nucleic acid recognition (Nielsen et al., 1991; Egholm et al., 1993; Demidov et al., 1994).

Since their discovery by Peter Nielsen and colleagues at the University of Copenhagen and Risø National Laboratory in Denmark in 1991, PNAs have evolved into promising therapeutic candidates. Early studies established that PNAs bind nucleic acids in a sequence-selective manner and often form more stable duplexes than comparable DNA/RNA systems (Nielsen et al., 1991; Egholm et al., 1993). This enhanced stability is largely attributed to the neutral PNA backbone, which avoids the electrostatic repulsion that occurs between negatively charged DNA or RNA strands. As a result, PNA/DNA hybrids typically exhibit substantially higher melting temperatures than equivalent DNA/DNA duplexes, with reported increases of approximately 7–20 °C under comparable experimental conditions (Schwarz, Robinson and Butler, 1999). However, PNA/DNA duplex stability remains dependent on sequence, length and base composition, meaning that T_m values must be interpreted in the context of the specific oligonucleotide and assay conditions used (Giesen et al., 1998).

These properties make PNAs particularly useful as antisense agents. By binding to target mRNA, PNAs can block gene expression by preventing translation or interfering with other nucleic-acid-dependent processes. In antibacterial applications, early studies showed that PNAs could inhibit bacterial gene expression, especially when conjugated to cell-penetrating peptides that improve transport across bacterial membranes (Good et al., 2001). More recent work has expanded this strategy to target essential bacterial genes, resistance genes and virulence genes, positioning PNAs as programmable, narrow-spectrum antimicrobial agents with potential value against antibiotic-resistant pathogens (El-Fateh, Chatterjee and Zhao, 2024).

Compared with other nucleic-acid-based therapeutic platforms, PNAs are distinguished by their neutral pseudopeptide backbone, high target affinity and resistance to nuclease and protease degradation. DNA antisense oligonucleotides, particularly phosphorothioate-modified ASOs, retain an anionic character while gaining improved nuclease stability and increased interactions with plasma and cellular proteins. Morpholinos, or phosphorodiamidate morpholino oligomers, also have neutral backbones, but they use morpholine rings and phosphorodiamidate linkages rather than the PNA pseudopeptide scaffold. Like PNAs, morpholinos are metabolically stable and primarily act through steric blocking; unlike many DNA gapmer ASOs, they do not recruit RNase H to degrade target RNA (Heo et al., 2024). These differences in backbone chemistry help explain the distinct binding, stability, pharmacokinetic and pharmacodynamic profiles of PNA, DNA antisense oligonucleotides and morpholinos.

The plasma pharmacokinetic profiles of these chemistries are also markedly different. Unmodified PNA appears to clear rapidly from plasma; in a rat intravenous study, a PNA targeting the rat neurotensin receptor had a distribution half-life of approximately 3 minutes and an elimination half-life of approximately 17 minutes (McMahon et al., 2002). Phosphorothioate DNA ASOs generally show more prolonged systemic exposure and extensive tissue distribution, driven largely by plasma protein binding and tissue uptake (Geary et al., 2015). Morpholinos such as eteplirsen also have relatively short plasma half-lives compared with many phosphorothioate ASOs; eteplirsen has a reported elimination half-life of approximately 3–4 hours, with renal clearance accounting for a substantial fraction of the administered dose within 24 hours (FDA, 2016).



Protein binding is one of the clearest differentiators between phosphorothioate DNA ASOs and neutral-backbone chemistries. Phosphorothioate ASOs interact extensively with proteins, and these interactions affect distribution, tissue delivery, cellular uptake, intracellular trafficking, potency and toxicity. PNA and morpholinos, because they are neutral molecules, generally show lower nonspecific electrostatic protein interactions than phosphorothioate ASOs. This lower protein binding can reduce plasma retention and contribute to more rapid renal clearance, particularly for unconjugated neutral oligonucleotide analogues (McMahon et al., 2002; Heo et al., 2024).

Although PNA, DNA ASOs and morpholinos all distribute beyond plasma, their dominant tissue patterns differ. In the rat PNA study, low but detectable PNA levels were observed across examined organs, with the highest concentrations in kidney, followed by liver, heart, brain and spleen (McMahon et al., 2002). Phosphorothioate DNA ASOs generally show broad systemic biodistribution, with highest tissue concentrations commonly observed in liver and kidney, followed by other tissues such as bone marrow, adipose tissue and lymph nodes (Geary et al., 2015). Morpholinos also show class-consistent distribution and elimination properties, with approved PMOs such as eteplersen, golodirsen and casimersen sharing broadly similar ADME/DMPK characteristics (Heo et al., 2024).

Despite their strong hybridization properties, PNAs have poor intrinsic cellular uptake, so their pharmacodynamic activity is often limited by delivery rather than target binding once inside the cell. Phosphorothioate ASOs are taken up mainly through endocytic pathways, although only a fraction of internalized oligonucleotide typically reaches productive intracellular compartments. Morpholinos face a similar delivery challenge,

especially in tissues such as skeletal muscle, which helps explain why high systemic doses or conjugated delivery approaches are often needed to achieve robust pharmacodynamic effects (Heo et al., 2024).

Pharmacodynamic duration is therefore not determined by plasma half-life alone. PNA can bind targets with high affinity and is biologically stable, but in vivo duration is often constrained by delivery, systemic exposure and tissue retention (McMahon et al., 2002). Phosphorothioate ASOs often show durable pharmacodynamic effects because they accumulate in tissues and are cleared slowly from intracellular and tissue compartments; consequently, tissue PK is often more informative than plasma PK for predicting response (Geary et al., 2015). Morpholinos may have relatively short plasma exposure but can still produce sustained mechanistic effects with repeated dosing, as reflected by the weekly dosing schedules used for approved DMD PMO drugs such as eteplirsen (FDA, 2016).

These pharmacological differences also shape the translational challenges of each platform. For PNA, the major challenge is converting excellent binding affinity into reliable in vivo activity, because unconjugated PNA has poor cellular uptake and limited systemic exposure. For DNA ASOs, the challenge is to balance potency, tissue distribution, RNase H activity, protein binding, immune stimulation and toxicity within a clinically tractable chemistry design (Geary et al., 2015). For morpholinos, the principal challenge is achieving sufficient intracellular concentrations in target tissues, particularly muscle, without requiring impractically high doses or introducing toxicity through delivery-enhancing conjugates (FDA, 2016; Heo et al., 2024).

PNAs are typically synthesized using solid-phase peptide synthesis. Unlike DNA and RNA synthesis, PNA assembly involves the stepwise coupling of protected PNA monomers, each composed of a nucleobase attached to an N-(2-aminoethyl)-glycine backbone, onto a solid support. The growing oligomer is extended through amide bond formation, usually using Boc- or Fmoc-protected monomers, followed by cleavage from the resin and purification by chromatographic methods. This peptide-like synthetic route provides considerable chemical flexibility, allowing functional groups, fluorophores, cell-penetrating peptides, intercalators and other delivery motifs to be introduced during or after synthesis to improve binding, detection, solubility or uptake (Nielsen et al., 1991; Egholm et al., 1993; El-Fateh, Chatterjee and Zhao, 2024).

This synthetic flexibility has enabled the development of chemically modified and conjugated PNAs designed for specific biological and biophysical applications. Bruce Armitage and collaborators expanded the chemical and supramolecular applications of PNAs through PNA-containing

hybrids, dye-PNA assemblies, strand invasion and quadruplex-targeting PNAs. For example, Smith, Olson and Armitage demonstrated that PNA-containing hybrids could template the spontaneous assembly of helical cyanine dye aggregates, highlighting the use of synthetic PNAs as programmable scaffolds for molecular recognition. Later work further developed PNAs as tools for targeting structured nucleic acids, including G-quadruplexes and mixed-sequence duplex DNA (Datta and Armitage, 2001; Datta, Schmitt and Armitage, 2003; Rapireddy et al., 2007). Together, these studies show the versatility of PNA synthesis for broader chemical biology toolkits.

Although PNAs have long been viewed as attractive therapeutic molecules, their clinical translation has been slow largely because unmodified PNAs have poor cellular uptake and require chemical modification or delivery strategies to reach intracellular targets. The most advanced example is OLP-1002, an OliPass PNA-derived antisense compound targeting SCN9A/Nav1.7 for osteoarthritis pain; a Phase 2a trial evaluating subcutaneous OLP-1002 enrolled 134 participants and was completed in 2023 (ClinicalTrials.gov, 2023).

OLP-1002, an OliPass PNA-derived antisense compound designed to inhibit SCN9A/Nav1.7, was evaluated in a two-stage Phase 2a trial for moderate-to-severe osteoarthritis pain in the knee and/or hip. The study enrolled 134 participants and was completed in mid-2023, with no formal results yet posted on ClinicalTrials.gov. Although an early interim analysis suggested a potential analgesic signal for the 2µg dose, with greater pain reduction than placebo at Day 43, the final sponsor-reported outcome did not confirm efficacy. In the expanded placebo-controlled stage, placebo responses exceeded those seen with OLP-1002: OliPass reported mean six-week pain reductions of 41% for placebo compared with 22% and 25% for the 1 µg and 2 µg OLP-1002 groups, respectively, and stated that OLP-1002 was “inferior to placebo” in this Phase 2a trial (OliPass Corporation, 2023b).

Another example is EOM613, described in its trial record as a peptide nucleic acid with immune-modulating properties, which entered a proof-of-concept study for severe COVID-19, although the trial status remains unclear (ClinicalTrials.gov). EOM613 is being developed by EOM Pharmaceuticals and appears to be the same historical product family previously known as Product R, OHR118, AVR118 or OHR/AVR118. Unlike a conventional single-molecule antisense PNA drug, EOM613 is best understood as a complex peptide–nucleotide solution containing peptide and nucleoside/nucleotide components, including a peptide–oligonucleotide conjugate. Its proposed pharmacology is “dynamic dual-action” immunomodulation: it is reported to regulate monocyte/macrophage cytokine responses, with both anti-inflammatory and pro-inflammatory

effects depending on immune-cell activation state. Early human data include open-label rheumatoid arthritis and cancer-cachexia studies, and an open-label Phase 1/2a COVID-19 study in hospitalized patients; these studies generally suggested tolerability and possible clinical or biomarker activity, but most efficacy evidence remains exploratory, small, open-label or sponsor-reported. In 2026, EOM reported FDA clearance for a Phase 2a cancer-cachexia study and plans for an additional study on Crohn's disease.

Therapeutic Modalities and Mechanisms

RNA sits at the interface of genetic information and biological function, enabling control over protein expression without permanently altering the genome. Despite its transient nature, targeting RNA can produce robust and often long-lasting clinical effects, making it a versatile platform for a wide spectrum of therapies.

One of the major advantages of RNA therapeutics is their ability to precisely and programmatically regulate gene expression. Depending on the therapeutic objective, these molecules can either enhance protein production through mechanisms such as microRNA inhibition and correction of RNA splicing or suppress the expression of harmful genes using strategies including RNase H-mediated transcript degradation and siRNA-directed RNA cleavage. Unlike many conventional small-molecule drugs, which often rely on broad chemical interactions with proteins, RNA therapeutics function through predictable sequence complementarity, enabling highly selective targeting of specific transcripts. This sequence-driven precision not only improves specificity but also supports a more rational drug discovery process, where candidate therapies can be identified and optimized through precision-based genetic and molecular screening approaches.

Beyond gene silencing and activation, RNA therapeutics can reprogram RNA processing and function. Technologies such as exon skipping, cryptic exon modulation, and isoform switching can alter the proteins produced by cells without permanently changing the genome itself. More advanced strategies employ enzymes such as Adenosine Deaminase Acting on RNA (ADAR) to perform site-specific RNA base editing, enabling transient and potentially reversible correction of pathogenic mutations. Collectively, these approaches highlight how RNA therapeutics move medicine closer to a rational and modular framework for drug development, where treatments are engineered based on precise molecular knowledge of disease pathways rather than discovered primarily through empirical screening.

The first ASO clinically approved



Fomivirsen (Vitravene™) was the first gene-silencing antisense therapy approved by the FDA, receiving approval in 1998. Originally discovered by a group at the NIH, it was first licensed by Isis Pharmaceuticals and later sublicensed to Novartis. The compound consists

of a simple 21-base, DNA oligonucleotide in which a sulfur atom replaces one of the non-bridging oxygen atoms in the phosphate backbone, forming phosphorothioate linkages between nucleosides. Vitravene was developed to treat cytomegalovirus (CMV) retinitis in immunocompromised patients and functions by binding to mRNA transcripts that encode IE2 proteins, essential for viral replication (Geary, Henry, and Grillone, 2002).

Preclinical and in vitro studies demonstrated potent antiviral activity, with mean 50% inhibitory concentrations (IC₅₀) ranging from 0.03 to 0.2 μM across multiple cell lines and a median effective concentration (EC₅₀) of approximately 0.34 ± 0.25 μM in human fibroblasts (Azad et al., 1993; Anderson et al., 1996).

Fomivirsen (ASO)	0.03–0.3 μM	Very potent, sequence-specific
Ganciclovir	1–10 μM	Antiviral comparator
Ibuprofen	1–20 μM	Moderate potency
Aspirin	10–1000 μM	Low potency
Paracetamol	100–1000 μM	Very low potency
Imatinib	~0.3 μM	High potency targeted drug

Vitravene demonstrated striking potency with EC₅₀ values in the range of 0.03–0.3 μM, highlighting the robustness of a sequence-specific mechanism of action. In comparison, traditional antivirals like ganciclovir showed lower potency (EC₅₀ ≈ 1–10 μM), while commonly used small molecules such as ibuprofen (1–20 μM) and aspirin (10–300 μM) exhibited only moderate to low activity. Paracetamol was even less potent (100–1000 μM), underscoring the substantial gap between conventional drugs and targeted nucleic acid therapeutics. Thus, reinforcing the promise of rational, sequence-driven drug design. Despite its scientific significance, it was ultimately a commercial failure. The drug was withdrawn in 2000, largely due to the success of highly effective antiretroviral therapies for HIV, which dramatically reduced the incidence of CMV retinitis, the very condition Vitravene was designed to treat.

The clinical landscape of RNA therapeutics has expanded substantially, encompassing a diverse and increasingly sophisticated set of approaches that act across multiple layers of gene regulation and cellular function. Established commercial therapies include RNase H–mediated knockdown (e.g., inotersen for transthyretin amyloidosis), siRNA-driven

RISC-mediated degradation (e.g., patisiran and inclisiran), aptamers that bind and inhibit protein targets (e.g., pegaptanib/Macugen), RNA splicing modulation (e.g., nusinersen), and mRNA-based strategies for both protein replacement and vaccination (e.g., mRNA COVID-19 vaccines) (Adams et al., 2018; Ray et al., 2017; Hoy, 2018; Finkel et al., 2017; Polack et al., 2020).

Beyond these established modalities, additional and conceptually distinct mechanisms of action are emerging. Among them, microRNA (miRNA)-based therapies enable network-level regulation by modulating entire gene programs either through inhibition (anti-miRs; e.g., miravirsen targeting miR-122 in HCV) or restoration (miRNA mimics; e.g., MRX34 in oncology) rather than acting on single transcripts (van der Ree et al., 2017; Beg et al., 2017). In parallel, RNA activation (RNAa) has emerged as a clinically validated strategy for upregulating endogenous gene expression using small activating RNAs (saRNAs).

Complementing activation-based approaches, RNA editing therapeutics particularly those leveraging adenosine-deaminases-acting-on-RNA (ADARs) introduce the ability to precisely modify RNA sequences at the transcript level. These systems enable site-specific A-to-I base conversion, which is interpreted as guanosine during translation, allowing correction of pathogenic mutations without permanent genomic alteration. Early clinical programs, such as WVE-006 for alpha-1 antitrypsin deficiency, demonstrate the translational potential of this approach by recruiting endogenous ADAR enzymes to restore functional protein expression in patients (Rainaldi, Mali and Nourredine, 2025).

Beyond editing and activation, additional emerging modalities further broaden the mechanistic scope of RNA therapeutics. Non-degradative antisense oligonucleotides can function through steric blocking to inhibit translation or disrupt RNA–protein interactions without inducing RNA cleavage, as seen in splice-switching therapies. RNA molecules can also act as innate immune agonists, for example synthetic RNAs that activate TLR7/8 or RIG-I pathways in oncology. Advances in expression platforms, including self-amplifying RNA (e.g., saRNA vaccine candidates such as ARCT-021) and circular RNA constructs, enhance the magnitude and durability of protein production (Blakney et al., 2021). Finally, RNA decoys and scaffolds can sequester proteins or regulatory RNAs, expanding RNA therapeutics into modulation of protein–RNA interaction networks.

These developments reflect a clear shift from first-generation paradigms centered on gene silencing or replacement toward a more expansive framework in which RNA therapeutics can activate, edit, regulate, and reprogram cellular behavior with increasing precision. Clinical examples spanning RNAi, antisense, RNAa, miRNA modulation, and RNA

editing underscore that this mechanistic diversity is already being realized in the clinic, marking a transition toward a more versatile and programmable class of medicines (Adams et al., 2018; Reebye et al., 2018; Rainaldi, Mali and Nourreddine, 2025).

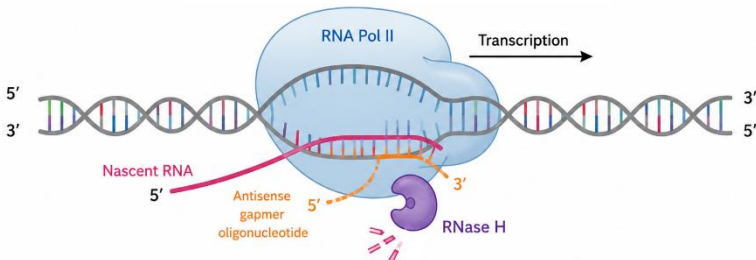
RNase H–Mediated Gene Silencing

“Die Nucleinsubstanzen liefern bei ihrer Spaltung verschiedene stickstoffhaltige Spaltungsprodukte, welche zu den wichtigsten Bestandteilen des Zellkerns gehören.”

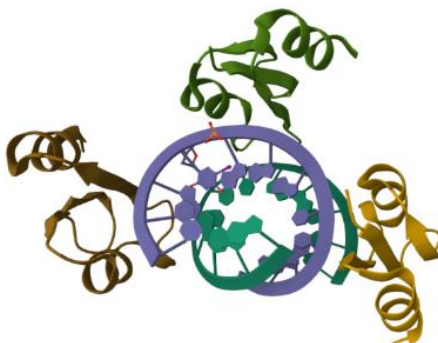
--Kossel, A., 1910. Die chemische Zusammensetzung der Zellkerne

Albrecht Kossel working at the University of Heidelberg first described nucleic substances in terms of their “cleavage products” (*Spaltungsprodukte*), illustrating that nucleic acids could be broken down into distinct nitrogen-containing bases. For this work Dr. Kossel was awarded the Nobel Prize in Physiology or Medicine. Kossel’s observations laid the foundation for the discovery of nucleases; enzymes specialized in the cleavage and degradation of DNA and RNA. During the mid-twentieth century, the identification of bacterial restriction endonucleases transformed molecular biology by enabling precise DNA manipulation for cloning, sequencing and recombinant DNA technology (Arber, 1979). Since then, nucleases including TREX1, EXO1 and MRE11 have been recognized as essential regulators of DNA replication, repair, apoptosis and innate immune defense in human systems, while also becoming indispensable tools in biotechnology, genome engineering and CRISPR-based applications (Roberts, 2005; Mazur and Perrino, 2022).

In 1976, venture capitalist Robert Swanson and biochemist Dr. Herbert Boyer founded Genentech to commercialize recombinant DNA methods pioneered by Boyer and Stanley Cohen, demonstrating that human genes could be inserted into bacteria to produce therapeutic proteins. This work led to the first successful production of human proteins in bacteria, including synthetic human insulin, which became the first recombinant pharmaceutical approved for clinical use in 1982. Genentech had established a new model for advancing therapeutics; one that deeply integrated our understanding of biology with our ability to use it for human therapeutics; a field we call Biotechnology.



RNase H is an endonuclease that plays a fundamental role in maintaining genome stability by selectively degrading the RNA strand of RNA–DNA hybrids. During DNA replication, it removes RNA primers required for initiation, ensuring their replacement with DNA and preserving genomic integrity. It also contributes to DNA repair by removing ribonucleotides, preventing structural instability and mutations (Eder, Walder and Walder, 1993). In addition, RNase H resolves R-loops that form during transcription, which, if accumulated, can interfere with replication and lead to DNA damage. Collectively, these functions establish RNase H as a key enzyme linking nucleic acid metabolism with genomic integrity (White, R.L. and Davis, R.W., 1976).



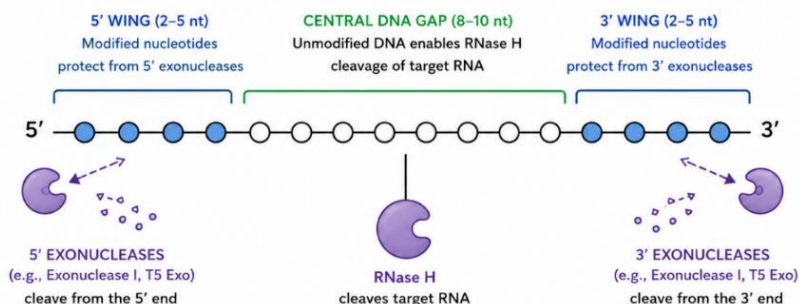
X-RAY DIFFRACTION Hybrid-binding domain of human RNase H1 in complex with 12-mer RNA/DNA (PDB 3BSU) (Nowotny, 2008)

Antisense oligonucleotide (ASO) therapeutics exploit these natural biological mechanisms for therapeutic purposes. When an ASO hybridizes to a complementary RNA transcript, it forms an RNA–DNA duplex that serves as a substrate for RNase H. The enzyme subsequently cleaves the phosphodiester backbone of the RNA strand, generating RNA fragments and preventing translation of downstream protein products.

To apply this mechanism therapeutically, scientists needed to overcome the challenge of delivering DNA-like molecules to their RNA targets while protecting them from rapid nuclease degradation and preserving their ability to recruit RNase H. This led to the development of chemical modifications that maintain the DNA-like properties required for

RNase H activity while substantially improving oligonucleotide stability in biological systems. Early unmodified oligonucleotides were highly susceptible to enzymatic degradation, severely limiting their therapeutic potential. This limitation was first addressed through backbone modifications, most notably phosphorothioate linkages, which confer resistance to nuclease-mediated degradation while retaining compatibility with RNase H activity. In parallel, sugar modifications such as 2'-O-methyl and 2'-O-methoxyethyl groups were introduced to enhance binding affinity, increase metabolic stability, and improve overall pharmacological properties.

These are often incorporated into “gapmer” designs, where a central DNA region capable of recruiting RNase H is flanked by modified nucleotides for protection. Collectively, these innovations enable antisense oligonucleotides to form stable RNA–DNA hybrids, activate RNase H, and achieve sufficient stability, bioavailability, and specificity for effective therapeutic application.



These advances in antisense chemistry and RNase H–mediated mechanisms have transformed ASOs from experimental molecular tools into precision therapeutics capable of targeting disease-causing transcripts with remarkable specificity. What began with Kossel’s early characterization of nucleic substances and the later discovery of nucleases has culminated in a new era of individualized genetic medicine, where pathogenic mutations can be selectively silenced at the RNA level while preserving normal cellular function.

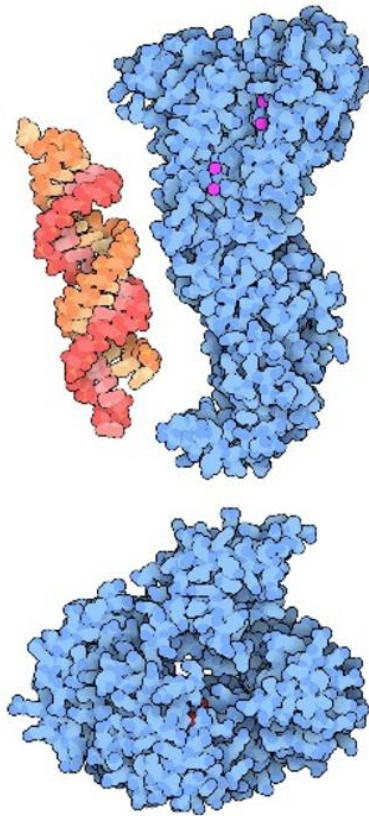
La Jolla Labs, a neurogenetics therapeutics company recently identified and developed an allele-selective antisense oligonucleotide targeting a familial form of amyotrophic lateral sclerosis (ALS). By leveraging advances in ASO chemistry the program moved from mutation identification to clinical therapeutic development in less than six months, an unprecedented timeline in the field of precision neurogenetics. The ASO

was specifically engineered to selectively suppress the mutant allele while sparing expression of the wild-type transcript, thereby minimizing the risk of disrupting essential physiological protein function.



The rapid development of this therapy illustrates how decades of fundamental discoveries in nuclease biology, RNA–DNA hybrid metabolism, and oligonucleotide engineering are now enabling highly personalized interventions for previously untreatable neurodegenerative diseases. The patient is currently being dosed at the University of Heidelberg, bringing this scientific trajectory full circle: from Kossel’s foundational investigations of nucleic acids in Heidelberg more than a century ago to the modern application of RNase H–dependent antisense therapeutics in the treatment of human disease at the same institution. This convergence of historical discovery, molecular biology, and translational medicine highlights the extraordinary progress of nucleic acid therapeutics and underscores the growing potential of allele-selective ASOs as rapid, adaptable treatments for rare genetic disorders.

RISC-Mediated Gene Silencing: RNA Interference (siRNA)



DAVID S. GOODSSELL

Small interfering RNA (siRNA) is a powerful cellular defense system that helps protect organisms from viral infections and regulate gene expression. Cells are particularly sensitive to double-stranded RNA, which is often produced during viral replication and signals potential danger. In response, specialized proteins such as dicer (see top blue protein in David Goodsell's watercolor illustration) cut these double-stranded RNA into short fragments. These are then incorporated into argonaut proteins (bottom protein in Goodsell's illustration; PDB 248S), which use one strand as a guide to identify and destroy matching messenger RNA (mRNA) molecules. Through this process, known as RNA interference, cells can selectively silence specific genes or eliminate viral RNA (Fire, A., Xu, S., Montgomery, M.K., Kostas, S.A., Driver, S.E. and Mello, C.C. 1998).

Andrew Z. Fire and Craig C. Mello received the 2006 Nobel Prize in Physiology or Medicine for this discovery of RNA interference (RNAi). Their landmark experiments, published in *Nature* in 1998, fundamentally

changed the understanding of gene regulation in eukaryotic organisms and opened entirely new avenues in molecular biology, biotechnology, and medicine.

Fire and Mello's work was conducted using the nematode *Caenorhabditis elegans* (*C. elegans*), a small transparent roundworm widely used as a model organism in genetics. At the time researchers knew that introducing "antisense" RNA could suppress gene expression. However, the effects were weak and inconsistent, and the mechanism remained unclear. Fire and Mello hypothesized that the true silencing trigger might not be single-stranded antisense RNA itself, but rather a "contaminating" double-stranded RNA formed during their experimental preparation.

To test this idea, they designed a series of carefully controlled experiments targeting the *unc-22* gene in *C. elegans*. This gene is responsible for normal muscle function; when it is disrupted, worms exhibit a characteristic twitching movement that can easily be observed under a microscope. The researchers synthesized three types of RNA corresponding to the *unc-22* sequence:

Sense RNA (matching the mRNA sequence)

Antisense RNA (complementary to the mRNA)

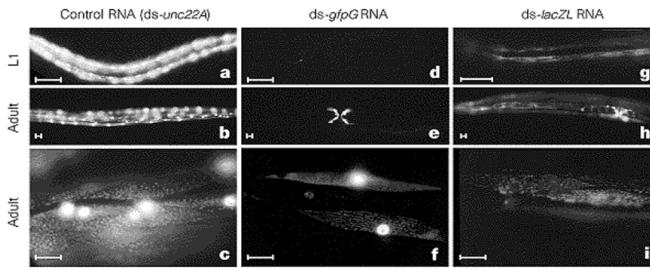
Double-stranded RNA created by hybridizing the sense and antisense strands

The results were striking; injection of either sense RNA or antisense RNA alone produced little or inconsistent gene silencing. In contrast, injections of tiny amounts of double-stranded RNA caused potent and highly specific suppression of the *unc-22* gene. The treated worms displayed the expected twitching phenotype, indicating that expression of the target gene had been effectively shut down. Importantly, Fire and Mello demonstrated that the effect was:

Sequence-specific: only genes matching the dsRNA sequence were silenced.

Highly potent: extremely small amounts of dsRNA were sufficient.

Systemic: the silencing effect spread throughout the organism.



The figure demonstrates that double-stranded RNA (dsRNA) can specifically silence gene expression in animal offspring through RNA interference. Control animals injected with ds-unc22A showed normal GFP expression in both nuclei and mitochondria, while animals injected with ds-gfpG displayed a dramatic loss of GFP expression, with only a few cells retaining activity. In contrast, ds-lacZL selectively eliminated nuclear GFP–LacZ expression without affecting mitochondrial GFP, showing that RNA interference can precisely target specific genes while leaving others unchanged.

Fire, A., Xu, S., Montgomery, M. *et al.* Potent and specific genetic interference by double-stranded RNA in *Caenorhabditis elegans*. *Nature* **391**, 806–811 (1998).
<https://doi.org/10.1038/35888>

Heritable: gene silencing persisted in subsequent generations of worms.

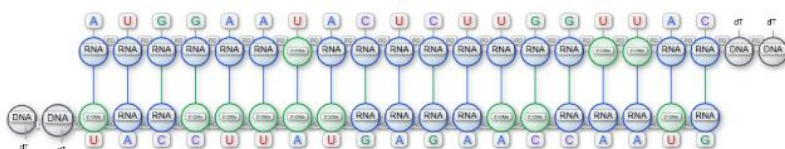
They repeated similar experiments with other genes and obtained comparable results, confirming that RNAi represented a general biological mechanism rather than a gene-specific anomaly. Their findings suggested that cells possess machinery capable of recognizing dsRNA and use it as a signal to destroy complementary mRNA molecules. Subsequent research revealed the molecular basis of this pathway. Double-stranded RNA is processed by an enzyme called Dicer into short interfering RNAs (siRNAs), typically around 21–23 nucleotides long. These siRNAs are incorporated into the RNA-induced silencing complex (RISC), which uses the siRNA sequence as a “guide” to identify and cleave matching mRNA targets, thereby preventing protein production.

RNAi has since become an important therapeutic strategy; one that enables precise silencing of disease-associated genes, including targets often considered “undruggable” by conventional therapies (Hu et al., 2020). Since the first FDA approval of Patisiran in 2018 for hereditary transthyretin-mediated amyloidosis, the field has advanced into a clinically validated therapeutic modality. Subsequent approvals of Givosiran, Lumasiran, Inclisiran, Vutrisiran, and Nedosiran expanded its reach into metabolic, cardiovascular, and rare genetic diseases. As of 2025, more than 260 siRNA drug candidates were in preclinical or clinical development. Research now spans major therapeutic areas, including oncology, neurodegenerative disease, infectious disease, and diabetes.

Despite its therapeutic promise, RNA interference (RNAi) therapy continues to face significant challenges related to the efficient delivery of

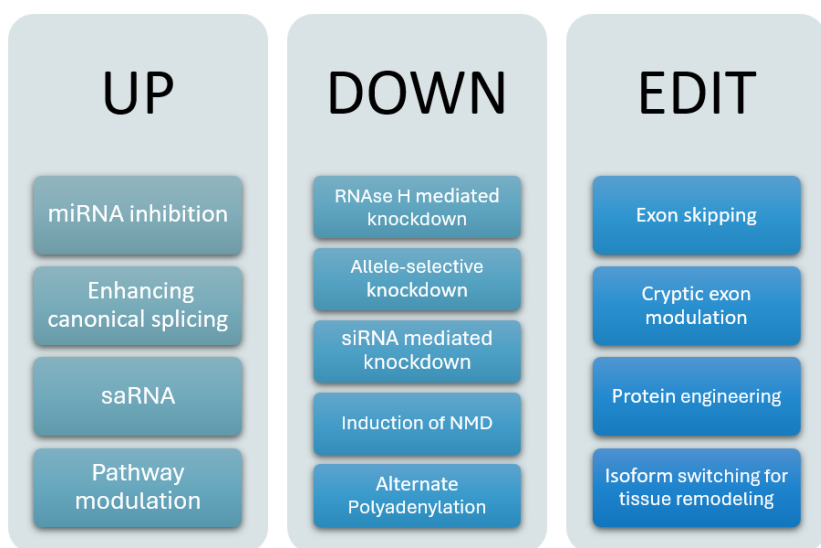
siRNAs to their intended targets. Unmodified siRNAs are highly unstable in vivo and are rapidly degraded by endogenous ribonucleases, resulting in extremely short circulation half-lives. In addition, their negatively charged, hydrophilic nature severely limits passive diffusion across cellular membranes, making intracellular delivery highly inefficient. Even when cellular uptake occurs, most siRNAs become trapped within endosomal compartments, preventing access to the cytoplasm where gene silencing takes place. Consequently, endosomal escape is widely regarded as the rate-limiting barrier restricting broader therapeutic applications of RNAi. Although numerous delivery strategies have been developed, including lipid conjugates, cationic peptides, phosphorothioate modifications, and viral delivery systems, many of these approaches are associated with toxicity caused by nonspecific endosomal disruption (Dowdy, 2023).

Onpatro (Patisiran): FDA-approved siRNA therapeutic indicated for the treatment of hereditary transthyretin-mediated amyloidosis (hATTR amyloidosis)



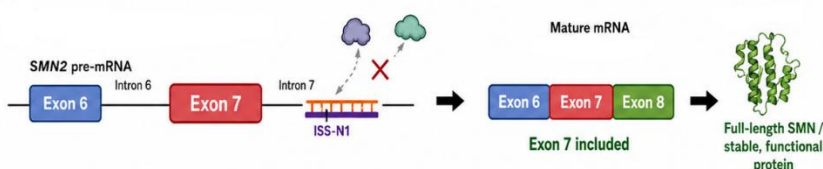
ONPATTRO (patisiran) was the first FDA-approved siRNA therapeutic indicated for the treatment of hereditary transthyretin-mediated amyloidosis (hATTR amyloidosis) with polyneuropathy. It functions through RNA interference by selectively degrading transthyretin (TTR) mRNA in hepatocytes, thereby reducing production of the pathogenic TTR protein and slowing progression of neuropathic disease.

In addition to delivery challenges, naked siRNAs can activate innate immune sensors, triggering inflammatory responses and unintended off-target gene silencing. To overcome these limitations, researchers have developed a range of sophisticated chemical modifications designed to improve siRNA stability, specificity, and pharmacokinetic behavior. Common modifications include 2'-O-methyl, 2'-fluoro, and phosphorothioate substitutions, while more advanced structural designs such as glycol nucleic acids (GNAs) further enhance binding affinity and target selectivity. Terminal conjugations, including cholesterol, GalNAc ligands, antibodies, and aptamers, have also enabled tissue-specific targeting and improved biodistribution. These advances in chemical engineering and delivery technology are transforming siRNA therapeutics into an increasingly versatile and clinically mature platform.



Steric-Blocking and Splice-Switching Oligonucleotides

Steric-blocking antisense oligonucleotides (ASOs) sit within a broader RNA-targeting therapeutic landscape in which short, programmable oligonucleotides are used to increase, decrease, or remodel gene expression. Unlike gapmer ASOs, which recruit RNase H to cleave target RNA, steric-blocking ASOs are typically fully chemically modified and act by physical occupancy of an RNA sequence, thereby preventing access by RNA-binding proteins, spliceosome components, ribosomes, microRNAs, or 3'-end processing factors (Bennett and Swayze, 2010; Havens and Hastings, 2016). This seemingly simple mode of action has proven unusually versatile: depending on where the ASO binds, it can enhance productive gene output, suppress pathogenic RNA or protein production, or edit the architecture of the mature transcript.



One major opportunity is gene or isoform UP-regulation, where ASOs are used to increase the production or activity of a beneficial protein. For example, nusinersen binds to the ISS-N1 element downstream of SMN2 exon 7 and blocks inhibitory splicing factors, thereby promoting exon 7 inclusion and increasing full-length SMN protein in spinal muscular atrophy (Hua et al., 2008; Finkel et al., 2017).

Related concepts include enhancing canonical splicing by masking cryptic splice sites or silencer elements, as well as inhibiting microRNAs through antagomirs or steric-blocking oligonucleotides that prevent miRNA–mRNA interactions (Krützfeldt et al., 2005). ASOs have also been explored as translational activators by targeting inhibitory elements in 5' UTRs, including upstream open reading frames (uORFs), with the goal of relieving translational repression and increasing downstream protein production (Liang et al., 2016). However, uORF-directed activation remains an emerging and context-dependent strategy, with recent replication studies highlighting that not all reported uORF-targeting effects are readily generalizable (Roberts et al., 2024). More broadly, these “UP” approaches suggest that ASOs can be used not only to silence genes, but also to restore insufficient protein expression or rebalance disease-associated pathways.

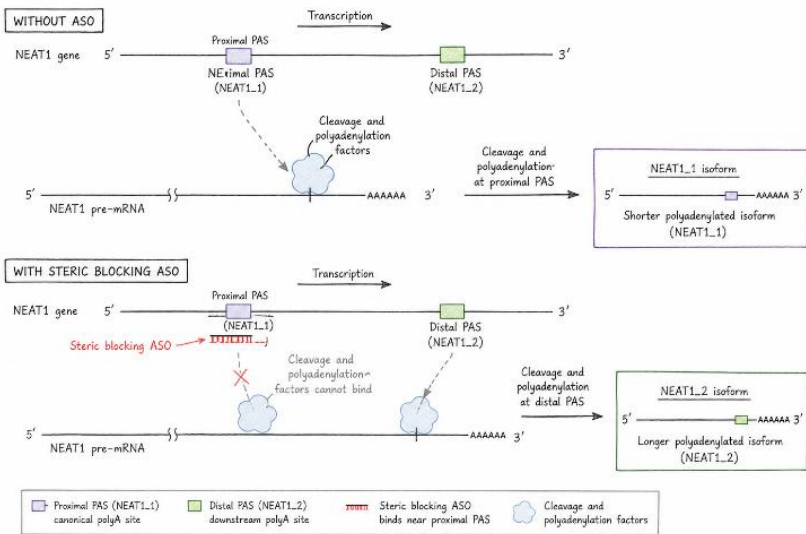
Steric blocking ASO's can also be used for gene DOWN-regulation, where oligonucleotides reduce harmful RNA or protein output. Classical antisense gapmers achieve this through RNase H–mediated cleavage of RNA–DNA hybrids, however, Steric-blocking ASOs can also reduce gene output indirectly by redirecting splicing to introduce premature termination codons and trigger nonsense-mediated decay, or by modulating alternative polyadenylation to alter transcript stability, localization, coding potential, or 3' UTR regulatory architecture (Havens and Hastings, 2016; Tian and Manley, 2017).

A third and particularly creative mode is “editing”, in which ASOs reshape the transcript rather than simply increasing or decreasing its abundance. Exon skipping in Duchenne muscular dystrophy is the most well-known clinical example: phosphorodiamidate morpholino oligomers such as eteplirsen are designed to mask splice elements in the DMD pre-mRNA, skip selected exons, restore the reading frame, and permit production of an internally deleted but partially functional dystrophin protein (Mendell et al., 2016; Aartsma-Rus and Corey, 2020). Similar splice-switching strategies can suppress cryptic exons, as shown by milasen, a personalized ASO designed to correct a patient-specific splicing defect in CLN7 disease (Kim et al., 2019). Beyond correction of single pathogenic events, ASOs can be used for protein engineering at the RNA level by controlling exon inclusion, exclusion, or isoform balance, potentially enabling tissue-specific remodeling of protein domains, signaling outputs, or developmental programs (Havens and Hastings, 2016; Lim et al., 2020).

One particularly creative transcript-editing strategy uses steric-blocking ASOs to remodel isoform output of NEAT1 with the goal to shift expression away from the shorter NEAT1_1 isoform and toward the longer

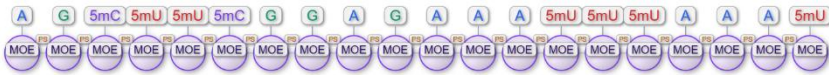
NEAT1_2 isoform. The two isoforms have different biological functions: NEAT1_2 acts as an architectural long non-coding RNA that scaffolds paraspeckles, nuclear bodies that can sequester RNA-binding proteins and regulate gene-expression programs. By contrast, NEAT1_1 does not efficiently form paraspeckles and can support paraspeckle-independent functions that, in some cancer contexts, are associated with increased cell proliferation. In high-risk neuroblastoma models, this isoform balance appears functionally important where NEAT1_1 is linked to a more proliferative cancer-cell state; increased NEAT1_2 and paraspeckle formation are associated with reduced neuroblastoma cell viability and activation of differentiation-related pathways.

In 2021, Naveed and colleagues at the University of Western Australia described a transcript-editing strategy that used steric-blocking ASOs to manipulate alternative polyadenylation of the long non-coding RNA NEAT1. Working in neuroblastoma models, the researchers designed ASOs that bind sequences surrounding the canonical polyadenylation site of the short NEAT1_1 isoform. By occupying this region, the ASOs physically blocked cleavage and polyadenylation factors from processing the transcript at the proximal polyadenylation site. This forced transcription to continue beyond the usual NEAT1_1 termination site and promoted production of the longer NEAT1_2 isoform through a downstream processing pathway. The study showed that steric-blocking ASOs can redirect RNA 3'-end formation, rather than simply alter splicing, and demonstrated how isoform switching can be used to probe opposing biological functions of long non-coding RNA isoforms in cancer cells (Naveed et al., 2021).



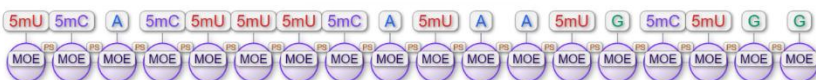
Naveed, A., Cooper, J.A., Li, R., Hubbard, A., Chen, J., Liu, T., Wilton, S.D., Fletcher, S. and Fox, A.H. (2021) 'NEAT1 polyA-modulating antisense oligonucleotides reveal opposing functions for both long non-coding RNA isoforms in neuroblastoma', *Cellular and Molecular Life Sciences*, 78, pp. 2213–2230. Available at: <https://doi.org/10.1007/s00018-020-03632-6>

Splice switching ASO for Usher syndrome type 2



Ulteversen (QR-421a) is a 21-mer chemically modified splice-altering ASO for the treatment of USH2A-associated retinitis pigmentosa and Usher syndrome type 2. The molecule incorporates a fully phosphorothioate backbone together with all nucleosides modified with 2'-O-(2-methoxyethyl; aka MOE) sugars (Girach et al., 2022). Ulteversen binds selectively to USH2A pre-mRNA and induces exon 13 skipping, thereby restoring production of a shorter but functional usherin protein (Girach et al., 2022). This mechanism is particularly relevant for USH2A disease because the large size of the USH2A gene limits conventional AAV-mediated gene replacement strategies. Intravitreal administration allows broad retinal distribution without the need for subretinal surgery or viral vectors; reducing procedural and inflammatory risks associated with gene therapy (Girach et al., 2022). Preclinical studies demonstrated sustained exon skipping, restoration of usherin expression, retinal uptake across photoreceptor layers, and recovery of functional electroretinographic responses (Dulla et al., 2021). Clinically ulteversen demonstrated a favorable tolerability profile with no serious ocular adverse events or treatment discontinuations, alongside trends toward stabilization of retinal structure and visual function compared with sham treatment (Yang et al., 2025). Preservation of ellipsoid zone area, together with stabilization of visual acuity and retinal sensitivity measures, suggests that ulteversen may slow progressive photoreceptor degeneration in USH2A-associated RP, supporting continued evaluation in ongoing Phase 2b/3 studies (Yang et al., 2025).

Splice modulating ASO to generate a productive protein product

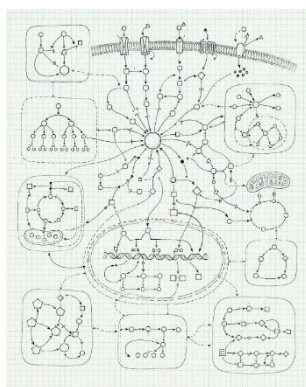


Another splicing example, Spinraza (nusinersen) is also a fully modified antisense oligonucleotide with a phosphorothioate backbone and fully modified 2'-O-methoxyethyl sugar 18mer. This ASO enhances the stability of the SMN2 pseudogene and is clinically indicated for the treatment of spinal muscular atrophy (SMA) in both pediatric and adult patients. The dosing regimen consists of four loading doses (three given at 14-day intervals followed by a fourth after 30 days), followed by a 12 mg

maintenance dose every four months (Biogen, 2016) illustrating the therapeutic robustness of this compound in the CNS.

Beyond the approved therapies for SMA and Duchenne muscular dystrophy, many splice-switching antisense oligonucleotides (ASOs) are now in preclinical and clinical development across a wide range of diseases. Active programs target rare neuromuscular and neurodevelopmental disorders, including Dravet syndrome, Batten disease, ataxia-telangiectasia, Huntington's disease, as well as emerging applications in cancer and immune-mediated disorders.

Splice-Switching for Pathway Modulation



Splice-switching ASOs can also be used as pathway-modulating therapeutics, where the objective is to reshape the behavior of a disease-relevant signaling network rather than simply increase or decrease expression of a single gene. This strategy is especially powerful when systems-level analysis identifies a regulatory node whose activity depends on isoform structure, protein–protein interactions, feedback regulation, or subcellular recruitment. In these cases, the most therapeutically useful intervention may not be gene silencing, but isoform rebalancing: shifting RNA splicing toward a version of the protein that preserves essential function while reducing pathogenic pathway output. Such approaches align naturally with systems biology, because they treat disease pathways as dynamic circuits with tunable control points. Instead of removing a component from the network, splice-switching can alter how that component transmits signal through the pathway. The SOS1/RAS axis provides a useful example of this logic, because SOS1 sits upstream of RAS activation and its splice isoforms appear to differ in Ras-stimulatory activity. A splice-switching strategy that favors the less activating SOS1 isoform could therefore reduce RAS–MAPK signaling while maintaining overall

SOS1 expression and avoiding the broader consequences of complete SOS1 knockdown.

Splice-switching to attenuate RAS activity through SOS1 isoform switch

RAS signaling is one of the central pathways controlling cell growth, survival, differentiation, and proliferation. In cancer, this pathway is frequently hyperactivated, either through activating mutations in RAS itself or through upstream regulators that increase the amount of active, GTP-bound RAS. One such upstream regulator is SOS1, a guanine nucleotide exchange factor that promotes conversion of inactive RAS-GDP into active RAS-GTP. Because SOS1 sits upstream of RAS activation, changing SOS1 function offers a potential way to reduce signaling through the RAS–MAPK pathway.

A group of scientists at La Jolla Labs designed a splice-switching strategy around a naturally occurring alternative splicing event in SOS1. SOS1 encodes a widely expressed guanine nucleotide exchange factor (GEF) that activates RAS-family GTPases by promoting GDP–GTP exchange, thereby increasing the pool of active RAS-GTP and downstream signaling through pathways such as RAS–RAF–MEK–ERK/MAPK (Baltanás et al., 2020). Human cells express two major SOS1 splice isoforms that differ by inclusion or exclusion of a short 45 bp exon, referred to here as exon 21. The exon-included isoform is referred to as IsfII, while the exon-skipped isoform is referred to as IsfI. Prior work identified these two human SOS1 isoforms and reported that they differ by a 15-amino-acid segment in the C-terminal proline-rich region, a regulatory region involved in adaptor-protein binding and control of SOS1 activity (Baltanás et al., 2020). That work suggested that inclusion of this segment enhances the Ras-stimulatory activity of SOS1. In this model, the +exon 21 isoform, IsfII, is more active, whereas the –exon 21 isoform, IsfI, is predicted to be less activating toward RAS.

The therapeutic idea, therefore, is not to knock down SOS1 entirely, but to bias splicing away from the more RAS-activating isoform and toward the less activating isoform. Splice-switching oligos were designed to sterically block inclusion of exon 21, forcing the transcript to skip this exon. By reducing production of IsfII and increasing production of IsfI, the oligos are expected to attenuate SOS1-mediated RAS activation while preserving overall SOS1 expression.

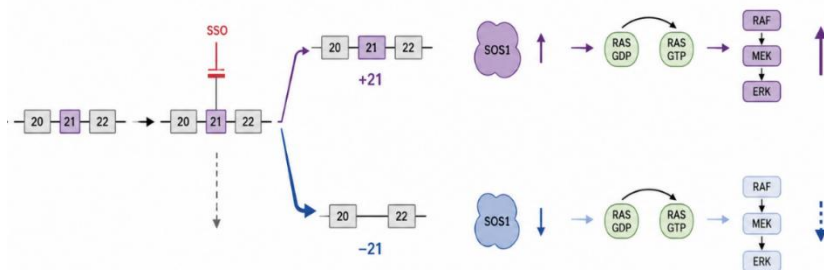
This distinction is important because SOS1 is not simply an abundance-driven target. Its activity is controlled by domain architecture, membrane recruitment, adaptor binding, autoinhibition, and allosteric

activation. SOS1 contains N-terminal regulatory domains, including histone-like, Dbl homology and pleckstrin homology domains, a central catalytic module containing REM and CDC25H domains, and a C-terminal proline-rich region that mediates interactions with SH3-domain-containing adaptor proteins such as GRB2 and E3B1 (Baltanás et al., 2020). Total SOS1 abundance may therefore remain largely unchanged while the functional composition of SOS1 isoforms changes; i.e. to alter the ratio of SOS1 splice isoforms in favor of a form with lower Ras-stimulatory activity. This makes the strategy conceptually different from siRNA, CRISPR knockout, or degradation-based approaches. It is a functional isoform switch, not a simple loss-of-expression strategy.

Mechanistically, SOS1 promotes RAS signaling by acting as a GEF. It helps exchange GDP for GTP on RAS, thereby increasing active RAS-GTP. Active RAS then propagates signaling through downstream pathways, including the RAS–MAPK cascade. SOS1 activity is also shaped by membrane localization, GRB2-mediated recruitment to activated receptor tyrosine kinases, allosteric stimulation by RAS-GTP, and negative feedback through ERK- and RSK-mediated phosphorylation of the SOS1 C-terminal region (Baltanás et al., 2020). In resting cells, SOS1 is maintained in an autoinhibited state through regulatory contributions from both the N-terminal and C-terminal regions. Recruitment to the membrane and interaction with adaptor proteins can relieve this inhibition, allowing SOS1 to engage membrane-localized RAS and catalyze nucleotide exchange.

The C-terminal proline-rich region is especially relevant to the splice-switching strategy because it participates in adaptor binding and regulatory control. GRB2–SOS1 complexes recruit SOS1 to activated receptor tyrosine kinases, enabling RAS activation at the inner plasma membrane. Alternative splicing within this region could therefore alter SOS1 signaling output since it targets a regulatory region rather than the catalytic core itself.

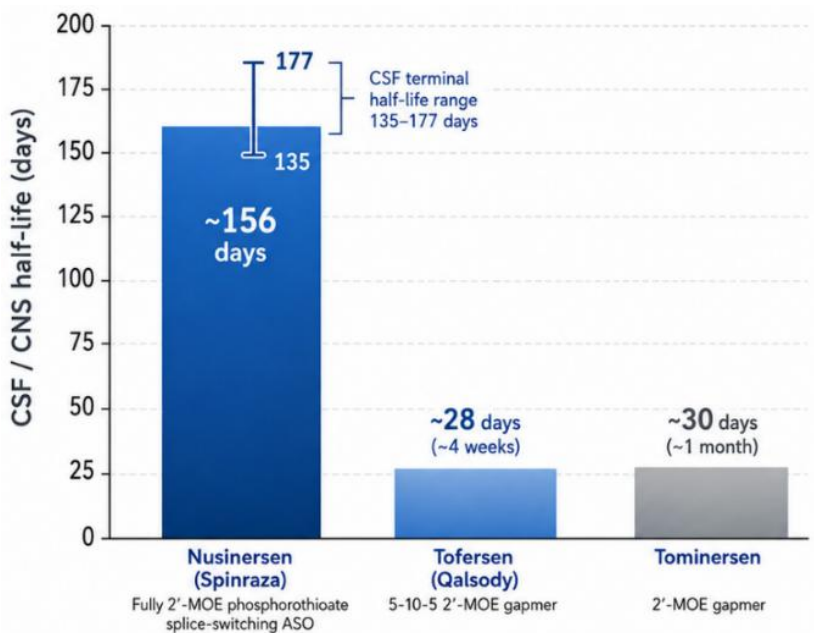
The splice-switching strategy: reduce RAS pathway activation by shifting SOS1 splicing from the exon 21-included isoform to the exon 21-skipped isoform and decreased generation of active RAS-GTP with further reduced signal flow through the RAS–MAPK pathway, while avoiding complete loss of SOS1 (Baltanás et al., 2020).



Experimental results support the feasibility of this splicing approach. The lead oligos showed upregulation of the *IsfI* isoform and loss or strong reduction of detectable *IsfII*. Importantly, general *SOS1* abundance was largely maintained, except for some exon 22-targeting designs that increased total *SOS1* signal. Sanger sequencing was consistent with the qPCR results, supporting the interpretation that the observed changes reflected genuine splice switching rather than assay artefact.

The main unresolved biological question is whether forcing *SOS1* toward *IsfI* is sufficient to produce a meaningful reduction in RAS signaling in disease-relevant cells. Key next steps include dose-response testing of lead oligos, validation in additional cancer cell lines, and protein-level readouts of RAS pathway activity. Useful assays could include measurements of RAS-GTP, phospho-MEK, phospho-ERK, or other downstream MAPK signaling markers. It will also be important to determine whether this strategy works similarly across *KRAS*-, *HRAS*-, and *NRAS*-driven contexts, and whether it remains effective in cells carrying activating RAS mutations. Because *SOS1* can also influence *RAC1*-dependent processes through adaptor-specific signaling complexes, assays of migration, cytoskeletal phenotypes, or *RAC1* activation may also be useful in selected contexts.

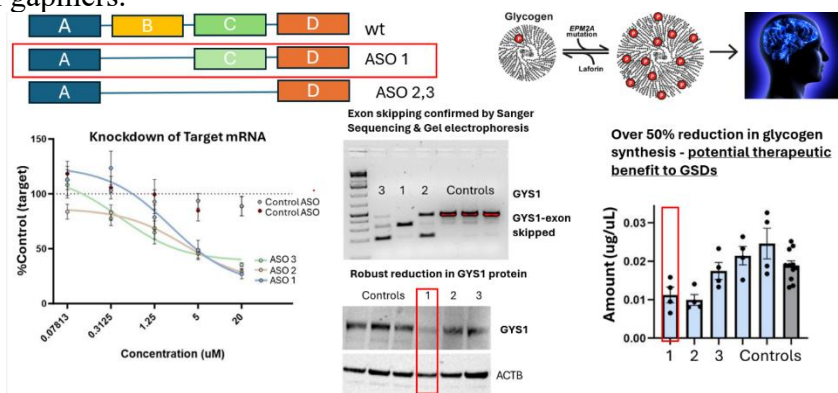
expression. This makes SOS1 exon 21 skipping an attractive model for splice-switching therapeutic development: the biology is tied to a clear functional isoform change, the mechanism acts upstream of RAS activation, and the pharmacologic intervention is well matched to the splicing-based mode of action. More broadly, the strategy fits with the emerging view of SOS1 as both a regulator of normal RAS signalling and a therapeutic target in RAS-dependent disease, but it seeks to modulate SOS1 function through isoform switching rather than direct enzymatic inhibition or protein depletion.



Gene Knockdown Using Nonsense Mediated Decay

RNase H-mediated knockdown via gapmer technologies such as tofersen use a chimeric architecture, typically with modified 2'-MOE “wings” flanking a central DNA gap, which enables RNase H recruitment and degradation of the target RNA. This design is potent, but the central DNA gap is less nuclease-protected than a fully modified steric-blocking ASO. Tofersen and Tominersen are 20-mer 5-10-5 MOE gapmers, with an estimated effective CSF half-life of approximately four weeks when intrathecally delivered (FDA, 2023; Goyal et al., 2023). By contrast, Nusinersen, a fully modified 2'-MOE phosphorothioate splice-switching ASO that does not require a DNA gap or RNase H recruitment has a substantially greater nuclease resistance and a much longer CSF terminal half-life of approximately 135–177 days (FDA, 2026).

This distinction suggests an attractive therapeutic strategy for targets where knockdown can be achieved by splice modulation rather than RNase H cleavage: a fully modified ASO that redirects splicing to create a premature termination codon and thereby induces nonsense-mediated decay (NMD). The drug behaves pharmacokinetically like nusinersen, but pharmacodynamically it reduces target protein by forcing the transcript into a non-productive, NMD-sensitive isoform. Therefore, when the biology of the target permits NMD-mediated silencing, a fully modified splice-switching ASO could offer a superior profile for chronic neurological disease: durable CNS exposure, less frequent intrathecal dosing, and avoidance of the shorter half-life liability inherent to DNA-gap-containing RNase H gapmers.



Fully modified steric-blocking ASO chemistry offers a compelling advantage when the therapeutic mechanism does not require RNase H recruitment. By eliminating the central DNA gap, these molecules are inherently more resistant to nuclease degradation, supporting longer residence time in the CNS and potentially enabling less frequent intrathecal dosing. The contrast between gapmer ASOs such as tofersen and tominersen, which rely on a nuclease-vulnerable DNA gap, and fully modified splice-switching ASOs such as nusinersen highlights the pharmacological value of chemical stability in chronic neurological indications (FDA, 2023; Goyal et al., 2023; FDA, 2026). For targets amenable to splice redirection and NMD-mediated silencing, this approach could combine the durability of fully modified 2'-MOE phosphorothioate chemistry with the functional outcome of protein knockdown. Thus, fully modified ASOs may provide a particularly attractive therapeutic architecture: nuclease-resistant, long-lasting, tunable, and mechanistically capable of reducing pathogenic protein expression without the shorter half-

life liability and tighter therapeutic window associated with DNA-gap-containing RNase H gapmers.

Optimize Productive Splicing for Gene Upregulation

Many genetic diseases arise not from toxic gain-of-function mutations, but from insufficient production of a normal or functional protein. These haploinsufficient disorders occur when one working copy of a gene does not produce enough protein to maintain normal cellular function. Traditional therapeutic strategies such as gene replacement, genome editing, recombinant protein delivery, or modified RNA can be powerful, but they may be limited by gene size, tissue targeting, immune responses, durability, or difficulty controlling expression levels. These challenges have created interest in approaches that can increase expression of a patient's own endogenous gene in a selective and titratable manner.

Antisense oligonucleotides, or ASOs, provide one such strategy. ASOs are short, chemically modified nucleic acids designed to bind specific RNA sequences through base pairing. Depending on their chemistry and design, they can reduce RNA levels, alter RNA processing, or redirect pre-mRNA splicing. While many ASO therapies have historically focused on silencing toxic transcripts or correcting mutation-specific splicing defects, a newer concept uses ASOs to increase gene expression by preventing naturally occurring non-productive splicing events.

This approach was demonstrated by Stoke Therapeutics in 2020 (Lim et al., 2020). The company introduced a platform called TANGO, or targeted augmentation of nuclear gene output, which uses ASOs to block alternative splicing events that normally generate premature-termination-codon-containing transcripts targeted for nonsense-mediated mRNA decay. By preventing these non-productive splice forms, the cell produces more productive mRNA, leading to increased full-length protein from the native gene locus.

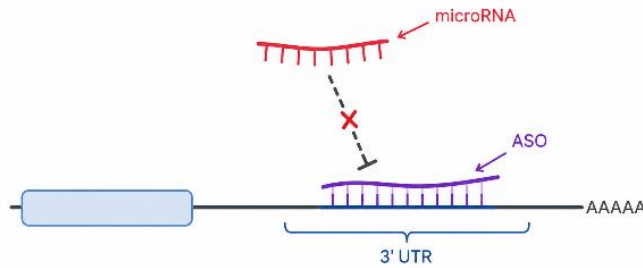
In the study, Stoke Therapeutics analyzed human RNA-seq datasets from ASO-accessible tissues, including liver, kidney, central nervous system, and eye, and identified non-productive splicing events in 7,757 protein-coding genes. Of these, 1,246 were associated with genetic disease. The authors then tested the strategy across multiple genes and splicing-event classes, including PCCA, SYNGAP1, CD274, and SCN1A. These genes represented different biological functions, disease contexts, and coding-sequence sizes, ranging from CD274 at 870 bp to SCN1A at 6,015 bp.

The work showed that ASOs could dose-dependently reduce non-productive splicing and increase productive mRNA in vitro. Importantly, this mRNA increase translated into higher protein levels, with reported

protein upregulation of approximately 1.5- to 2.6-fold at the highest ASO concentrations. Stoke also demonstrated in vivo proof of concept by delivering SCN1A-targeting ASOs into neonatal mouse brain, where treatment increased productive Scn1a mRNA and NaV1.1 protein in a dose-dependent manner.

Together, these findings established ASO-mediated upregulation as a mutation-independent strategy for increasing endogenous protein expression. Rather than replacing a gene or editing DNA, this approach leverages natural RNA-processing biology to shift splicing away from non-productive transcripts and toward productive mRNA.

Blocking miRNA Binding Sites for Gene Upregulation

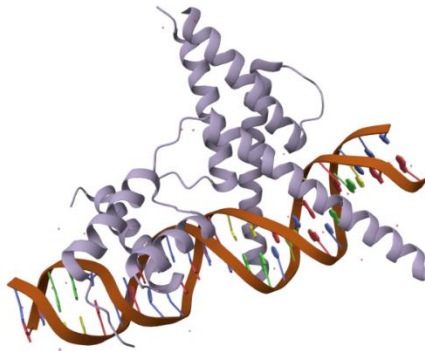


MicroRNAs are small endogenous non-coding RNAs that usually repress gene expression by binding target transcripts and reducing mRNA stability or translation (Bartel, 2018; Shang et al., 2023). MicroRNAs often have multiple targets, can act differently depending on cell state, and may produce indirect effects through regulatory networks. microRNA-based upregulation strategies require careful validation at multiple stages before their effects can be interpreted therapeutically. First, the candidate microRNA must be shown to be expressed in the disease-relevant tissue or cell type, since tissue-specific microRNA expression and tissue-restricted target availability strongly influence whether a given microRNA–mRNA interaction is biologically active (Sood et al., 2006; Liang et al., 2009; Liang and Li, 2011). This is particularly important because a microRNA that produces an apparent regulatory effect in one tissue may have little or no activity in another if the microRNA itself, its cofactors, or its relevant targets are absent. Second, during in vitro screening, the selected cell lines must be confirmed to express the microRNA of interest at physiologically

meaningful levels, rather than if a general cellular model recapitulates the relevant tissue context. Even when expression is confirmed, quantifying microRNA-mediated upregulation without bias remains challenging, because observed increases in target gene expression may reflect direct activation, indirect network effects, differences in baseline transcript abundance, or cell-state-dependent shifts in microRNA function. These considerations make it difficult to assign a precise and generalizable magnitude of upregulation during the screening phase and underscore the need for orthogonal validation in tissue-relevant models

microRNA targeting for up regulation of TCF4 haploinsufficiency

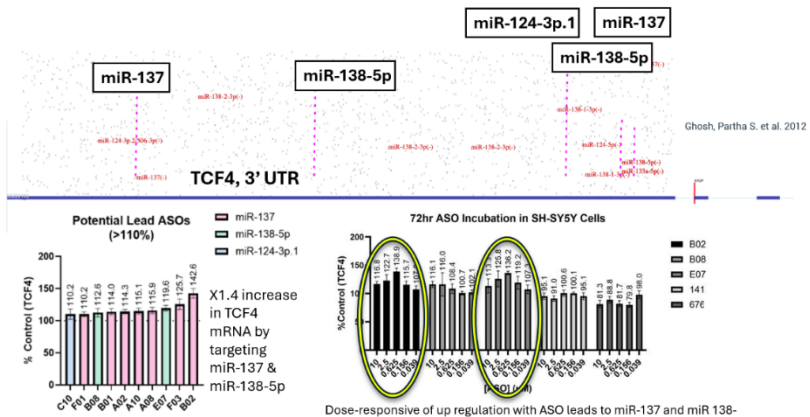
Pitt-Hopkins syndrome, or PTHS, is a rare neurodevelopmental disorder most often caused by a heterozygous pathogenic variant in TCF4 or by a deletion of chromosome 18q21.2 involving TCF4 (Zollino et al., 2019; Whalen et al., 2012). Clinically, PTHS is associated with developmental delay, moderate-to-severe intellectual disability, severely limited or absent speech, motor delay, seizures, constipation, sleep disturbance, and characteristic breathing abnormalities such as episodic hyperventilation or breath-holding



(Zollino et al., 2019; Bedeschi et al., 2017). Molecular diagnosis is established by identifying a heterozygous pathogenic or likely pathogenic TCF4 variant, small intragenic deletion, or larger 18q21.2 deletion involving the gene (Sweetser et al., 2024).

The core disease mechanism in many cases is TCF4 haploinsufficiency. This means that one functional copy of TCF4 does not produce enough *transcription factor 4* protein to support normal neurodevelopmental gene-regulatory programs. TCF4 encodes a class I

basic helix-loop-helix transcription factor (PDB: 8OSB), and disruption of its dosage affects broad downstream transcriptional networks relevant to neuronal development and function (Sepp et al., 2012; Dennis et al., 2019). Because PTHS is caused by reduced TCF4 function rather than by a toxic gain of function, a central therapeutic idea is to increase expression from the remaining functional allele. Preclinical work supports this logic: restoration of Tcf4 expression in a mouse model improved behavioral, electrophysiological and transcriptional phenotypes, suggesting that normalizing TCF4 dosage could be therapeutically meaningful (Kim et al., 2022).



This dosage-correction model makes PTHS an attractive candidate for RNA-based upregulation strategies. However, because TCF4 is a transcription factor with widespread effects, the therapeutic goal is not uncontrolled overexpression. The goal is to control a physiological range. ASOs, are well suited to this type of approach because they can be designed to bind RNA and alter gene expression through steric blocking (Bennett and Swayze, 2010; Ward et al., 2014).

La Jolla Labs, in collaboration with the Pitt Hopkins Research Foundation, screened antisense oligonucleotides to assess whether they could restore TCF4 protein levels by upregulating the functional copy of TCF4 in Pitt-Hopkins syndrome (PTHS) (Pitt Hopkins Research Foundation, 2023). La Jolla Labs' strategy was to disrupt microRNA-mediated repression as a way to increase endogenous TCF4 expression. MicroRNAs are short non-coding RNAs that bind target sites, often within the 3' untranslated regions of mRNAs, and reduce gene expression by promoting transcript destabilization or inhibiting translation (Bartel, 2009). If specific miRNAs repress TCF4, then blocking those miRNAs or their binding sites could relieve repression and increase TCF4 expression. Steric-blocking ASOs can, in principle, reduce miRNA activity either by binding the miRNA itself or by blocking a miRNA binding site on a particular target

mRNA, thereby preventing the miRNA-loaded silencing complex from engaging that transcript (Roberts et al., 2024).

The La Jolla Labs approach could therefore be framed as a target-site validation and pharmacological de-repression strategy. Candidate miRNA binding sites in the TCF4 transcript were first identified, but these candidates were not advanced based solely on computational prediction. Each candidate miRNA was validated for expression in the relevant tissue or cellular model. This expression filter was essential because a predicted miRNA–TCF4 interaction was only therapeutically useful if the miRNA was present in the disease-relevant context and computationally the 3' UTR contains over 1000 miRNA hits.

After tissue-expression validation, the next step was functional validation. Candidate miRNAs were tested for interaction with TCF4 to determine whether they measurably repressed TCF4 RNA or protein. Inhibition experiments then assessed whether blocking the miRNA or its binding site produced TCF4 upregulation. A dose-responsive upregulation was measured to distinguish specific effects from noise and provided a foundation potency and therapeutic-window optimization.

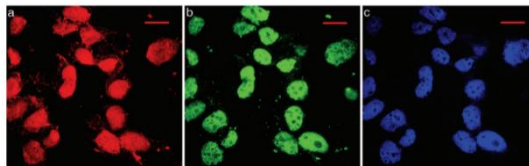
Pitt-Hopkins syndrome is a genetic disorder in which understanding the underlying disease mechanism can guide therapeutic design. Because PTHS results from TCF4 haploinsufficiency, increasing TCF4 expression represents a potential therapeutic goal. However, TCF4 is dosage-sensitive, and excessive upregulation could potentially be harmful. Unlike gene therapy, ASO-based approaches may offer a more tunable strategy for achieving an optimal level of TCF4 restoration. La Jolla Labs' miRNA-site-directed approach aimed to identify endogenous miRNA-mediated repression of TCF4 and relieve that repression using oligonucleotides. By integrating tissue-relevant miRNA expression, experimental validation of miRNA–TCF4 interaction, and dose-responsive TCF4 upregulation after inhibition, this strategy provided a rational framework for controlled restoration of TCF4 protein levels in PTHS.

MicroRNAs may also upregulate gene expression at the transcriptional level through promoter-associated mechanisms. Place et al. (2008) demonstrated that miR-373 could induce expression of CDH1, encoding E-cadherin, and CSCD2 by targeting complementary promoter sequences. This form of microRNA-mediated gene activation overlaps conceptually with RNA activation, or RNAa, in which small RNAs directed to promoter or promoter-associated sequences can increase transcription rather than silence gene expression (Place et al., 2008; Li et al., 2025). These observations expanded the view of microRNAs from purely repressive regulators to context-dependent modulators capable of either decreasing or increasing gene output.

Short Activating RNAs (saRNA)

Short activating RNAs (saRNAs) are a class of small double-stranded RNA molecules that upregulate gene expression by targeting promoter regions and triggering RNA activation (RNAa). In contrast to siRNAs, saRNAs work within the nucleus, often involving Argonaute proteins, to enhance transcription of specific genes.

A conundrum might seem that siRNAs are understood to function in the cytoplasm, whereas saRNAs, with the same size act within the nucleus to regulate transcription. Early studies of RNA activation infer nuclear localization indirectly, showing that promoter-targeted dsRNAs induce locus-specific chromatin modifications (Li et al., 2006; Janowski et al., 2007). Direct imaging studies demonstrate that siRNAs can enter the nucleus in a selective, target-dependent manner, localizing to nuclear RNAs such as 7SK (RNA polymerase II transcription elongation) and U6 (an RNA that is part of the spliceosome) (Berezghna et al., 2006). However, this is balanced by evidence that small RNAs are not freely distributed and may be actively excluded from parts of the nucleus, indicating regulated trafficking rather than constitutive nuclear access (Ohrt et al., 2006). Together, these findings suggest saRNA activity is context-dependent and maintains selective localization, rather than an absolute compartmental separation.



7SK siRNA accumulates in the nucleus in Huh-7 cells 2 h after transfection. siRNA antisense (red) and sense (green) strands colocalize primarily in the nucleus (blue) and seen in the cytoplasm only in small amounts

Berezghna, S.Y., Supekova, L., Schultz, P.G. and Deniz, A.A. (2006) 'siRNA in human cells selectively localizes to target RNA sites', *Proceedings of the National Academy of Sciences*, 103(20), pp. 7682–7687.

Because of this unique ability, saRNAs are emerging as promising tools for therapeutic gene activation (Desaulniers *et al.*, 2025). MTL-CEBPA (CEBPA-51/AW1-51), a liposomal encapsulated 21-nt saRNA targeting the transcription factor CEBPA, which is frequently silenced in hepatocellular carcinoma. In a first-in-human Phase I study (NCT02716012), MTL-CEBPA demonstrated an acceptable safety profile alongside evidence of target gene activation, antitumor activity, including

disease stabilization and occasional objective responses. These results supported advancement into randomized Phase II studies and combination trials with immunotherapies, establishing MTL-CEBPA as a prototype RNAa therapeutic and demonstrating the feasibility of systemic gene activation in humans (Reebye et al., 2018).



Another cancer-targeting RNAa therapeutic is RAG-01 (Ractigen Therapeutics), which targets the tumor suppressor CDKN1A (p21). RAG-01 is delivered via intravesical administration, a method where the drug is delivered directly into the bladder through a catheter, to patients with non-muscle invasive bladder cancer (NCT06351904). This compound attempts to directly activate p21, a historically “undruggable” pathway. Early Phase I data indicate favorable safety, absence of dose-limiting toxicities, and encouraging preliminary efficacy signals, including complete responses in carcinoma in some patients (Kang et al., 2024).

miR-551b-3p is a human microRNA which binds to a promoter and activates the expression of the STAT3 gene. miR-551b-3p activates, rather than represses, gene expression through promoter-associated RNA activation. Experimental studies in ovarian and triple-negative breast cancer models demonstrated that miR-551b-3p translocates to the nucleus and interacts with the STAT3 promoter, leading to recruitment of transcriptional machinery including RNA polymerase II and TWIST1, and increasing STAT3 transcription and downstream oncogenic signaling (Parashar et al., 2016; Parashar et al., 2019). The nuclear localization of miR-551b-3p depends on Importin-8 (IPO8), a member of the karyopherin/importin family previously identified as a nuclear transport mechanism for Argonaute-microRNA complexes (Wei et al., 2014; Parashar et al., 2019). Promoter-directed activation by RNA remains a relatively specialized and emerging area of RNA therapeutics, however the nucleotide chemistry modifications to date have not deviated from those commonly used by other platforms like siRNA.

The origins of the field can be traced to work on catalytic RNA (ribozymes), particularly by Professor Gerald Joyce and colleagues at the Salk Institute in La Jolla, California. Joyce argued that RNA possessed both “genotypic and phenotypic properties” and was therefore “a suitable substrate for molecular evolution in the laboratory” (Joyce, 1989). In his 1989 paper *Amplification, mutation and selection of catalytic RNA*, Joyce outlined a conceptual framework in which RNA molecules could undergo iterative cycles of amplification, mutation and selection, effectively creating artificial Darwinian evolution in a test tube (Joyce, 1989).

In 1990, Debra Robertson and Gerald Joyce at the Scripps Research Institute demonstrated one of the earliest laboratory evolution experiments involving RNA enzymes. Their paper, *Selection in vitro of an RNA enzyme that specifically cleaves single-stranded DNA*, showed that ribozymes could undergo repeated rounds of selection and amplification to evolve new catalytic activities (Robertson and Joyce, 1990). Importantly, they proposed that “controlled evolution experiments” could be used to generate RNAs with “novel catalytic function” (Robertson and Joyce, 1990). This work demonstrated the experimental feasibility of iterative molecular selection and helped establish the conceptual foundation for later aptamer selection technologies. Building on these advances, researchers soon began exploring whether similar evolutionary strategies could be applied not only to catalytic RNAs, but also to nucleic acids capable of highly specific molecular recognition. These efforts ultimately led to the development of in vitro selection methods capable of screening vast random-sequence nucleic acid libraries for ligand-binding activity.

In 1990 Andrew Ellington and Jack Szostak, working at the Department of Molecular Biology at Massachusetts General Hospital and Harvard Medical School in Boston, demonstrated that vast libraries of random RNA molecules could be screened to isolate sequences capable of binding small organic ligands with high specificity (Ellington and Szostak, 1990). In their landmark Nature paper, *In vitro selection of RNA molecules that bind specific ligands*, Ellington and Szostak showed that RNA populations containing approximately 10^{13} – 10^{15} distinct sequences could yield rare molecules that folded into highly specific ligand-binding structures.

At almost the same time, Craig Tuerk and Larry Gold at the University of Colorado developed the methodology that became known as SELEX (Systematic Evolution of Ligands by EXponential enrichment), formally establishing iterative selection and amplification as a general platform for evolving high-affinity nucleic acid ligands in vitro (Tuerk and

without disrupting the core folding architecture. As highlighted in broader aptamer reviews, pegaptanib exemplifies how structured oligonucleotides can achieve antibody-like specificity through well-defined three-dimensional folds, despite being identified and optimized primarily through functional selection rather than structure-first design (Keefe, Pai and Ellington, 2010).



Izervay (avacincaptad pegol) is a pegylated RNA aptamer developed for the treatment of geographic atrophy secondary to age-related macular degeneration (AMD). Unlike earlier retinal therapies that primarily targeted abnormal blood vessel growth, Izervay works by inhibiting complement component C5, a key mediator in the inflammatory complement cascade implicated in retinal cell degeneration. By selectively blocking C5 activation, the drug is designed to reduce downstream inflammatory damage and slow the progressive loss of retinal tissue associated with geographic atrophy. Its approval marked an important advance in ophthalmology, representing one of the first therapies specifically aimed at modifying the complement-driven pathology underlying dry AMD.

Aptamer Bio-tools

In addition to therapeutic aptamers, fluorogenic RNA aptamers have emerged as transformative tools for visualizing RNA localization and dynamics in living cells, enabling direct observation of RNA trafficking, translation, and phase separation with high spatiotemporal resolution. Early studies of mRNA localization using in situ hybridization and live-cell tagging systems established that asymmetric RNA distribution is a conserved mechanism for spatial regulation of gene expression across organisms ranging from yeast to neurons. Aptamer technology was quickly recognized as a tool for visualizing these dynamic processes.

The development of genetically encoded RNA imaging systems, particularly the MS2 bacteriophage platform, enabled the first “live” visualization of endogenous RNA transport dynamics in yeast and mammalian cells, revealing that localized mRNAs undergo directed, motor-dependent transport, anchoring, and translational repression during transit. Subsequent fluorogenic aptamer technology called Spinach and Mango introduced “conditional” fluorescence activation, greatly expanding utility to specific cellular contexts (Paige et al., 2011; Dolgosheina et al., 2014).

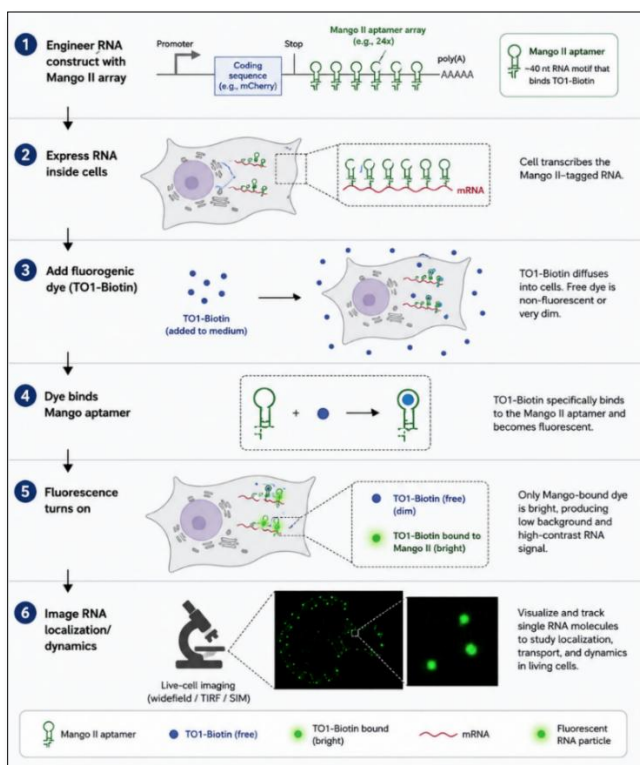
Fluorophore aptamers have also been instrumental in uncovering principles of RNA-mediated phase separation and stress granule biology. Using live-cell imaging approaches, these tools enabled visualization of mRNAs dynamically shuttling into and out of stress granules and demonstrated that RNAs themselves actively contribute to biomolecular condensate assembly rather than passively accumulating within them (Khong et al., 2017). Such studies established that selective RNA recruitment, transport kinetics, and localized translation are tightly coupled to condensate dynamics and cellular stress responses. Importantly, aberrant stress granule assembly and defective RNA trafficking have been directly linked to neurodegenerative diseases including amyotrophic lateral sclerosis (ALS) and frontotemporal dementia through pathological aggregation of RNA-binding proteins such as TDP-43 and FUS, which disrupt normal bidirectional RNA transport and granule dynamics (Williams et al., 2016; Khong et al., 2017).

Like Spinach, Mango II functions through a structurally preorganized RNA fluorophore pocket built around a compact G-quadruplex scaffold, but it achieves fluorescence activation through binding of thiazole orange derivatives such as TO1-biotin with exceptionally high affinity (~1 nM). Structural studies revealed that Mango II forms a three-tiered G-quadruplex in which the fluorophore stacks directly on the terminal G-quartet while surrounding adenine “propeller loops” create a constrained and highly complementary binding environment that suppresses torsional motion of the dye and thereby reduces nonradiative decay pathways. In the bound state, the benzothiazole and methylquinoline rings of TO1-biotin become nearly coplanar and rigidified against the quadruplex surface, dramatically increasing fluorescence quantum yield and lifetime. Importantly, the Mango II scaffold is highly preorganized even before ligand binding, contributing to both its thermostability and unusually high fluorophore affinity. Crystal structures further demonstrated that fluorescence efficiency correlates strongly with ligand–RNA shape complementarity and conformational homogeneity: mutations such as A22U reduce structural plasticity within the binding pocket, improve fluorophore packing, increase fluorescence

enhancement, and lengthen excited-state lifetimes by further restricting dye flexibility. Together, these findings establish Mango II as an RNA analog of protein fluorophore cages, where precise steric confinement and π -stacking interactions within a G-quadruplex architecture stabilize the emissive state and enable bright RNA imaging in living cells.



Spinach fluoresces because its G-quadruplex architecture locks DFHBI into a rigid, planar, hydrogen-bonded configuration that suppresses nonradiative relaxation and allows efficient photon emission essentially functioning as an RNA version of GFP's chromophore pocket.
PDB: 4TS0, PTS2 -- Crystal structure of the Spinach RNA



Fluorogenic RNA aptamer systems are valuable tools in cancer as they allow researchers to see where specific RNA molecules are located inside cells. For example, the Mango aptamer can be genetically attached to an RNA of interest, and when a dye such as SALAD1 binds to the aptamer, the RNA becomes fluorescent. This can help cancer researchers study whether an oncogenic transcript is concentrated in the nucleus, exported to the cytoplasm, or gathered in RNA granules that may affect tumor-cell survival.

Applications of aptamers bio-tools

Fluorescent RNA aptamers have important applications in studying disease pathophysiology; they allow researchers to monitor RNA activity, localization and abundance inside cells. In cancer research, these tools are especially useful because many disease mechanisms are controlled by RNA, including microRNAs, long noncoding RNAs and messenger RNAs that regulate oncogenes or tumor suppressor genes.

In 2021, Dou and colleagues developed a genetically encoded dual-color light-up RNA sensor for ratiometric imaging of microRNA. Their RNA-based fluorescent sensor allowed researchers to monitor microRNA activity inside cells (Dou et al., 2021). This is relevant to cancer research because microRNAs often regulate tumor suppressor genes and oncogenes. Similar aptamer-based reporters could be used to study whether a cancer-associated microRNA reduces the amount of a target RNA in the cytoplasm, redirects that RNA to processing bodies, or changes its localization after treatment with a targeted therapy. This would help researchers connect microRNA activity to cancer processes such as altered gene expression, uncontrolled proliferation, invasion and drug resistance.

Fluorescent aptamers are also useful in neurodegenerative disease research, where abnormal RNA repeats or RNA aggregates can disrupt normal cell function. In 2013, Strack, Disney and Jaffrey used the fluorescent RNA aptamer Spinach2 to study toxic repeat-containing RNA in mammalian cell-culture models. They engineered a reporter RNA containing expanded CGG repeats, which are associated with fragile X-associated tremor/ataxia syndrome, and attached Spinach2 to the repeat-containing RNA. When Spinach2 bound its fluorophore, the tagged RNA became fluorescent, allowing researchers to observe CGG-repeat RNA foci inside living cells and monitor how those foci changed after drug treatment (Strack, Disney and Jaffrey, 2013).

Spinach2-based microRNA assays also have value for studying cancer pathophysiology. In 2016, Aw, Tang, Teo and Cohen developed Pandan, a Spinach2-based fluorescent microRNA sensor, at research institutions in Singapore and Denmark. They redesigned the Spinach2 RNA aptamer so that it regained fluorescence only when it bound a specific target microRNA. This allowed the researchers to detect microRNAs using a visible fluorescence signal rather than relying only on indirect amplification-based methods (Aw et al., 2016).

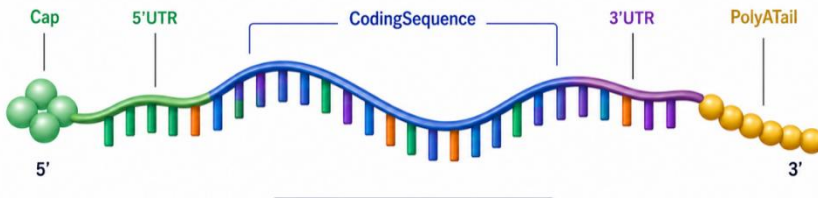
The value of this type of assay is that microRNAs are not simply biomarkers; they can actively shape disease mechanisms. In cancer, some microRNAs suppress tumor suppressor genes, while others reduce oncogene expression. For example, if a microRNA such as miR-21 is overactive in a cancer model, researchers could use a related fluorescent aptamer sensor to measure its activity and link that activity to downstream effects such as reduced tumor suppressor expression, increased proliferation, invasion or resistance to therapy.

These assays can also show how microRNA activity changes under different disease conditions. Researchers could compare fluorescence signals between healthy cells, early-stage cancer cells and aggressive cancer cells to determine whether a specific microRNA becomes more active as disease progresses. They could also treat cancer cells with chemotherapy, targeted therapy or a microRNA inhibitor and then measure whether the fluorescent signal decreases. This would help show whether treatment changes the RNA regulatory pathways that drive disease.

In 2024, Spence, Waldron and colleagues at the University of Aberdeen used a TDP-43 RNA aptamer, called TDP-43APT, to detect pathological TDP-43 in formalin-fixed paraffin-embedded human ALS tissue. The researchers applied the aptamer to post-mortem tissue from a deeply phenotyped ALS cohort and used it alongside single-molecule in situ hybridization to measure STMN2 cryptic splicing, a marker of TDP-43 loss of function. This allowed them to compare where abnormal TDP-43 appeared in cells with where RNA-splicing defects appeared in the same disease context. Their study showed that TDP-43APT detected pathological TDP-43 changes that standard antibody staining did not detect. In particular, the aptamer revealed abnormal nuclear TDP-43 pathology before obvious cytoplasmic TDP-43 accumulation. The researchers also found that this early nuclear pathology coincided with STMN2 cryptic splicing, suggesting that TDP-43 dysfunction begins before the later cytoplasmic aggregates usually associated with ALS pathology become prominent. This finding is valuable for understanding ALS pathophysiology because it changes how researchers think about the timing of TDP-43 disease mechanisms. Traditional pathology often focuses on cytoplasmic TDP-43 inclusions, but

Spence and colleagues showed that earlier nuclear changes may already reflect TDP-43 loss of function. Their work highlights that aptamer-based staining can reveal subtle early aggregation events, link those events to RNA-processing defects, and help identify disease changes that may occur before clinical symptoms or advanced pathology are obvious (Spence and Waldron et al., 2024).

mRNA Vaccines



For much of the 1960s and early 1970s, messenger RNA (mRNA) was viewed primarily as a passive intermediary between DNA and protein synthesis, with research focused largely on coding sequences while non-coding regions were considered biologically insignificant. Early studies of bacterial translation established foundational principles of translational initiation, including identification of AUG as the primary start codon and discovery of the Shine–Dalgarno sequence, a ribosome-binding motif that positioned prokaryotic ribosomes at initiation sites (Shine and Dalgarno, 1974). However, the absence of comparable motifs in eukaryotic mRNAs, together with the presence of long and structurally complex untranslated regions, suggested that eukaryotes used fundamentally different mechanisms to regulate translation. Subsequent discoveries progressively redefined mRNA as a highly organized regulatory molecule, beginning at the 5' end with identification of the 7-methylguanosine cap and the ribosomal scanning model proposed through the pioneering work of Marilyn Kozak, which established the importance of sequence context surrounding the AUG initiation codon (Kozak, 1984; Kozak, 1987). Later investigations extended this framework across the full length of the transcript, revealing regulatory roles for 5' untranslated regions (5' UTRs), coding sequence optimization, RNA secondary structure, modified nucleosides, poly(A) tails, and 3' untranslated regions (3' UTRs) in controlling mRNA stability, localization, and translational efficiency. Together, these discoveries transformed understanding of gene expression from a simple linear flow of information into a coordinated system of post-transcriptional regulation extending from the 5' to the 3' end of the molecule. These discoveries provided the technology stack for synthetic mRNA engineering modern mRNA therapeutics, including vaccines and programmable gene-delivery platforms.

The 5' Cap

In the 1970s, Aaron Shatkin and colleagues discovered that eukaryotic messenger RNAs contain an unusual, methylated guanosine linked to the 5' end through a 5'–5' triphosphate bridge (Shatkin, 1976).

They subsequently identified this structure as 7-methylguanosine (m7G) and demonstrated that the cap plays essential roles in mRNA stability and translational efficiency.

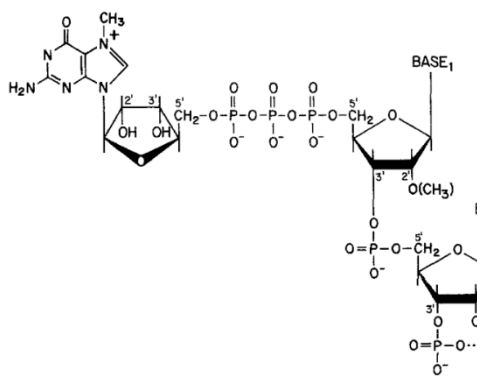


Figure 1. Structure of 5' Cap

Shatkin, A.J. (1976) 'Capping of eucaryotic mRNAs', *Cell*, 9(4 Pt 2), pp. 645–653. doi:10.1016/0092-8674(76)90128-8.

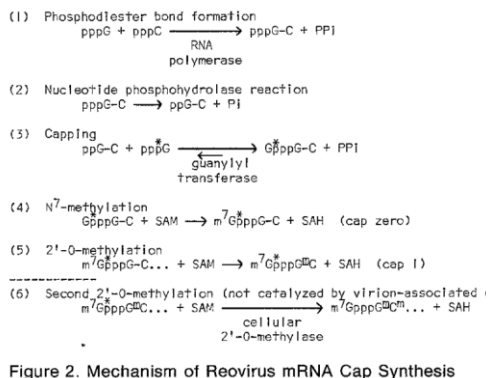
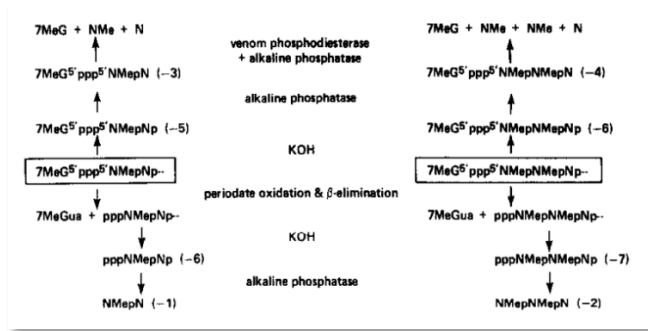


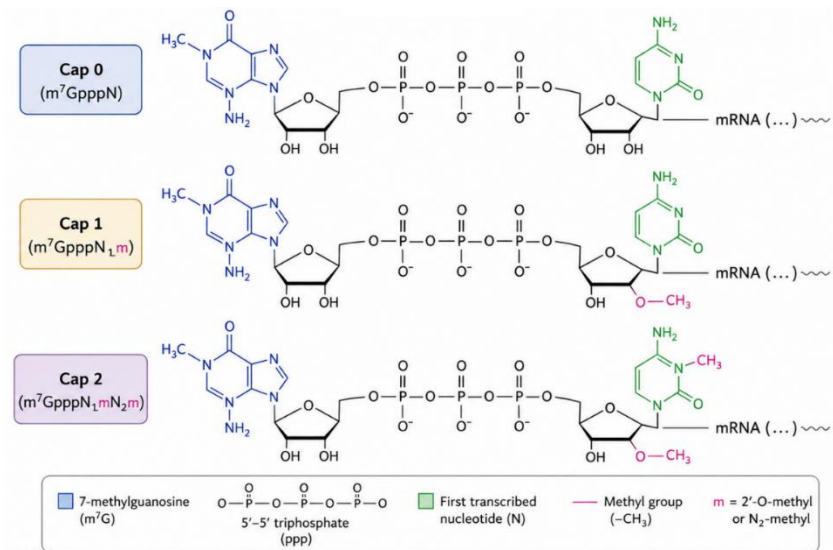
Figure 2. Mechanism of Reovirus mRNA Cap Synthesis

Over the following decades, researchers established that the 5' cap functions not only as a structural modification, but as a regulatory element coordinating transcription, RNA processing, nuclear export, translation, and RNA degradation (Furuichi and Shatkin, 2000; Topisirovic et al., 2011); the cap is a structurally distinct feature of eukaryotic mRNA and suggested that it played specialized roles. During the following decades, researchers discovered that eukaryotes frequently modify the first and second transcribed nucleotides through ribose 2'-O methylation.



IN THIS EXPERIMENT, RESEARCHERS CAREFULLY BROKE APART MESSENGER RNA FROM HUMAN HELa CELLS AND TRACKED THE CHEMICAL PIECES AT THE VERY FRONT END OF THE MOLECULE. BY USING A SERIES OF CHEMICAL TREATMENTS AND SEPARATION METHODS, THEY DISCOVERED THAT THE RNA CARRIED A SPECIAL MODIFIED "CAP" STRUCTURE.

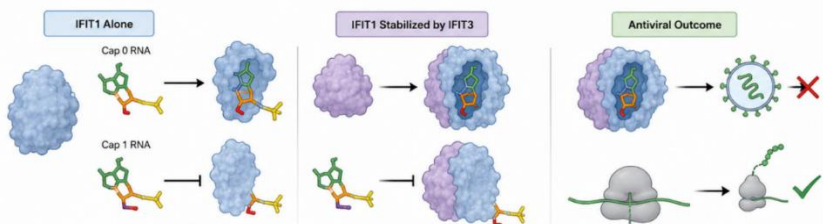
In 1975, Cha-Mer Wei, Alan Gershowitz, and Bernard Moss at the National Institute of Allergy and Infectious Diseases discovered that messenger RNA in human cells carries a previously unknown methylated structure at its 5' end; this was later termed "Cap 1". Using radiolabeled HeLa cell poly(A)+ mRNA together with enzymatic digestion, DEAE-cellulose chromatography, and periodate oxidation experiments, researchers identified a blocked 5' terminal sequence containing 7-methylguanosine linked through multiple phosphates to 2'-O-methylated nucleotides. Their work provided the first evidence that eukaryotic mRNAs possess a specialized methylated cap structure that likely plays an important regulatory role in mRNA function and stability.



Decades later, biochemical and virus studies clarified the immunological significance of 2'-O-methylation within the mRNA cap structure. Experiments comparing Cap 0 and Cap 1 RNAs demonstrated that interferon-induced proteins such as IFIT1 selectively bind and suppress translation of RNAs lacking 2'-O-methylation, whereas properly methylated Cap 1 RNAs evade this restriction and undergo efficient translation (Daffis et al., 2010; Habjan et al., 2013). Viral genetic studies further showed that viruses lacking functional 2'-O-methyltransferase activity, and therefore unable to generate proper Cap 1 structures, were highly sensitive to innate immune responses, confirming that Cap 1 methylation functions as a molecular signature of self RNA (Züst et al., 2011). Together, these studies established that 2'-O-methylation of the mRNA cap constitutes a critical mechanism by which eukaryotic cells distinguish self from non-self RNA.

Building on these findings, researchers at the Institute of Immunobiology, Kanton Hospital St. Gallen, Switzerland investigated how ribose 2'-O-methylation regulates innate immune recognition during coronavirus infection. Using genetically engineered human and mouse coronaviruses lacking functional 2'-O-methyltransferase enzymes, investigators demonstrated that the cytoplasmic RNA sensor Mda5 specifically recognizes viral RNAs lacking proper 2'-O-methylation. By infecting macrophages with methylation-deficient coronaviruses, they observed activation of interferon regulatory factor 3 (IRF3), elevated IFN- β production, and enhanced antiviral signaling pathways. In contrast, macrophages deficient in Mda5 failed to mount these interferon responses, confirming that Mda5 functions as a key sensor of improperly methylated viral RNA (Züst et al., 2011).

These findings were further validated in mouse models of coronavirus infection. When they infected wild-type mice with methylation-deficient coronaviruses, the viruses replicated poorly and exhibited markedly reduced pathogenicity. However, viral replication was restored in mice lacking interferon receptors or the RNA sensors Mda5 and TLR7. These experiments demonstrated that Cap 1 methylation enables coronaviruses to evade innate immune recognition and establish productive infection within the host (Züst et al., 2011).



Several groups determined the structural basis of IFIT-mediated RNA discrimination through structure-based studies. Using high-resolution crystal structures of human IFIT1 bound to capped viral RNA, researchers demonstrated that IFIT1 forms a binding pocket that preferentially accommodates Cap 0 RNAs lacking ribose 2'-O-methylation, whereas Cap 1 methylation sterically interferes with stable IFIT1 engagement. They further showed that IFIT3 stabilizes IFIT1 and enhances its selective recognition of unmethylated viral RNAs, thereby strengthening antiviral restriction of viruses deficient in 2'-O-methyltransferase activity (Johnson et al., 2018). Together, these structural and biochemical studies provided direct mechanistic validation that 2'-O-methylation functions as a molecular signature of self RNA by preventing IFIT-mediated recognition and translational suppression of properly methylated host transcripts.

In 2025 Rahul Singh and collaborators in the laboratory of structural immunologist Yorgo Modis at the MRC Laboratory of Molecular Biology and the University of Cambridge in the United Kingdom investigated the molecular basis by which MDA5 distinguishes self from non-self RNA. Using cryo-electron microscopy, ATPase assays, RNA filament reconstruction, they demonstrated that MDA5 employs ATP-dependent proofreading and cooperative filament formation along double-stranded RNA to discriminate endogenous cellular RNA from viral RNA. The investigators further showed that disease-associated MDA5 variants altered RNA binding, ATPase activity, and interferon signaling, thereby directly linking defective RNA recognition to dysregulated antiviral immunity and susceptibility to autoimmune disease (Singh et al., 2025). Together, these findings reinforced the concept that innate immune sensors rely on highly precise structural recognition mechanisms to identify aberrant viral RNA while avoiding inappropriate immune responses against host RNA.

Ribose 2'-O-methylation is a critical molecular signature of self RNA. By mimicking host mRNA cap structures, viruses and synthetic therapeutic mRNA evade innate immune detection and escape interferon-mediated antiviral restriction.

Translation Initiation

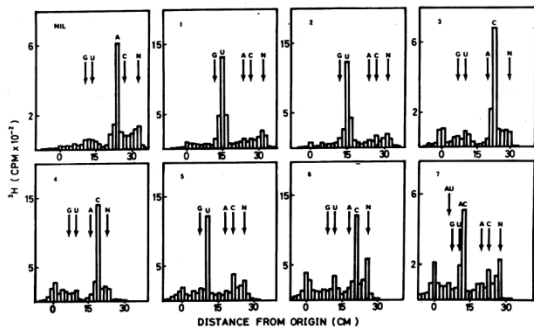
The identification of AUG as the universal initiation codon emerged from early efforts to decipher the genetic code and understand how ribosomes establish the correct reading frame during protein synthesis. In 1966, T. A. Sundararajan and Robert E. Thach, working in the Department of Chemistry at Harvard University under the broader influence of the pioneering molecular biologist Paul Doty, demonstrated that the triplet AUG

strongly promoted the binding of formylmethionine-tRNA to bacterial ribosomes and functioned as a translational “phaser” that fixed the correct reading frame during initiation. Their experiments with synthetic messenger RNAs showed that AUG was far more effective than other candidate initiation codons such as UUG, providing one of the first mechanistic explanations for how ribosomes recognize translational start sites and establishing AUG as the principal bacterial start codon.

During the late 1960s and early 1970s, studies of prokaryotic translation initiation revealed that ribosomes recognize specific nucleotide sequences upstream of the AUG start codon. At the John Curtin School of Medical Research at the Australian National University, John Shine and Lynn Dalgarno identified a purine-rich sequence upstream of bacterial initiation codons that was complementary to the 3' terminus of 16S ribosomal RNA (Shine and Dalgarno, 1974). Through comparative sequence analysis and ribosome-binding experiments, they demonstrated that base pairing between mRNA and rRNA positioned the ribosome at the translational start site and promoted efficient initiation of protein synthesis.

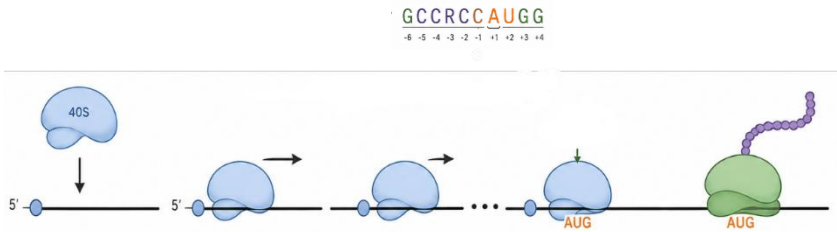
The Shine–Dalgarno model

16S RNA was stepwise degraded, terminally labeled with radioactive [^3H]iNzd, digested with pancreatic ribonuclease, and analyzed by paper electrophoresis using nucleotide standards. After seven degradation cycles, the digestion product was identified as ApC-iNzd, revealing part of the terminal RNA sequence.



The discovery established the Shine–Dalgarno sequence and provided one of the first mechanistic explanations for how ribosomes identify initiation codons in prokaryotes. Because bacterial systems dominated early molecular biology, the Shine–Dalgarno model strongly influenced assumptions about translation initiation in all organisms. However, the absence of comparable motifs in eukaryotic mRNAs, together with the presence of long and structurally complex 5' untranslated regions (5'UTRs), suggested that eukaryotes used fundamentally different mechanisms for start codon recognition. These observations ultimately motivated the work of Marilyn Kozak and others, leading to the ribosomal scanning model and recognition of the 5'UTR as a critical regulatory element in eukaryotic gene expression.

The scanning model of ribosomal initiation



Marilyn Kozak's pioneering studies in molecular biology fundamentally transformed understanding of eukaryotic translation initiation by revealing how ribosomes recognize the correct start codon on messenger RNA. Through systematic mutational analyses in the late 1970s and 1980s, Kozak demonstrated that nucleotide sequences flanking the AUG initiation codon strongly influence translational efficiency, culminating in the definition of the canonical "Kozak sequence" (GCCRCCAUGG), now recognized as a central determinant of translational initiation in eukaryotes (Kozak, 1984; Kozak, 1987). Her work established the scanning model of ribosomal initiation, in which the 40S ribosomal subunit binds near the 5' cap of mRNA and scans downstream to identify an appropriate AUG codon embedded within an optimal sequence context. These discoveries provided a mechanism for understanding gene regulation at the translational level and implications for virology, developmental biology, cancer research, and biotechnology, and ultimately the design of synthetic expression constructs and mRNA therapeutics (Hinnebusch, 2011). Kozak's contributions remain foundational to contemporary RNA biology and continue to shape investigations into translational control and post-transcriptional gene regulation.

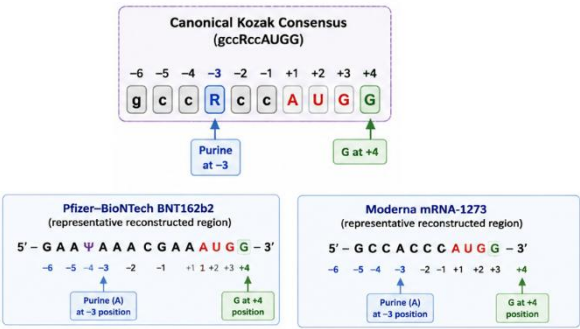
By the late 1980s, strong experimental evidence supported the ribosomal scanning hypothesis. In The scanning model for translation, Kozak and others consolidated findings showing that ribosomes bind near the capped 5' terminus and scan linearly through the 5' UTR until encountering an AUG codon in a favorable sequence context. The model integrated evidence regarding upstream AUGs, RNA secondary structure, and inefficient initiation sites, demonstrating that the 5'UTR functions as a dynamic regulatory region controlling initiation fidelity and ribosome movement.

Functional importance of the 5'UTR transformed our understanding of RNA. Regions once dismissed as non-coding “space” became recognized as central regulatory platforms governing translation initiation, ribosomal fidelity, and protein expression. Kozak’s work established the mechanistic foundations of eukaryotic translation initiation and demonstrated that small variations within non-coding RNA regions can profoundly influence gene expression. These discoveries continue to shape modern biotechnology, synthetic biology, and mRNA therapeutic design, where optimization of the 5' UTR and Kozak sequence remains essential for efficient protein production.

In 1995, Amy E. Pasquinelli, James E. Dahlberg, and Elsebet Lund identified a critical limitation in synthetic mRNA production while studying RNAs generated by bacteriophage RNA polymerases. They demonstrated that conventional cap analogs such as m7GpppG could be incorporated during in vitro transcription in both the correct and reverse orientations, resulting in a substantial proportion of “reverse-capped” transcripts (Pasquinelli, Dahlberg and Lund, 1995). Unlike endogenous eukaryotic mRNAs, whose 5' caps are enzymatically added in a single defined orientation during transcription in the nucleus, synthetic cap dinucleotides can initiate transcription bidirectionally because both guanosine residues possess reactive 3'-OH groups. Reverse-oriented caps bind poorly to the cap-binding protein eIF4E and consequently impair translation efficiency and RNA function. To overcome this problem, Janusz Stepinski and colleagues developed Anti-Reverse Cap Analogs (ARCAs), which contain chemical modifications at the 3' position of the m7G moiety that prevent reverse incorporation while preserving efficient cap recognition (Stepinski et al., 2001). Subsequent studies by Jacek Jemielity and colleagues demonstrated that ARCA-capped mRNAs exhibit substantially enhanced translational efficiency and improved intracellular stability, establishing ARCAs as a foundational advancement in synthetic mRNA technology (Jemielity et al., 2003; Grudzien-Nogalska et al., 2007).

Although ARCAs marked a major advance in synthetic mRNA engineering, newer capping methods such as CleanCap more closely mimic endogenous mammalian mRNA cap structures. ARCAs use chemically modified dinucleotide cap analogs to prevent reverse incorporation during transcription, whereas CleanCap uses a trinucleotide-based co-transcriptional capping approach that ensures correct cap orientation without anti-reverse chemical modifications. ARCAs mainly generate Cap 0 structures, while CleanCap efficiently produces Cap 1 mRNAs with 2'-O-methylation on the first transcribed nucleotide. While ARCAs laid the conceptual and technical foundation for modern synthetic mRNA capping,

CleanCap and related next-generation systems are now widely used in therapeutic mRNA applications.

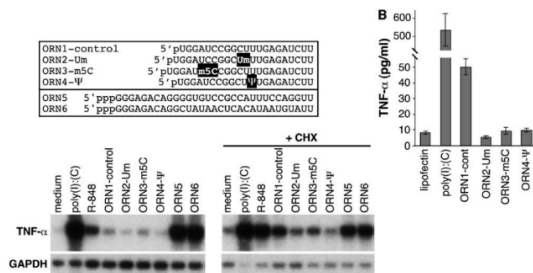


The scanning model established that efficient eukaryotic translation initiation depends on ribosome recruitment at the capped 5' terminus and subsequent scanning of the mRNA by the 40S ribosomal subunit to locate the start codon. This mechanistic framework became essential for modern synthetic mRNA engineering because it revealed that both cap structure and 5' UTR architecture directly determine translational efficiency. Modern mRNA vaccines and protein replacement therapies incorporate optimized UTRs to enabling robust protein expression for therapeutic benefit.

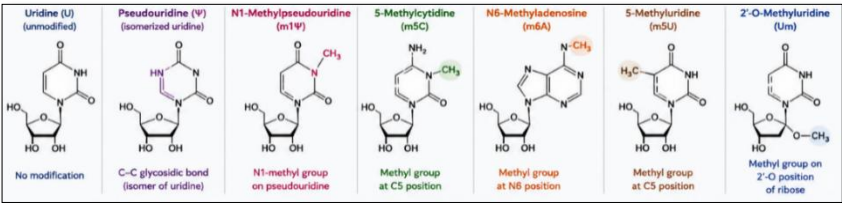
Kozak's discovery that nucleotide context surrounding the AUG codon strongly affects initiation efficiency also directly shaped therapeutic RNA sequence engineering. Nearly all modern therapeutic mRNAs incorporate an optimized Kozak consensus sequence surrounding the start codon to maximize translational output. In mRNA vaccines, for example, inclusion of a strong Kozak context substantially increases antigen production and immune potency. This principle is now routine in mRNA construct design for vaccines, CAR-T engineering, genome editing systems, and protein replacement therapies.

While optimization of the Kozak sequence and 5' UTR architecture greatly improved translation initiation, researchers recognized that efficient therapeutic mRNA expression also depends on the properties of the coding sequence itself. This realization shifted mRNA engineering beyond initiation-site optimization toward broader control of transcript stability, ribosomal elongation kinetics, and innate immune recognition. Modern therapeutic mRNAs therefore incorporate extensive coding-sequence engineering, including codon optimization, modulation of GC content, reduction of inhibitory secondary structures, and incorporation of modified nucleotides such as pseudouridine and N1-methylpseudouridine. These

nucleotide modifications reduce activation of innate immune sensors such as Toll-like receptors and PKR while simultaneously enhancing ribosome processivity, transcript stability, and protein production. Together with optimized cap structures, Kozak contexts, and engineered untranslated regions, these advances transformed synthetic mRNA from an unstable experimental molecule into a clinically viable therapeutic platform.



In 2005, Katalin Karikó, Michael Buckstein, Houping Ni, and Drew Weissman at the University of Pennsylvania identified that synthetic RNA strongly activates innate immune pathways. They showed that unmodified RNA stimulates human Toll-like receptors TLR3, TLR7, and TLR8, triggering dendritic-cell activation and inflammatory cytokine production. Crucially, they demonstrated that naturally occurring nucleoside modifications, including pseudouridine, 5-methylcytidine, N6-methyladenosine, 5-methyluridine, and 2-thiouridine, suppress this immune recognition. This work explained why mammalian RNA, which is rich in modified nucleosides, is less immunostimulatory than bacterial or in vitro-transcribed RNA, and it established pseudouridine modification as a foundational strategy for making synthetic mRNA safer, more stable, and more suitable for therapeutic use (Karikó et al., 2005).



Subsequent work by Brian R. Anderson, Hisaaki Muramatsu, Katalin Karikó, Drew Weissman, and colleagues substantially expanded the mechanistic understanding of how modified nucleosides improve

therapeutic mRNA performance. Building on earlier findings that pseudouridine suppresses Toll-like receptor activation, the group demonstrated that modified nucleosides also inhibit intracellular antiviral defense pathways mediated by interferon-induced 2'–5' oligoadenylate synthetase (OAS) and RNase L (Anderson *et al.*, 2011). Under normal antiviral conditions, unmodified foreign RNA activates OAS enzymes, which synthesize short 2'–5'-linked oligoadenylates that subsequently activate the latent endoribonuclease RNase L. Activated RNase L rapidly degrades intracellular RNA, suppresses protein synthesis, and promotes apoptosis as part of the innate antiviral response. Anderson and colleagues showed that incorporation of modified nucleosides such as pseudouridine markedly reduces activation of OAS and downstream RNase L-mediated RNA cleavage, thereby protecting synthetic mRNAs from intracellular degradation (Anderson *et al.*, 2011). These findings revealed that nucleoside modification does not just reduce sensing through TLR pathways but also suppresses cytoplasmic antiviral surveillance systems that normally limit translation of exogenous RNA.

These advances were crucial to the development of clinically useful mRNA therapeutics. Nucleoside modification greatly improved the stability, translational efficiency, and tolerability of synthetic mRNA platforms. Building on the success of pseudouridine, the field progressed to next-generation modified nucleosides such as N1-methylpseudouridine (m1Ψ), which further increased protein expression while reducing inflammatory signaling and innate immune activation. Both the Pfizer–BioNTech BNT162b2 and Moderna mRNA-1273 SARS-CoV-2 vaccines incorporate m1Ψ in place of uridine throughout the coding region. This modification suppresses activation of Toll-like receptors, PKR, and the OAS/RNase L antiviral pathway while also improving ribosome processivity and intracellular mRNA stability, enabling the high levels of spike protein expression required for strong vaccine-induced immunity. Together, these innovations transformed synthetic mRNA from a highly immunostimulatory experimental molecule into the foundation of modern mRNA vaccines and therapeutic RNA technologies.

A-to-I RNA editing oligonucleotide (ADAR)



CRYO-EM STRUCTURE OF HUMAN ADAR1 IN COMPLEX WITH DSRNA DERIVED FROM HT2C GENE
HOMO SAPIENS
2025-03-26
AUTHORS: DENG, X., GAO, Y.

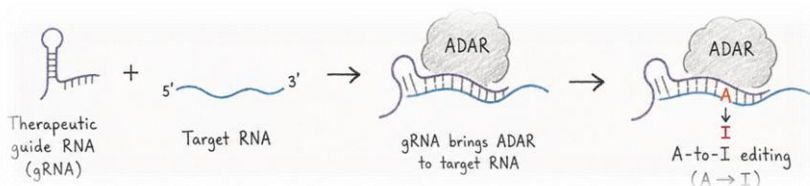
Early work on RNA processing had already shown that cells could alter RNA after transcription, raising the possibility that genetic information did not always pass unchanged from DNA to mature RNA. Researchers studying RNA structure then focused on double-stranded RNA since base-paired RNA molecules form stable duplexes that can influence translation, viral defense and RNA maturation. Against this background, Bass and Weintraub examined *Xenopus laevis* (frog) oocytes and embryos and identified an enzymatic activity that acted specifically on RNA duplexes. They initially described the activity as one that “unwound” double-stranded RNA because treated duplexes lost the behavior expected of fully base-paired RNA (Bass and Weintraub, 1987; Bass and Weintraub, 1988).

Subsequent biochemical work, undertaken to explain why dsRNA duplexes lost their normal base-paired behavior after treatment with the “unwinding” activity, showed that this apparent unwinding reflected chemical modification rather than conventional helicase activity. At The Wistar Institute and the University of Pennsylvania in Philadelphia, Wagner, Smith, Cooperman and Nishikura demonstrated that the activity introduced structural changes into double-stranded RNA through adenosine-to-inosine conversion in mammalian cells and *Xenopus* eggs (Wagner et al., 1989). Polson and colleagues at the University of Utah in Salt Lake City then confirmed the mechanism directly by mass spectrometry, showing that the double-stranded RNA unwinding/modifying activity converted adenosine to inosine (Polson et al., 1991). These findings linked the original dsRNA

observation to RNA editing and led to the definition of ADARs as adenosine deaminases acting on RNA.

Work on glutamate receptor pre-mRNAs then showed how endogenous transcripts create the dsRNA structures required for ADAR activity. Higuchi and colleagues showed that editing at the GluR-B/GluA2 Q/R site required base pairing between an exonic sequence and a downstream intronic editing-site complementary sequence, demonstrating that cellular pre-mRNAs can fold into the local duplexes needed to direct site-specific editing (Higuchi et al., 1993). ADAR1 and ADAR2 were subsequently cloned and characterized as RNA-editing enzymes (Kim et al., 1994; Melcher et al., 1996). ADAR2-mediated editing of the GluA2 Q/R site changes a genomically encoded glutamine codon into an arginine-read codon and limits calcium permeability through AMPA receptors, making this site a central example of physiologically important finding (Higuchi et al., 1993; Wright et al., 2023).

Genome-wide analyses then showed that ADAR activity extends far beyond a small set of coding sites. Levanon and colleagues mapped thousands of A-to-I editing sites in the human transcriptome and showed that most occur in non-coding regions, especially within Alu repeat-derived double-stranded RNA structures (Levanon et al., 2004). Thus, the field shifted from viewing ADARs primarily as enzymes that diversify proteins to recognizing them as broader regulators of non-coding RNA structure, repetitive-element-derived dsRNA and endogenous RNA surveillance.



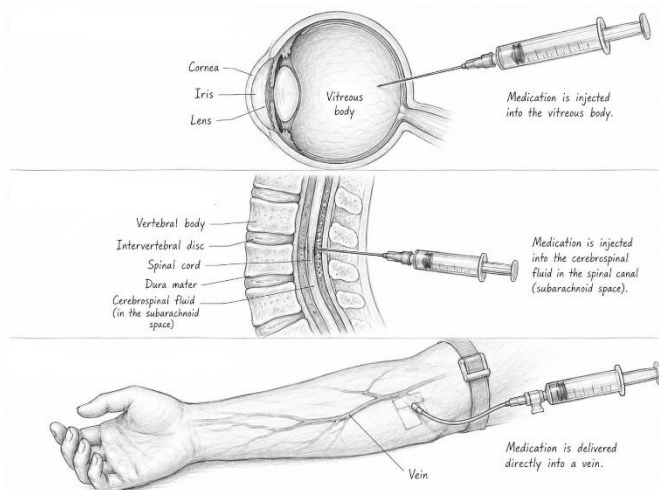
ADAR Rx

Industry efforts aim to redirect ADAR activity with engineered guide RNAs or oligonucleotides that create a programmable RNA duplex around a disease-relevant adenosine, allowing endogenous or engineered ADAR enzymes to install a targeted A-to-I change without permanently altering DNA. Wave Life Sciences has moved this concept into the clinic with WVE-006 for alpha-1 antitrypsin deficiency, reporting the first therapeutic RNA editing in humans in 2024 and subsequent phase 1b/2a data showing therapeutically relevant AAT levels across multiple dose cohorts. ADARx Pharmaceuticals is also pursuing an RNA editing medicines pipeline across

hepatic and extrahepatic diseases, including complement-mediated, cardiovascular, cardiometabolic and central nervous system indications.

While this technology has promise for context-relevant genetic therapeutics, it still faces many of the same practical barriers that have limited other genetic medicine platforms. Chief among these is delivery: getting the therapeutic payload to the right tissue, into the right cell type, at sufficient levels, and with enough durability to produce a meaningful clinical effect. This remains especially challenging for tissues that are difficult to access, such as the central nervous system, skeletal muscle, heart, and certain solid tumors. Even when systemic delivery is possible, biodistribution can be uneven, with uptake concentrated in organs such as the liver or spleen rather than the intended disease-relevant tissue.

Pharmacokinetics and pharmacodynamics of RNA Therapeutics



Pharmacokinetics and pharmacodynamics are central to the development of nucleic acid therapeutics because these modalities do not behave like conventional small molecules or typical biologics. For antisense oligonucleotides, siRNAs, lipid nanoparticle-formulated RNAs and related genetic medicines, the relationship between administered dose, systemic exposure, tissue distribution, intracellular trafficking, target engagement and biological response is often indirect, delayed and highly tissue-dependent. In many cases, plasma concentration is only a partial measure of pharmacologically relevant exposure; the more important question is

whether sufficient active drug reaches the correct tissue, the correct cell type and the correct intracellular compartment.

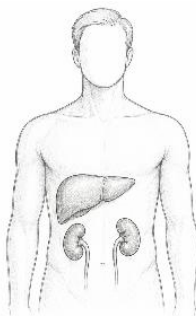
Antisense oligonucleotides are generally single-stranded, chemically modified nucleic acids designed to bind complementary RNA through sequence-specific hybridization. Depending on their design, ASOs may trigger RNase H-mediated degradation of target RNA, alter splicing, block translation or modulation of RNA processing. Their pharmacokinetics are strongly influenced by chemistry modifications, plasma protein binding, nuclease stability and tissue uptake. Phosphorothioate backbone modifications, for example, increase plasma protein binding and reduce rapid renal clearance, supporting broader tissue distribution and longer tissue residence (Geary et al., 2015). ASOs often distribute rapidly from plasma into tissues, followed by slow tissue elimination and prolonged pharmacodynamic activity, meaning that plasma PK alone can substantially underestimate the duration and relevance of tissue exposure (Geary et al., 2015; Burel et al., 2024). In practice, ASO PK/PD therefore requires attention to tissue concentration, intracellular availability, target RNA knockdown and the turnover kinetics of the downstream protein or pathway.

siRNAs are double-stranded RNA molecules that act through the RNA-induced silencing complex, or RISC, to degrade complementary mRNA. Their pharmacology depends on tissue distribution, cell uptake, endosomal trafficking, cytosolic release, guide-strand loading into RISC, target mRNA cleavage and downstream protein reduction. For GalNAc-conjugated siRNAs, hepatocyte delivery is mediated by binding to the asialoglycoprotein receptor, creating a highly targeted liver-delivery platform with distinctive PK/PD behavior (Ayyar and Song, 2024). Mechanistic models of GalNAc-siRNA disposition incorporate receptor-mediated uptake, intracellular sequestration, slow degradation and RISC-dependent mRNA silencing, reflecting the fact that the pharmacologically active species is not simply circulating siRNA but intracellular, functionally loaded guide strand (Ayyar and Song, 2024). This helps explain why siRNA pharmacodynamic effects can last weeks to months even after plasma concentrations have fallen.

A key pharmacology feature of these modalities is durability. Studies of GalNAc-siRNA conjugates suggest that chemically stabilized siRNA can persist in acidic intracellular compartments and be slowly released over time, enabling continued RISC loading and prolonged gene silencing (Brown et al., 2020). This creates an important PK/PD distinction between systemic exposure, tissue exposure and productive intracellular exposure. Two siRNAs may show similar early knockdown but differ markedly in duration depending on chemical stability, trafficking route and intracellular depot behavior. Delivery route matters: GalNAc conjugates and

LNP-formulated siRNAs can produce different onset and duration profiles, consistent with differences in intracellular trafficking and cytosolic release kinetics. A related durability principle is seen with Spinraza® (nusinersen), an intrathecally administered antisense oligonucleotide for spinal muscular atrophy. Although nusinersen acts through splice modulation rather than RNAi, its prolonged clinical dosing interval illustrates how oligonucleotide chemistry, tissue retention and intracellular persistence can uncouple short systemic exposure from sustained pharmacodynamic effect. Following loading doses, nusinersen is maintained with dosing every four months, supported by slow elimination from cerebrospinal fluid and distribution within central nervous system tissues (FDA, 2026). Its reported CSF terminal half-life of approximately 135–177 days further highlights that the durability of oligonucleotide therapeutics depends less on plasma PK alone and more on residence time in the relevant tissue compartment and continued productive engagement of intracellular RNA-processing machinery (FDA, 2026; Chiriboga et al., 2016).

Lipid nanoparticles add to another level of complexity because both the RNA payload and the delivery vehicle contribute to pharmacology, biodistribution and safety. LNPs are used to deliver mRNA, siRNA, gene-editing systems and other nucleic acid payloads by protecting RNA from degradation, promoting cellular uptake and enabling intracellular release. Their behavior is shaped by particle size, ionizable lipid chemistry, PEG-lipid content, cholesterol and helper lipid composition, surface charge, protein corona formation and route of administration. For LNP-formulated RNA therapeutics, PK/PD therefore includes not only where the nanoparticle goes, but also whether the payload escapes the endosome and becomes biologically active. For mRNA LNPs, the primary pharmacodynamic outcome is protein expression rather than target knockdown. The PK/PD sequence therefore links dose to LNP biodistribution, cellular uptake, endosomal escape, cytosolic mRNA availability, translation, protein expression, protein turnover, and the resulting biological effect.



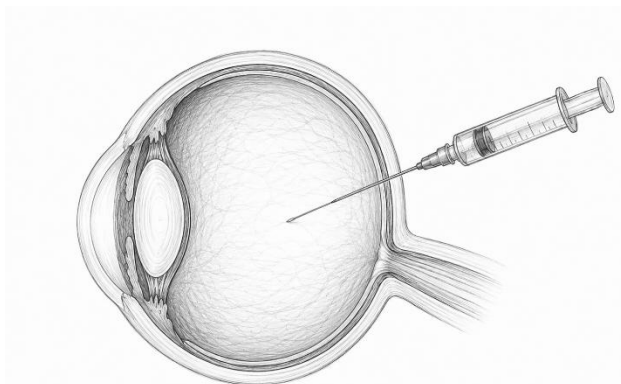
An example of how formulation can reshape oligonucleotide PK/PD is provided by Yu et al. (1999) at Isis pharmaceuticals (no called Ionis Pharmaceuticals), who compared a phosphorothioate antisense oligonucleotide against Ha-Ras delivered either as a non-liposomal saline formulation or encapsulated in long-circulating PEGylated “stealth” liposomes. The study showed that liposomal encapsulation prolonged systemic exposure: plasma half-life increased from approximately 1.1 hours for the saline formulation to approximately 57.8 hours for the PEGylated liposome formulation, while clearance fell from about 54.0 to 0.892 ml/hr/kg. The liposomal formulation also protected the oligonucleotide from nuclease-mediated metabolism, with mostly intact oligonucleotide compound detected in plasma and blood over time, whereas the saline formulation showed rapid disappearance and more chain-shortened metabolites. Tissue distribution was also altered: non-liposomal oligonucleotide showed high renal exposure, consistent with uptake and clearance by kidney pathways, while PEGylated liposomes shifted exposure toward reticuloendothelial-rich tissues such as liver, spleen and lymph nodes, with slower tissue uptake and longer residence. This illustrates a broader PK/PD principle for nucleic acid therapeutics: formulation can change not only circulation time, but also the dominant clearance route, metabolic stability, tissue exposure pattern and potential therapeutic index. In the context of oncology, the authors highlighted that long-circulating PEGylated liposomes may exploit leaky tumor vasculature and sites of inflammation, supporting improved delivery to tissues with increased vascular permeability (Yu et al., 1999).

PK/PD	Non-liposomal oligonucleotide	PEGylated stealth liposome
Plasma half-life	1.1 hours	~57.8 hours
Clearance	54.0 ml/hr/kg	0.892 ml/hr/kg
Metabolism	Rapid degradation; more shortened metabolites	Greater protection; more intact oligonucleotide
Distribution pattern	High kidney exposure	More liver, spleen and lymph-node exposure
Translational implication	Fast clearance, renal uptake, short exposure	Longer circulation, slower clearance, possible enhanced delivery to permeable tissues

Clinical compounds with route of administration

Compound	Base	Backbone	Sugar	Conjugation	Route
Fomivirsen	21-mer antisense DNA	Full phosphorothioate	DNA	None	Intravitreal injection
Mipomersen	Gapmer ASO	Phosphorothioate	2'-MOE + DNA gap	None	Subcutaneous injection
Eteplirsen	30-mer antisense	PMO	Morpholino	None	Intravenous infusion
Nusinersen	18-mer	Phosphorothioate	2'-MOE	None	Intrathecal injection
Inotersen	Gapmer ASO	Phosphorothioate	2'-MOE	None	Subcutaneous injection
Golodirsen	Antisense	PMO	Morpholino	None	Intravenous infusion
Volanesorsen	Gapmer ASO	Phosphorothioate	2'-MOE	None	Subcutaneous injection
Viltolarsen	Antisense	PMO	Morpholino	None	Intravenous infusion
Casimersen	Antisense	PMO	Morpholino	None	Intravenous infusion
Tofersen	20-mer gapmer	Mixed PS/PO	2'-MOE	None	Intrathecal injection
Eplontersen	Gapmer ASO	Mixed PS/PO	2'-MOE	GalNAc	Subcutaneous injection
Pegaptanib	RNA aptamer	Modified RNA	Modified ribose	PEG	Intravitreal injection
Avacincaptad pegol	RNA aptamer	Modified RNA	Modified ribose	PEG	Intravitreal injection
Patisiran	siRNA	Phosphodiester	2'-OMe	LNP	Intravenous infusion
Givosiran	siRNA	PS modified	2'-F/2'-OMe	GalNAc	Subcutaneous

Lumasiran	siRNA	PS modified	2'-F/2'-OMe	GalNAc	injection Subcutaneous injection
Inclisiran	siRNA	PS modified	2'-F/2'-OMe	GalNAc	Subcutaneous injection
Vutrisiran	siRNA	PS enhanced	2'-F/2'-OMe	GalNAc	Subcutaneous injection
Nedosiran	siRNA	PS/PO	2'-F/2'-OMe	GalNAc	Subcutaneous injection
Miravirsen	Anti-miR	Phosphorothioate	LNA/DNA	None	Subcutaneous injection
RG-101	Anti-miR	Modified ASO	Not specified	GalNAc	Subcutaneous injection



Pharmacokinetic studies showed that Fomivirsen distributes primarily within ocular tissues following intravitreal injection and had sustained concentrations with a vitreous half-life of approximately 55 hours in humans and an estimated retinal half-life of up to 78 hours in animal models (Geary, Henry and Grillone, 2002). Metabolism occurs via endo- and exonuclease-mediated degradation into shorter oligonucleotide fragments, with minimal systemic exposure due to localized administration.

For phosphorothioate (PS) ASOs, which are typically ≥ 85 –99% protein bound, renal clearance of intact drug is minimal, with studies showing that < 1 –4% of the administered dose is recovered in urine within 24 hours, reflecting predominant elimination via slow nuclease-mediated metabolism in tissues (Geary et al., 2024; mipomersen data). In contrast, low protein-binding chemistries such as phosphorodiamidate morpholino oligomers (PMOs) undergo substantial renal elimination, with a large fraction of the administered dose excreted unchanged in urine due to limited plasma protein binding and rapid glomerular filtration (Shadida et al., 2021). siRNA shows an intermediate and modality-dependent profile: unmodified siRNA is rapidly cleared by renal filtration, whereas clinically used GalNAc-conjugated siRNAs exhibit reduced renal clearance, with approximately 12–25% of the dose recovered unchanged in urine and the remainder undergoing tissue uptake and intracellular metabolism (Habtemariam et al., 2021; vutrisiran data).

Nucleases are fundamental to life. These enzymes catalyze the cleavage of DNA and RNA, enabling essential processes such as DNA replication (e.g., removal of RNA primers and proofreading), DNA repair

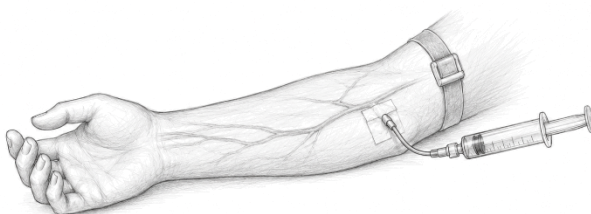
and recombination, RNA maturation, and cellular defense against foreign genetic material. For example, 3'–5' exonucleases such as DnaQ perform proofreading during DNA replication, while RNase H removes RNA primers from DNA–RNA hybrids to ensure proper genome duplication (Yang, 2011). In DNA repair, nucleases like Mre11 initiate double-strand break processing, and structure-specific nucleases such as FEN1 remove flap intermediates during lagging-strand synthesis (Yang, 2011). In RNA biology, nucleases including RNase E and RNase Z are essential for RNA processing and maturation, whereas Argonaute proteins mediate RNA interference by cleaving target mRNAs (Yang, 2011). Additionally, restriction endonucleases and CRISPR-associated nucleases provide defense against invading genetic elements, highlighting the diverse yet indispensable roles of nucleases across biological systems (Yang, 2011).



CRYSTAL STRUCTURE OF HUMAN FLAP ENDONUCLEASE FEN1 (D181A) IN COMPLEX WITH SUBSTRATE 5'-FLAP DNA AND K⁺; HUMAN FLAP ENDONUCLEASE STRUCTURES, DNA DOUBLE-BASE FLIPPING, AND A UNIFIED UNDERSTANDING OF THE FEN1 SUPERFAMILY. TSUTAKAWA, S.E., CLASSEN, S., CHAPADOS, B.R., ARVAI, A.S., FINGER, L.D., GUENTHER, G., TOMLINSON, C.G., THOMPSON, P., SARKER, A.H., SHEN, B., COOPER, P.K., GRASBY, J.A., TAINER, J.A. (2011) CELL 145: 198-211

Antisense oligonucleotides (ASOs) inevitably encounter nucleases immediately upon administration and cellular uptake. In the extracellular space and circulation, they are exposed to DNase I, a nonspecific endonuclease that cleaves both single- and double-stranded DNA into short oligonucleotides as part of normal extracellular DNA clearance and turnover. Following cellular uptake via endocytosis, ASOs enter endosomal and lysosomal compartments, where they encounter degradative enzymes including DNase II, and RNases involved in nucleic acid recycling. Within the intracellular environment, additional nucleases such as 3' exonucleases and endonucleases further metabolize oligonucleotides into shortened fragments, contributing to their overall clearance and activity profile (Jo et al., 2023). For single stranded ASOs that enter the cell nucleus there are numerous nucleases involved in managing various aspects of

genomic stability and replication (e.g. FEN1 see figure above). This widespread exposure to therapeutic-damaging enzymes highlights the need for protective strategies when attempting to deliver ASOs to their intracellular targets.



Systemic dosing

Systemic dosing of RNA therapeutics is governed by delivery, chemical modification, tissue targeting and productive intracellular uptake rather than by plasma exposure alone. Naked siRNA, PMO and related oligonucleotide therapeutics are large, highly polar and often negatively charged, which limits passive membrane permeability and makes them vulnerable to nuclease degradation or rapid clearance. As a result, systemically administered RNA therapeutics generally require chemical stabilization, conjugation or delivery systems such as lipid nanoparticles, ligand-directed constructs, cell-penetrating peptides or antibody–oligonucleotide conjugates to protect the RNA, promote tissue uptake and enable intracellular activity (Choi et al., 2025).

Route of administration can reshape the tissue PK/PD profile from the start. In one study (Bäckström et al.), subcutaneous dosing of non-conjugated Malat1 LNA ASO produced the highest concentrations in kidney and liver, whereas intratracheal dosing shifted the highest exposure to lung, followed by kidney and liver. Intratracheal dosing achieved similar lung knockdown at a much lower dose than subcutaneous dosing, illustrating how local or organ-directed administration can reduce the dose requirement for the target tissue while changing systemic exposure patterns. Systemic exposure after non-IV routes depend heavily on route, dose and tissue transit into circulation (Bäckström et al., 2024).

Chemistry is central to the PK/PD behavior of oligonucleotide therapeutics. Phosphodiester linkages are highly susceptible to nuclease degradation, so many antisense oligonucleotides use phosphorothioate backbones to improve nuclease resistance and reduce renal clearance through increased plasma protein binding. However, phosphorothioate modification can reduce target RNA affinity, so ribose modifications such as

2'-O-methoxyethyl, locked nucleic acid and constrained ethyl are used to improve binding affinity and potency. Gapmer ASOs typically combine a phosphorothioate backbone with chemically modified “wings” flanking a central DNA gap that supports RNase H1-mediated target RNA cleavage (Bäckström et al., 2024). PMO chemistries, by contrast, are neutrally charged morpholino oligomers used frequently for splice modulation, including exon skipping in Duchenne muscular dystrophy, but their clinical effect has historically been limited by inefficient delivery to skeletal and cardiac muscle (Yun et al., 2025; Boucher et al., 2025).

Plasma protein binding is another important modifier of systemic oligonucleotide disposition because it can influence circulation time, renal clearance, tissue distribution and the unbound drug fraction available for distribution. Protein binding is strongly chemistry dependent. In a study comparing sequence-matched antisense oligonucleotides, MOE/phosphorothioate-modified ASOs showed much higher plasma protein binding than phosphorodiamidate morpholino oligomers. At 1 μM in human plasma, the unbound fraction was only 4.4% for 20mer-MOE and 1.0% for 25mer-MOE, compared with 56% for 20mer-PMO and 36% for 25mer-PMO (Yun et al., 2025). These data show that backbone and sugar chemistry can materially alter systemic exposure and distribution, even when sequence and length are held constant.

This has direct implications for systemic dosing. Highly protein-bound oligonucleotides may have reduced filtration and slower renal elimination, supporting longer systemic persistence and greater opportunity for tissue uptake. By contrast, weakly protein-bound chemistries such as PMOs may remain more freely circulating and be more readily cleared unless conjugation improves tissue delivery. Yun et al. also found that MOE/PS ASOs showed evidence of saturable plasma protein binding above approximately 1 μM , whereas PMO ASOs did not show saturation up to 10 μM . At clinically relevant micromolar exposures, saturation of binding could increase the unbound fraction, alter tissue uptake and change the apparent PK/PD relationship (Yun et al., 2025).

The identity of plasma binding proteins may also matter. Although albumin is often assumed to dominate oligonucleotide binding, Yun et al. found that human γ -globulins made the largest contribution to binding for both tested PMO and MOE/PS ASOs. For 25mer-PMO, γ -globulins accounted for 58% of binding contribution, compared with 22% for albumin and 20% for α 1-acid glycoprotein. For 25mer-MOE, γ -globulins contributed 50%, with albumin contributing 13%, LDL 18%, HDL 17% and α 1-acid glycoprotein 2% (Yun et al., 2025). This suggests that disease states associated with altered immunoglobulin levels, such as chronic

inflammatory disease, liver disease, cancer or hypergammaglobulinaemia, could plausibly affect systemic oligonucleotide PK and PD.

Plasma concentrations may fall relatively quickly after systemic dosing, but the pharmacological effect can persist because intracellular RNA remains active after uptake into the target tissue. In non-clinical studies, intravenously administered lumasiran in monkeys had a plasma half-life of approximately 0.6 hours, whereas the liver tissue half-life was about 296 hours. Similarly, after IV administration of givosiran in rats, the plasma half-life was approximately 0.2 hours, while the liver half-life was 55 hours and the kidney half-life was 119 hours. Bäckström et al. similarly reported prolonged ASO tissue half-lives in mouse tissues, with non-conjugated Malat1 LNA ASO after subcutaneous administration showing tissue half-lives of approximately 152 hours in liver, 161 hours in kidney, 188 hours in heart and 179 hours in lung. These findings show that plasma exposure captures only the early distribution phase, while tissue retention and intracellular activity better reflect the duration of pharmacological effect (Bäckström et al., 2024; Choi et al., 2025).

These chemistry, conjugation and disposition differences have practical consequences for systemic dosing. Bäckström et al. compared Malat1-targeting ASOs with LNA, cEt and 2'-MOE ribose chemistries after subcutaneous and intratracheal administration in mice. After systemic subcutaneous dosing, tissue distribution was broadly similar across the three chemistries, with the highest concentrations generally observed in kidney, followed by liver, lung and heart. However, the degree of knockdown differed: LNA and cEt produced broadly similar Malat1 knockdown, whereas 2'-MOE generally produced less knockdown in liver, kidney, lung and heart despite measurable tissue exposure. This highlights that tissue concentration alone does not fully predict pharmacodynamic activity, because productive intracellular uptake and intrinsic potency also matter (Bäckström et al., 2024).

The disconnect between tissue concentration and pharmacodynamic effect is a recurring feature of RNA therapeutics. In the mouse study, kidney exposure after subcutaneous dosing was higher than liver exposure, yet liver knockdown was generally greater. This suggests that total tissue concentration includes both productive and non-productive uptake, and that the relevant pharmacological driver is not simply how much oligonucleotide enters a tissue, but how much reaches the correct cell type and intracellular compartment where it can engage the target RNA (Bäckström et al., 2024). The same principle applies to siRNA therapeutics, where the active antisense strand must escape the endosome, enter the cytosol, and load into RISC before durable target mRNA degradation occurs (Jing et al., 2023; Choi et al., 2025).

Tissue targeting can be a dominant determinant of response for systemically dosed RNA therapeutics. In liver-directed siRNA and ASO programs, GalNAc conjugation works not because it is the most powerful endosomal escape technology, but because it efficiently delivers oligonucleotides to hepatocytes through the asialoglycoprotein receptor. Similarly, in DMD, TfR1-targeted antibody–oligonucleotide conjugation may be more impactful than a generalized endosomal escape strategy because it addresses a central delivery barrier: broad, efficient and repeatable delivery to skeletal and cardiac muscle (Bäckström et al., 2024; Boucher et al., 2025).

GalNAc-conjugated siRNAs provide a useful benchmark for this delivery-first principle. GalNAc binds with high affinity and specificity to the asialoglycoprotein receptor, which is highly abundant on hepatocytes and undergoes rapid recycling. After subcutaneous administration, GalNAc–siRNA conjugates are absorbed into systemic circulation and rapidly taken up by hepatocytes through ASGPR-mediated endocytosis. Human plasma exposure is brief, with typical T_{max} values of approximately 3–4 hours and plasma half-lives generally under 10 hours, but pharmacodynamic effects can persist for weeks to months because the relevant driver is not plasma concentration or even total liver concentration. Instead, durable activity depends on the small fraction of intracellular siRNA that escapes endosomal trapping, enters the cytosol and loads into RISC.

Preclinical ADME data reinforce this distinction between exposure and productive pharmacology. GalNAc–siRNAs show extensive liver distribution, with liver-to-plasma ratios of approximately 5500 in rats and 6300 in monkeys, and recovery of up to 63% of dose in rat liver and 87% in monkey liver after pharmacologically relevant subcutaneous doses. Yet even in the liver, total tissue concentration does not fully explain the pharmacodynamic response. RISC-loaded siRNA better tracks the timing and durability of target mRNA knockdown, helping explain why GalNAc–siRNAs can show transient plasma PK but prolonged pharmacology. This disconnect between short systemic exposure and durable intracellular activity supports infrequent dosing schedules, including monthly dosing for givosiran and lumasiran, every three-month dosing for vutrisiran, and twice-yearly maintenance dosing for inclisiran.

The same distinction is evident with lipid nanoparticle-mediated siRNA delivery. The clearest clinical example of systemic intravenous siRNA delivery remains patisiran, an RNA interference therapeutic formulated in lipid nanoparticles and administered by IV infusion every three weeks. Unlike GalNAc-conjugated siRNAs, which are generally given subcutaneously and rely on receptor-mediated uptake, patisiran uses an LNP

formulation to support systemic administration and liver delivery. Following IV administration, LNPs accumulate predominantly in the liver, partly through interactions with serum apolipoprotein E and uptake by hepatocytes via low-density lipoprotein receptors (Choi et al., 2025).

Patisiran demonstrates how systemic dosing can produce durable pharmacodynamic effects despite complex disposition. After 0.3 mg/kg IV administration every three weeks in patients, patisiran had a reported plasma half-life of approximately 76.8 hours, with a C_{max} of 7150 ng/mL and AUC of 184,000 ng·h/mL. At steady state, the plasma half-lives of the LNP components DLin-MC3-DMA and PEG2000-C-DMG were approximately 10.9 and 3.76 days, respectively. Clinically, this exposure supported around an 80% reduction in serum transthyretin, demonstrating that systemic LNP delivery can translate into substantial and sustained target protein knockdown (Choi et al., 2025). Here again, plasma PK is not the main determinant of pharmacologic durability: the relevant sequence is liver deposition, hepatocyte uptake, endosomal escape, RISC loading and sustained target engagement.

Together, GalNAc conjugates and LNPs show that successful RNA pharmacology depends on more than systemic exposure. GalNAc has been successful because it solves the upstream targeting problem for hepatocytes: reliable, receptor-mediated delivery to the right liver cell type at sufficient scale. LNPs solve a related problem through formulation-driven liver tropism after IV administration. In both cases, productive pharmacology depends on the fraction of delivered oligonucleotide that reaches the appropriate intracellular compartment, not simply on the amount present in plasma or even in bulk tissue.

The recent clinical contrast between Entrada Therapeutics and Avidity Biosciences in Duchenne muscular dystrophy is consistent with this broader delivery principle. Entrada's ENTR-601-44 is an Endosomal Escape Vehicle-conjugated PMO designed for exon 44 skipping. In the first cohort of the Phase 1/2 ELEVATE-44-201 study, participants received three intravenous 6 mg/kg doses, and the company reported favorable safety and tolerability, but lower-than-expected plasma C_{max} and AUC in pediatric DMD participants compared with healthy adult volunteers. The molecular readout was also modest relative to expectations: ENTR-601-44 increased dystrophin by 2.36% over baseline, below the double-digit dystrophin production that investors and analysts had anticipated (Entrada Therapeutics, 2026; Taylor, 2026).

By contrast, Avidity's delpacibart zotadirsen, or del-zota, is an antibody–oligonucleotide conjugate that links an exon 44-skipping PMO to an antibody targeting transferrin receptor 1, a receptor expressed on skeletal and cardiac muscle. In the Phase 1/2 EXPLORE44 study, Avidity reported

skeletal muscle PMO concentrations of 200 nM and approximately 25% normal dystrophin production after three doses. The company also reported greater than 80% reductions in creatine kinase compared with baseline, consistent with a biologically meaningful reduction in muscle damage (Avidity Biosciences, 2025).

Together, these data suggest that endosomal escape is necessary but not sufficient. Once an oligonucleotide is internalized, escape into the cytosol or nucleus remains critical. However, the Entrada and Avidity data imply that, in DMD, an upstream delivery bottleneck may be more important: getting enough drug into the right skeletal and cardiac muscle cells in the first place. A platform optimized for cellular uptake and endosomal escape may still underperform if systemic exposure is lower than expected, if delivery to muscle is inefficient, or if a large fraction of total uptake is non-productive. By contrast, Avidity's TfR1-directed antibody conjugation may more directly address the upstream delivery problem by enriching PMO in skeletal and cardiac muscle, increasing the probability that enough oligonucleotide reaches the relevant intracellular compartment to drive exon skipping and dystrophin restoration (Avidity Biosciences, 2025; Boucher et al., 2025).

This delivery distinction becomes even more important as RNA therapeutics move toward larger and more structurally complex cargos. Short siRNAs, splice-switching ASOs, PMOs, mRNAs and gene-editing systems all face the same broad delivery problem, but they impose different intracellular requirements. A small siRNA must reach the cytosol and load into RISC; a splice-modulating oligonucleotide often needs access to nuclear pre-mRNA; an mRNA must escape the endosome intact and remain translationally competent; and gene-editing cargos may require coordinated delivery of larger nucleic acid, protein or ribonucleoprotein components. As cargo size and complexity increase, total tissue exposure becomes an even less reliable proxy for productive pharmacology. The key question shifts from whether a platform can deliver oligonucleotide mass to a tissue, to whether it can deliver the right molecular cargo, in the right form, to the right intracellular compartment, at sufficient levels to drive the intended biology.

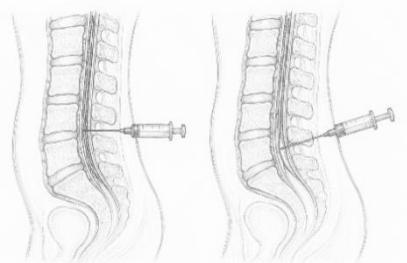
For PK/PD modelling, systemic RNA therapeutics therefore require approaches that go beyond conventional plasma exposure-response relationships. Models need to account for plasma distribution, protein binding, unbound fraction, target-organ uptake, conjugation chemistry, receptor-mediated targeting, endosomal escape, intracellular RISC or RNase H1 engagement, target RNA degradation and downstream biomarker change. This is why liver or RISC compartments have been incorporated into PK/PD models for several siRNA therapeutics, including givosiran and

lumasiran, while patisiran has been modelled using plasma and liver compartments linked to an indirect pharmacodynamic model of serum transthyretin reduction (Choi et al., 2025). For ASOs and PMOs, clinical experience in DMD now suggests that models should explicitly capture muscle delivery and productive uptake, not just plasma exposure or theoretical endosomal escape efficiency (Entrada Therapeutics, 2026; Avidity Biosciences, 2025).

Systemic dosing also has important safety implications. IV lipid-based formulations can trigger infusion-related reactions or innate immune activation, requiring attention to formulation composition, premedication and tolerability. Patisiran, for example, requires premedication to reduce infusion-related reactions. More broadly, systemic delivery can expose non-target tissues such as kidney, spleen and immune cells, so biodistribution, off-target hybridisation, immune stimulation, organ-specific accumulation, plasma protein binding and chemistry-specific tissue retention should be evaluated alongside efficacy biomarkers (Jing et al., 2023; Choi et al., 2025; Yun et al., 2025).

Overall, the PK/PD of systemically dosed RNA therapeutics is best understood as a multi-step delivery and disposition process that begins with administration and systemic exposure, passes through plasma binding, tissue distribution, tissue targeting, cellular uptake and intracellular trafficking, and ends with target engagement and durable pharmacodynamic response. Plasma PK describes systemic exposure, but pharmacodynamic durability depends on how much RNA remains protected in circulation, how it interacts with plasma proteins, how much reaches the target organ, how chemistry and conjugation affect productive uptake, and how long the active intracellular species persists. The Entrada and Avidity DMD data sharpen this conclusion: enhancing endosomal escape may help, but the more decisive variable may be tissue-targeted delivery to the right disease-relevant cells. In systemic RNA therapeutics, the winning platform is likely to be the one that delivers sufficient active oligonucleotide to the correct tissue first, and only then solves intracellular release and target engagement (Jing et al., 2023; Bäckström et al., 2024; Choi et al., 2025; Yun et al., 2025; Avidity Biosciences, 2025; Entrada Therapeutics, 2026).

Intrathecal administration



In 1846, William T.G. Morton demonstrated anesthesia, allowing surgeons to perform operations without causing severe pain; a revolutionary medical advancement that marked the beginning of modern anesthesiology. In 1898, August Bier administered cocaine into the intrathecal space of a patient, creating the first successful spinal anesthetic. His pioneering work laid the foundation for modern regional anesthesia and inspired further developments in both spinal and epidural techniques. Shortly thereafter, in 1901, the Romanian surgeon Racoviceanu-Pitesti reported the first use of opioids administered intrathecally, recognizing their potential to enhance analgesia through neuraxial pathways. In 1979, Behar and colleagues published the first report describing epidural morphine for pain management, demonstrating prolonged and effective analgesia. These discoveries transformed perioperative pain control and led to the widespread adoption of neuraxial opioids as part of routine intraoperative and postoperative analgesic regimens.

The historical progression of neuraxial anesthesia also established the intrathecal space as a valuable route for therapeutic delivery beyond anesthetics and analgesics. As understanding of cerebrospinal fluid dynamics, spinal anatomy, and drug distribution improved, clinicians and researchers increasingly recognized the potential of intrathecal administration to bypass systemic barriers and achieve targeted effects within the central nervous system. This concept has become particularly important in the field of genetic medicine. The blood-brain barrier, while essential for protecting neural tissue, presents a significant obstacle to the delivery of many biologic therapies, including viral vectors, nucleic acids, and gene-editing technologies. Intrathecal administration offers a means of circumventing this barrier by delivering therapeutic agents directly into the cerebrospinal fluid, thereby facilitating broader access to the brain and spinal cord.

Today, intrathecal delivery has emerged as a promising platform for gene replacement, gene silencing, and gene-editing therapies aimed at treating previously intractable neurological disorders. The foundations of this approach can be traced to the early development of adeno-associated virus (AAV) vectors for central nervous system (CNS) gene transfer. In 1994, Kaplitt and colleagues demonstrated long-term transgene expression and phenotypic correction in the mammalian brain following AAV-mediated gene transfer, providing some of the earliest evidence that genetic therapies could be delivered directly to neural tissues (Kaplitt et al., 1994). Subsequent studies confirmed persistent CNS transduction and tissue-specific expression patterns following AAV administration, establishing the feasibility of long-term gene expression within the nervous system (McCown et al., 1996). These advances ultimately led to the first clinical applications of CNS-directed gene therapy, including treatment of Canavan disease, a fatal leukodystrophy caused by mutations in the ASPA gene. In a landmark clinical study, AAV2-mediated gene transfer resulted in reduced levels of N-acetylaspartate, slower progression of cerebral atrophy, and stabilization of neurological function, demonstrating the therapeutic potential of direct CNS gene delivery (Leone et al., 2012).

More recently, the success of intrathecally administered antisense oligonucleotides for spinal muscular atrophy (SMA) has validated the cerebrospinal fluid as a route for repeated genetic intervention. Nusinersen, an antisense oligonucleotide that modifies SMN2 splicing to increase production of survival motor neuron protein, significantly improved motor function and survival in patients with SMA and established a new paradigm for neuraxial genetic medicine (Finkel et al., 2017). Building upon this success, intrathecal and intracisternal delivery approaches are being explored for a growing number of neurological disorders. In amyotrophic lateral sclerosis, tofersen has demonstrated the feasibility of suppressing expression of mutant SOD1 through repeated intrathecal administration (Miller et al., 2022). Similarly, AAV-mediated gene replacement strategies have been investigated for giant axonal neuropathy, where intracisternal administration of AAV9 vectors achieved widespread CNS distribution and substantial pathological improvement in preclinical models (Mussche et al., 2013).

Endosomal Escape

Early studies using electron microscopy with gold-labelled siRNA enabled direct visualization and counting of intracellular particles, revealing that the vast majority localize within endosomes, with only rare cytosolic events, corresponding to ~1–2% escape during a narrow window of

endosomal maturation (Gilleron et al., 2013). Quantitative approaches such as nanoscale secondary ion mass spectrometry (NanoSIMS) also provided high-resolution spatial and absolute measurements of oligonucleotide distribution and confirming dense accumulation within endosomal structures and minimal cytosolic presence; again, yielding estimates of ~1–2% productive escape in vivo (He et al., 2021; Dowdy, 2023). Biochemical fractionation methods, which physically separate cytosolic and membrane-bound compartments, further support these findings; despite a known bias toward overestimating cytosolic content due to endosomal disruption during lysis, these studies still report similarly low cytosolic fractions (~1%) (Juliano, 2018). Live-cell fluorescence imaging offers mechanistic insight into this inefficiency, demonstrating that escape is rare and transient (Wittrup et al., 2015).

Functional assays provide an additional layer of validation, as only cytosolic siRNA can engage the RNA-induced silencing complex; the observation that relatively few intracellular molecules are sufficient for robust gene knockdown implies that only a small proportion of the total internalized pool is productively delivered (Wittrup et al., 2015; Dowdy, 2023). Finally, in vivo pharmacokinetic and pharmacodynamic analyses, particularly of GalNAc–siRNA conjugates, reinforce this paradigm, indicating that only 0.3% of internalized siRNA is present in the cytosol at steady state, reflecting slow and continuous leakage from endosomal compartments rather than efficient release (Brown et al., 2020; Dowdy, 2023). Collectively these efforts provide strong evidence that virtually all internalized RNA remains sequestered within endosomal compartments and escape as the rate-limiting step in siRNA delivery.

To address this delivery bottleneck, a wide range of strategies have been developed that aim to enhance endosomal escape by transiently destabilizing membranes or altering intracellular trafficking pathways. Small-molecule agents like chloroquine and other cationic amphiphilic compounds can act by buffering endosomal acidification and inducing osmotic swelling, membrane perturbation, or vesicle rupture, thereby facilitating release of entrapped cargo (Grau and Wagner, 2024; Du Rietz et al., 2020). However, these approaches are inherently non-specific, often disrupting multiple endo-lysosomal compartments, and are limited by dose-dependent cytotoxicity and poor translational performance in vivo, where concentrations required for efficient escape are typically not well tolerated.

Peptide-based approaches such as arginine-rich cell-penetrating peptides and pH-responsive peptides exploit electrostatic interactions and acid-triggered conformational changes to promote membrane fusion, pore formation, or leakage from late endosomal compartments (Grau and Wagner, 2024; Mastrobattista et al., 2002; Wadia et al., 2004), but these

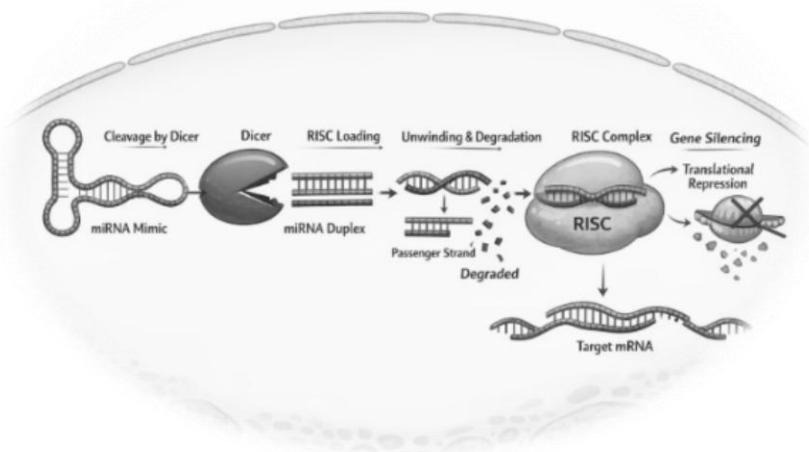
systems frequently suffer from serum instability and immunogenicity, as membrane-lytic activity can extend beyond endosomes to other cellular membranes.

Nanoparticle-based delivery systems, particularly ionizable lipid nanoparticles and cationic nanoparticles, are designed to exploit the endosomal environment by undergoing protonation at low pH and interact with anionic lipids to promote formation of non-bilayer structures that destabilize membranes and enable cytosolic release (Cullis and Hope, 2017; Hafez et al., 2001). However, despite their clinical success, these modalities retain intrinsically low escape efficiencies, are largely biased toward hepatic tissue, and can elicit inflammatory or infusion-related responses. These limitations are reflected in the strategic shift by several companies toward ligand–conjugate approaches like GalNAc. For example, Alnylam’s transition away from lipid nanoparticle–formulated siRNA (patisiran/Onpattro) toward GalNAc-conjugated platforms was driven by the need for more predictable pharmacokinetics, improved tolerability, and simpler subcutaneous dosing; therapeutic compatibility and commercial viability often relegate endosomal escape strategies to a lower priority.

Additional approaches, including photochemical internalization and emerging chemical biology strategies that modulate endosomal trafficking machinery, further underscore the breadth of mechanisms being explored to overcome this barrier (Grau and Wagner, 2024; Juliano, 2021). However, these strategies are often constrained by practical limitations, including restricted tissue penetration and the inherent complexity and consequences of perturbing intracellular trafficking networks. These challenges highlight that enhancing endosomal escape is not simply a question of increasing membrane disruption, but rather a fundamentally constrained optimization problem that requires balancing delivery efficiency with cellular integrity and therapeutic compatibility. Despite extensive innovation across multiple modalities, substantial improvements beyond the low baseline escape efficiency observed across systems have remained elusive.

	ASO (Antisense Oligonucleotides)	siRNA (non-conjugated / baseline)
Absorption (SC → plasma)	Rapid absorption via lymphatics; Tmax 2–6 h (<i>Geary et al., 2015</i>)	Rapid absorption; Tmax 1–4 h, but often lower bioavailability (<i>Ayyar & Song, 2023</i>)
Plasma exposure	Moderate Cmax; declines within hours (≤ 24 h) due to rapid tissue uptake (<i>Geary et al., 2015</i>)	Low and transient plasma exposure; often cleared within hours (<i>Ayyar & Song, 2023</i>)
Primary tissues (SC dosing)	Liver, kidney, spleen; broad distribution via protein binding (<i>Geary et al., 2015</i>)	Limited distribution without targeting; some liver/kidney uptake but inefficient (<i>Ayyar & Song, 2023</i>)
Time to tissue distribution	Rapid uptake into tissues within hours post-dose (<i>Geary et al., 2015</i>)	Within hours, but lower fraction reaches productive intracellular compartments
Tissue half-life	Long: 1–4 weeks (liver/kidney) due to intracellular retention (<i>Geary et al., 2015</i>)	Functional half-life depends on RISC; days–weeks if successfully loaded, otherwise short
Intracellular trafficking	Endosomal uptake → gradual release; relatively higher intracellular accumulation	Endocytosis → endosomal escape is limiting ($<1\%$ escape) (<i>Ayyar & Song, 2023</i>)
Onset of PD effect	Typically, 1–3 days post-dose (depends on RNA turnover)	Typically 2–7 days post-dose (processing + RISC loading + RNA turnover)
Peak PD effect	Often ~3–7 days post-dose	Often ~5–10 days post-dose
Duration of PD effect	Sustained for 1–4+ weeks depending on tissue retention (<i>Geary et al., 2015</i>)	Sustained for weeks–months if sufficient RISC loading occurs (<i>Hammond, 2005</i>)
PK–PD relationship	More tissue concentration-driven (liver/kidney levels correlate with effect) (<i>Geary et al., 2015</i>)	Decoupled: small intracellular active pool drives prolonged effect (<i>Hammond, 2005</i>)
Efficiency of tissue use	Higher fraction of dose becomes pharmacologically active	Very small fraction ($<1\%$) becomes active (RISC-loaded)

Design Strategies & Delivery



miRNA mimics

A microRNA (miRNA) mimic is typically designed as a synthetic double-stranded RNA that resembles the endogenous miRNA duplex. It contains a guide strand (Antisense) corresponding to the mature miRNA sequence and a complementary passenger strand (Sense), forming a short duplex with characteristic features such as 2-nucleotide 3' overhangs generated by Dicer processing. After introduction into the cell, the miRNA mimic duplex is unwound during RISC loading, the passenger strand is degraded, and the guide-strand is incorporated into the RNA-induced silencing complex (RISC). Within RISC, the guide strand directs gene silencing by partial base pairing with target mRNAs, leading to translational repression or a catalytic mRNA degradation.

miRNAs are powerful regulators that can control multiple genes and entire cellular pathways, making them promising targets for therapy as well as biomarkers. However, despite strong preclinical progress and growing interest, especially following the success of mRNA vaccines, the field remains in early clinical stages, with no FDA-approved miRNA therapies and several trials halted due to toxicity and immune-related side effects. It remains unclear whether the observed toxicities in miRNA therapeutics arise from the miRNA or the delivery system.

MRX34 was a first-in-class microRNA therapeutic composed of a synthetic double-stranded miR-34a mimic formulated within a lipid nanoparticle delivery system to restore the tumor-suppressive miR-34 pathway downstream of p53 and repress multiple oncogenic targets. Preclinical studies and early Phase 1 clinical data demonstrated successful systemic delivery, target engagement, and limited antitumor activity, supporting proof-of-concept for miRNA replacement therapy. However, the trial was terminated in 2016 due to severe immune-related toxicities, including cytokine release syndrome and treatment-related deaths, likely caused by innate immune activation triggered by the nanoparticle. This outcome highlighted key challenges in RNA therapeutics particularly the balance between structural design, delivery efficiency, and immunogenicity despite the strong biological rationale for miR-34-based therapy (Zhang et al., 2019; Hong et al., 2020)

MRX34 is an unmodified RNA formulated using a liposomal nanoparticle delivery system (NOV340) composed of amphoteric lipids designed to encapsulate and protect the double-stranded miR-34a mimic during systemic administration. The lipid formulation typically included components such as phospholipids, cholesterol, and PEGylated lipids to enhance stability, prolong circulation time, and facilitate cellular uptake and endosomal release of the RNA cargo.

In the Phase 1 clinical trial, MRX34 was administered intravenously twice weekly (commonly on days 1 and 4 of a 3-week cycle), with dose escalation up to a maximum tolerated dose of approximately 110 mg/m² (about 40mg per dose) in non-hepatocellular carcinoma and HCC patients, before immune-related toxicities limited further dosing (Hong et al., 2020; Beg et al., 2017). The trial was terminated due to serious immune-mediated toxicities, including fatal adverse events in multiple patients (Kim and Croce, 2023). These toxicities were characterized by strong immune activation and while MRX34 successfully downregulated oncogenic target genes, the severity of these adverse effects led to trial discontinuation (Kim and Croce, 2023).

The toxicity and clinical outcomes observed with MRX34 cannot be attributed to a single factor, as multiple confounding elements were involved. The lipid nanoparticle delivery system facilitated broad systemic distribution, potentially exposing diverse tissues to the therapy, while frequent and high dosing likely led to excessive off-target accumulation. The complexity of the drug product complicates efforts to determine whether toxicity was primarily driven by the delivery system, payload miRNA, or the intrinsic biology of miRNA regulation.

An alternative to traditional, repeated systemic delivery of synthetic miRNA mimics and lipid nanoparticles (LNPs) involves a vector-mediated

strategy for sustained, intracellular expression. Researchers at UCL (University College London) have utilized viral vector gene therapy to achieve this, specifically through the AAV5-delivered miRNA construct underlying AMT-130. This approach employs an engineered pri-miR-451a scaffold to express an anti-huntingtin microRNA directly within transduced neurons.

Unlike MRX34, which required frequent intravenous dosing due to the transient nature of synthetic RNA, the AAV-mediated delivery enables long-term, endogenous production of the therapeutic miRNA following a single administration. In this approach, the therapeutic sequence is not delivered as a mature, fragile duplex RNA; it is instead delivered as a DNA cassette encoding a pri-miRNA transcript, which is subsequently processed by the host native machinery into the active guide strand.

AMT-130 incorporates a CAG promoter, engineered pri-miR-451 flanking regions, a pre-miRNA hairpin containing the HTT-targeting guide strand, and an hGH polyadenylation signal cassette to support stable neuronal expression and accurate RNA processing (Evers et al., 2018). Importantly, the miR-451 scaffold was selected because it undergoes Ago2-dependent, Dicer-independent maturation, reducing passenger-strand accumulation and potentially improving strand specificity compared with conventional shRNA or duplex miRNA approaches (Cheloufi et al., 2010; Yang et al., 2010). Although AAV-mediated delivery introduces its own challenges particularly the permanence of transgene expression, limited reversibility, and concerns regarding long-term safety and immune responses to viral capsids, it avoids several limitations associated with systemic nanoparticle delivery, including repeated high-dose exposure, broad biodistribution of synthetic RNA, and dose-limiting innate immune activation. Consequently, the transition from MRX34-like nanoparticle systems to AAV-expressed miRNA therapeutics reflects a broader evolution in the RNA therapeutics field toward more durable, tissue-targeted, and biologically integrated delivery platforms.



The therapeutic cassette described in the patent consists of a fully engineered adeno-associated viral (AAV) expression designed to achieve long-term suppression of huntingtin (HTT) expression in neurons through RNA interference. In the disclosed pVD-CAG-miH12-451 configuration, the cassette begins with the synthetic CAG promoter, a strong hybrid promoter combining the cytomegalovirus (CMV) enhancer and chicken β -actin promoter elements, included to drive high-level transgene expression

in neuronal tissue following intracerebral AAV delivery (Evers et al., 2018). Downstream of the promoter is the 5' pri-miRNA flanking region derived from endogenous miR-451a. These flanking sequences are necessary because they provide the structural and sequence context recognized by the endogenous microRNA-processing machinery, particularly the Drosha/DGCR8 complex, enabling correct cleavage of the primary transcript into a precursor miRNA hairpin (Cheloufi et al., 2010). Embedded within this framework is the engineered pre-miRNA hairpin itself, which constitutes the active therapeutic silencing element. The hairpin contains the HTT-targeting guide strand sequence 5'-AAGGACUUGAGGGACUCGAAG-3' (SEQ ID NO.5), which is fully complementary to the selected exon-1 huntingtin target sequence and mediates sequence-specific degradation of HTT mRNA through incorporation into the RNA-induced silencing complex (RISC) (Evers et al., 2018). Opposite the guide strand is the engineered passenger strand, designed not only to stabilize the stem-loop structure but also to minimize passenger-strand loading into RISC and thereby reduce off-target silencing effects. The use of the miR-451a scaffold is particularly important because miR-451 follows a non-canonical Ago2-dependent, Dicer-independent maturation pathway that preferentially produces a single dominant guide strand, improving strand specificity and reducing unintended RNAi activity (Cheloufi et al., 2010; Yang et al., 2010). Following the pre-miRNA region, the cassette contains the 3' pri-miRNA flanking sequence, which further supports accurate Drosha processing, transcript stability, and nuclear export of the precursor hairpin. Finally, the construct terminates with the human growth hormone (hGH) polyadenylation signal, which directs efficient transcriptional termination and polyadenylation, thereby stabilizing the therapeutic transcript after expression from the AAV genome. Collectively, these engineered sequence components form a highly optimized neuronal RNAi cassette intended to provide durable expression, efficient miRNA biogenesis, selective guide-strand loading, and sustained suppression of mutant and wild-type huntingtin expression within the central nervous system (Evers et al., 2018).

The role of Dicer

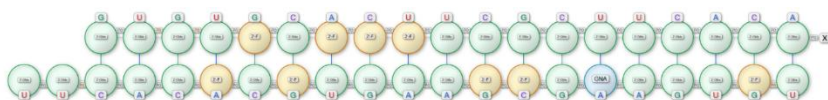
RNA interference (RNAi) is a highly conserved mechanism for post-transcriptional gene regulation. In mammalian systems, this process is mediated by a series of coordinated molecular interactions involving the RNase III enzyme Dicer, which cleaves longer dsRNA into short interfering RNAs (siRNAs), and the RNA-induced silencing complex (RISC); uses the guide strand to recognize and cleave target transcripts (Hefner et al., 2008).

While standard (Alnylam-style) synthetic 21–23 nucleotide siRNAs mimic the “products” of Dicer processing and can directly engage RISC, they bypass the natural Dicer processing step. This may lead to reduced efficiency and less controlled strand selection. In contrast, longer Dicer-substrate RNAs (e.g., 27mers), designed to engage the endogenous RNAi machinery more fully, may enable Dicer-mediated cleavage and directional handoff to RISC and potentially leading to enhances potency, durability, and specificity of gene silencing (Hefner et al., 2008). This is the theory behind the platform technology developed by a company called Dicerna which sold to Novo Nordisk acquired in 2021 for \$3.3 billion.



A research group from Bio-Rad Laboratories in collaboration with Integrated DNA Technologies (IDT) performed a series of experiments comparing Dicer-substrate 27mer RNAs with conventional 21mer siRNAs to evaluate the impact of engaging the Dicer processing step

on RNAi performance. In HeLa cell models across multiple gene targets (including CDK2, TP53, RAF1, and ACTB), the Dicer-substrate 27mers produced greater gene knockdown in dose–response studies, particularly at lower concentrations, indicating enhanced intrinsic potency (Hefner, E., Clark, K., Whitman, C., Behlke, M.A., Rose, S.D., Peek, A.S. and Rubio, T. (2008). They also demonstrated more sustained silencing over time, whereas the activity of 21mer siRNAs declined more rapidly. The authors attributed these improvements to Dicer-mediated processing and efficient handoff to RISC, which promotes accurate guide-strand selection and improved complex loading. Notably, the ability to achieve effective silencing at lower doses suggests a potential reduction in off-target effects and toxicity, supporting both improved efficacy and safety relative to traditional siRNA designs (Hefner et al., 2008).



siRNA

The typical design of small interfering RNAs (siRNAs) as 21–23 nucleotide double-stranded duplexes with 2-nucleotide 3' overhangs is considered optimal because it closely mimics the natural products of Dicer processing and maximizes compatibility with the RNA interference (RNAi) machinery. Elbashir et al. (2001) demonstrated that synthetic 21-nt siRNA duplexes are sufficient to induce efficient and sequence-specific gene silencing in mammalian cells while avoiding the nonspecific responses triggered by longer double-stranded RNAs. The presence of 2-nt 3' overhangs, typically composed of UU or TT residues, further enhances silencing efficiency by facilitating Dicer recognition and promoting effective loading into the RNA-induced silencing complex (RISC), as these structural features resemble endogenous Dicer cleavage products. Additionally, the requirement for a double-stranded RNA duplex is critical for RISC assembly, as strand selection and guide strand incorporation depend on duplex structure and thermodynamic asymmetry (Sioud, 2025).

21–23 nucleotides

2-nt 3' overhangs (usually “UU” or “TT”)

Double-stranded RNA duplex required for RISC loading

Strand Selection and Thermodynamic Asymmetry

Efficient siRNA function depends on strand selection driven by thermodynamic asymmetry within the duplex. Specifically, the 5' end of the antisense (guide) strand is designed to have lower thermodynamic stability, while the 5' end of the sense (passenger) strand is relatively more stable, creating an asymmetry that directs RISC to preferentially incorporate the correct guide strand (Schwarz et al., 2003; Sioud, 2025). This asymmetry is further reinforced by nucleotide composition, where an A/U-rich region at the 5' end of the antisense strand reduces stability, and a G/C-rich region at the 5' end of the sense strand increases stability (Ui-Tei et al., 2004). Together, these features promote accurate strand selection, ensuring efficient guide strand loading into RISC and enhancing overall gene silencing specificity and efficacy.

Lower stability at the 5' end of antisense (guide) strand

Higher stability at the 5' end of sense (passenger) strand

This promotes correct guide strand loading into RISC

A/U-rich at antisense 5' end
G/C-rich at sense 5' end

GC Content

Optimal siRNA efficiency is strongly influenced by GC content, with a moderate range of approximately 30–50% providing the best balance between stability and functionality. Excessively high GC content increases duplex stability, which can hinder unwinding by RISC and reduce silencing efficiency, whereas very low GC content weakens target mRNA binding and compromises activity (Reynolds et al., 2004). Consequently, functional siRNAs typically exhibit moderate GC levels combined with relatively lower internal stability, facilitating efficient strand separation and target recognition within the RNAi pathway (Sioud, 2025).

Optimal 30–50% GC

Too high GC leads to overly stable duplex and poor RISC unwinding

Too low GC results in weak target binding

Functional siRNAs tend to have moderate GC and lower internal stability

Nucleotide Preferences

In the antisense (guide) strand, the presence of A/U at position 1 (the 5' end) reduces thermodynamic stability and promotes its incorporation into RISC, while a corresponding G/C-rich 5' end in the sense (passenger) strand increases stability and supports proper strand discrimination (Ui-Tei et al., 2004). Additionally, the seed region (positions 2–8), which is critical for target mRNA recognition, tends to be more effective when A/U-rich, improving binding specificity and silencing efficiency (Reynolds et al., 2004). Conversely, long stretches of G/C bases or repetitive sequences are generally avoided, as they can increase internal stability and reduce siRNA functionality (Reynolds et al., 2004).

Antisense strand:

A/U at position 1 (5' end)

Sense strand:

G/C at 5' end

Seed Region

The seed region (positions 2–8) is crucial for target recognition and should ideally be A/U-rich while avoiding long GC stretches or repetitive sequences. Because this region largely determines binding specificity, it is also the main source of off-target effects: even partial complementarity can unintentionally silence other genes in a miRNA-like manner. To minimize this risk, sequences should be designed to avoid matching the 3' UTRs of unrelated genes.

Seed (positions 2–8):

Prefer A/U

Avoid long GC stretches

Main source of off-target effects

Chemical modifications (e.g. 2'-O-methyl) can reduce off-target effects (Lennox and Behlke, 2016)

Target Site Considerations

Target sites should be accessible regions with minimal secondary structure, while also avoiding areas like the start codon and highly structured regions can hinder binding. Additionally, sequence design should exclude problematic motifs such as long repeats, palindromes, internal hairpins, and immune-stimulatory elements (e.g., GU-rich sequences) to ensure stability and reduce unwanted cellular responses.

Gapmer (RNase-H) ASOs

Gapmer antisense oligonucleotides are single-stranded sequences typically 16–20 nucleotides long, designed with a central DNA “gap” (8–10 nt) flanked by chemically modified wings that enhance stability and binding (Bennett and Swayze, 2010; Crooke, 2017; Lennox and Behlke, 2016). The DNA gap is essential for forming an RNA–DNA hybrid, which recruits RNase H to cleave the target RNA. Chemical modifications such as LNA, 2'-O-methyl (2'-OMe), or 2'-O-methoxyethyl (2'-MOE) in the wings and a predominantly phosphorothioate (PS) backbone improve nuclease resistance, cellular uptake, and protein binding, while limited incorporation of phosphodiester linkages can fine-tune these properties without

compromising activity (Kurreck, 2003; Bennett and Swayze, 2010; Geary et al., 2015). Effective gapmer design maintains 40–60% GC content and an appropriate melting temperature (55–65°C), targets accessible RNA regions, and avoids highly structured, repetitive, or protein-bound sites (Watts and Corey, 2012; Lennox and Behlke, 2016).

GC content: 40–60%

Melting temperature: 55–65°C

Avoid:

Highly structured regions

Repetitive sequences

Protein-bound sites

Off-target & Toxicity Considerations

Avoid sequence motifs:

CpG motifs

Long G runs

Palindromes & Self-complementarity

To minimize off-target effects and toxicity, sequences should avoid motifs such as CpG, long G runs, palindromes, and self-complementarity, be screened for transcriptome-wide specificity, and be tested as multiple candidates (Krieg, 2006; Crooke, 2017; Bennett and Swayze, 2010). Gapmers function in both the nucleus and cytoplasm, leveraging RNase H's natural role in degrading RNA–DNA hybrids to achieve targeted gene silencing (Crooke, 2017; Wahba et al., 2011).

Steric-Blocking single-stranded ASOs

Steric-blocking antisense oligonucleotides are single-stranded sequences typically 15–25 nucleotides long with fully modified backbones, as they do not rely on RNase H activity (Bennett and Swayze, 2010; Lennox and Behlke, 2016). Instead, they function by physically blocking molecular interactions such as ribosome binding, spliceosome assembly, or protein–RNA interactions, enabling translation inhibition or splice modulation (Watts and Corey, 2012). These ASOs use fully modified chemistries such as 2'-OMe, 2'-MOE, LNA, or PMOs with phosphorothioate or neutral backbones to enhance stability, binding affinity, and cellular uptake (Kurreck, 2003; Summerton, 2007; Geary et al., 2015). Effective design requires moderate GC content (40–60%) and strong binding affinity, targeting accessible RNA regions such as start codons or splice regulatory elements while avoiding structured or protein-bound regions (Watts and Corey, 2012; Bennett and Swayze, 2010). To reduce toxicity and immune activation, sequences should avoid motifs like CpG, long G runs, and self-

complementarity, and carefully balance high-affinity modifications (Krieg, 2006; Lennox and Behlke, 2016). These ASOs can function in both the nucleus and cytoplasm, with splice-switching applications primarily occurring in the nucleus.

Compare Splicing Technology: PMO vs 2'-MOE

	PMO (Morpholino; e.g., Eteplirsen, Golodirsen, Viltolarsen)	2'-MOE PS ASO (e.g., Nusinersen)
Backbone	Phosphorodiamidate (neutral)	Phosphorothioate (negatively charged)
Sugar/Scaffold	Morpholine ring (no ribose)	Ribose with 2'-MOE modification
Charge	Neutral	Negative
Protein binding	Low	High
Nuclease resistance	Very high	High
Binding affinity (Tm)	High	High–very high
Cellular uptake	Poor (often needs high dose or conjugates)	Better (PS-driven uptake)
Distribution	Limited without conjugates	Broad tissue distribution
Toxicity profile	Generally low	Moderate (PS-related effects possible)
Mechanism	Steric blocking (splice modulation)	Steric blocking (splice modulation)

Design implications: - PMOs prioritize safety and stability, with tradeoffs in delivery efficiency. MOE-PS ASOs balance delivery, potency, and pharmacokinetics, with increased biological interactions.

Conjugates and Delivery Strategies

RNA therapeutics have emerged as programmable medicines capable of modulating gene expression through Watson–Crick base pairing with their genetic targets. However, their clinical performance is strongly shaped by delivery barriers, including nuclease degradation, poor passive membrane permeability, rapid clearance and inefficient endosomal escape. To address these challenges, modern oligonucleotide drug design combines chemical stabilization with delivery-enabling conjugation strategies, such as phosphorothioate backbones, 2'-sugar modifications and ligand-directed targeting (2021; Roberts, Langer and Wood, 2020). Among these approaches, N-acetylgalactosamine (GalNAc) conjugation has become a leading strategy for hepatocyte delivery, as multivalent GalNAc ligands bind the asialoglycoprotein receptor (ASGPR) and promote receptor-mediated uptake into liver cells (Nair *et al*., 2014). Beyond GalNAc, RNA therapeutics are increasingly being engineered with lipids, peptides, antibodies, aptamers, polymers and nanoparticle systems to improve biodistribution, cellular uptake and tissue selectivity, expanding the potential of ASOs and other RNA-targeting agents beyond the liver while highlighting delivery as a central determinant of therapeutic index (Benizri *et al*., 2020; Dowdy, 2017; Roberts, Langer and Wood, 2020).

Feature	Small Molecules	GalNAc–siRNA Conjugates	Therapeutic Antibodies
1. Size	<1000 Da	~16,000 Da	>150,000 Da
2. Route of delivery	Oral, IV, IM, SC, inhalation	Subcutaneous (SC)	Mostly IV and SC
3. Plasma protein binding	Variable (high for lipophilic, low for hydrophilic)	High (~80–90%)	Low
4. Tissue distribution	Often large volume of distribution	Extensive liver distribution	Mostly confined to circulation & extracellular fluid
5. Metabolism	CYP450 enzymes (drug-metabolizing enzymes)	Exonucleases & endonucleases (no CYP involvement)	Proteolytic catabolism (receptor & non-specific)
6. Renal clearance	Often important pathway	Minor (<25% for intact siRNA)	None for intact antibodies
7. Plasma half-life	Short (hours)	Short (<10 hours)	Long (weeks)
8. PK linearity	Usually linear (can be nonlinear)	Linear	Often nonlinear at low doses (TMDD), linear at high doses
9. PK/PD relationship	PK → PD (PD depends on PK)	PK independent of PD; PD driven by RISC-loaded siRNA	PK often depends on PD (TMDD)

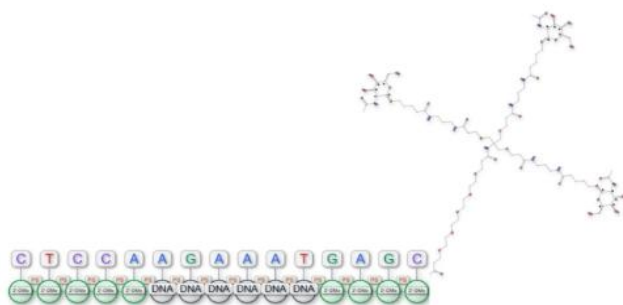
For many years, N-acetylgalactosamine, or GalNAc, was understood primarily as an amino sugar related to galactose. GalNAc helped form blood group A structures, appeared in mucin-type O-glycosylation, and marked important cancer-associated glycans such as the Tn antigen (Brockhausen and Stanley, 2015; Rodríguez et al., 2023). At this stage, GalNAc belonged mainly to glycobiology. It helped scientists understand how cells labelled themselves, communicated with their environment, and changed during disease.

In the 1960s at the National Institutes of Health, Gilbert Ashwell and Anatol Morell studied how the liver cleared circulating glycoproteins from the blood. They worked on plasma glycoprotein metabolism and noticed when glycoproteins lost terminal sialic acid and exposed galactose or GalNAc residues, the liver removed them rapidly. Their work led scientists to identify the hepatic asialoglycoprotein receptor (Ashwell and Morell, 1974; Stockert, 1995). This discovery gave GalNAc a new identity. Hepatocytes, the main functional cells of the liver, displayed the asialoglycoprotein receptor on their surface. That receptor recognized terminal galactose and GalNAc residues and internalized bound molecules. In practical terms, Ashwell and Morell had shown the field that the liver had a molecular doorway, and GalNAc could help unlock it.

This discovery became especially important when RNA medicines entered the scene. Small interfering RNAs, or siRNAs, could silence disease-causing genes, but they faced a delivery problem. They were large, charged, and vulnerable to degradation. Scientists could design powerful RNA molecules in the laboratory, but they still needed to get those molecules into the right cells in the body.

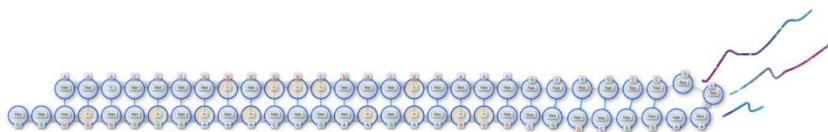
In the late 1990s, Erik A. L. Biessen and colleagues at the Division of Biopharmaceutics, Leiden Center for Drug Research, Leiden University, in the Netherlands, conjugated antisense oligodeoxynucleotides to a multivalent sugar ligand designed to bind the asialoglycoprotein receptor and redirect oligonucleotide uptake into parenchymal liver cells. This was an important proof of concept: an oligonucleotide that would otherwise be cleared or distributed inefficiently could be chemically attached to a hepatocyte-targeting ligand and pulled into the liver through receptor-mediated endocytosis (Biessen et al., 1999). These early glycoconjugates improved hepatocyte accumulation, but they did not yet produce the full pharmacological benefit that later GalNAc platforms would deliver. They showed both the promise and the problem: GalNAc-like receptor targeting could bring RNA drugs to the right tissue, but chemistry, linker design, oligonucleotide stability and intracellular trafficking still had to catch up.

Muthiah Manoharan and colleagues at Alnylam Pharmaceuticals then helped translate GalNAc conjugation into a delivery technology for RNAi therapeutics. Building these earlier studies, the Alnylam team developed multivalent GalNAc ligands that could be attached directly to chemically modified siRNAs, allowing the RNA duplex to reach hepatocytes after subcutaneous administration without the need for lipid nanoparticles. Their GalNAc-siRNA work showed that these conjugates localized to hepatocytes and produced robust RNAi-mediated gene silencing in vivo (Nair et al., 2014). Shortly after, T. P. Prakash and colleagues at Isis Pharmaceuticals took this principle to gapmer ASOs, showing that triantennary GalNAc conjugation improved ASO potency and hepatocyte delivery (Prakash et al., 2014). Together, these studies turned GalNAc from a glycobiology motif into one of the central delivery technologies for liver-targeted RNA medicines.



There are currently seven FDA-approved siRNA therapeutics on the market, reflecting the rapid clinical maturation of RNA interference as a therapeutic modality. Among these, patisiran remains the only approved siRNA drug formulated with a lipid nanoparticle (LNP) delivery system. In contrast, the other approved siRNA therapeutics, givosiran, lumasiran, inclisiran, vutrisiran, nedosiran, and fitusiran, use GalNAc conjugation to enable targeted delivery to hepatocytes. There are also approximately thirteen FDA-approved ASO therapeutics, including splice-modulating ASOs such as nusinersen and the Duchenne muscular dystrophy exon-skipping drugs eteplirsen, golodirsen, viltolarsen, and casimersen, as well as RNase H-mediated ASOs such as inotersen, tofersen, eplontersen, olezarsen, and donidalorsen. Unlike siRNA therapeutics, where GalNAc conjugation now dominates liver-directed delivery, most approved ASOs are not formulated in lipid nanoparticles; most are administered chemically “naked”, using backbone and sugar modifications to improve stability, potency, and tissue exposure. However, the field is increasingly moving toward targeted conjugation strategies: eplontersen was the first FDA-

approved GalNAc-conjugated ASO, and newer liver-directed ASOs such as olezarsen also use GalNAc-based targeting.



Dicerna's GalXC platform

Dicerna's GalXC platform (acquired by Novo Nordisk) is a liver-targeted RNA interference (RNAi) technology designed to silence disease-associated genes at the mRNA level. The platform uses double-stranded RNA molecules conjugated to GalNAc, enabling selective uptake by hepatocytes through GalNAc-mediated receptor binding and internalization into liver cells (Dicerna Pharmaceuticals, 2021). Belcesiran, also known as DCR-A1AT, applied this platform to alpha-1 antitrypsin deficiency-associated liver disease (AATLD) by targeting hepatic production of alpha-1 antitrypsin (AAT), including the misfolded Z-AAT protein that accumulates in hepatocytes and contributes to liver injury, fibrosis, cirrhosis and hepatocellular carcinoma (Dicerna Pharmaceuticals, 2021).

The program was mechanistically supported by early clinical data. In a Phase I, double-blind, placebo-controlled, single-ascending-dose study in healthy volunteers, Belcesiran produced dose-dependent reductions in circulating serum AAT after subcutaneous administration. Interim data showed mean maximum serum AAT reductions of approximately 50%, 69% and 80% at the 1.0, 3.0 and 6.0 mg/kg dose levels, respectively, while individual participants in the 6.0 mg/kg cohort achieved maximum reductions ranging from 62% to 91% (Dicerna Pharmaceuticals, 2021). These results provided evidence of target engagement and supported progression into the Phase II ESTRELLA study. The early safety profile also appeared acceptable: no serious adverse events were reported in the interim Phase I analysis, most treatment-emergent adverse events were mild, and no clinically significant changes in laboratory parameters or lung function were observed during the treatment periods assessed (Dicerna Pharmaceuticals, 2021).

However, the subsequent Phase II ESTRELLA study did not generate the level of clinical evidence required to support continued development. ESTRELLA was designed as a randomized, double-blind, placebo-controlled, multiple-dose study evaluating the safety, tolerability, pharmacokinetics and pharmacodynamics of Belcesiran in adults with AATD-associated liver disease (ClinicalTrials.gov, 2026). The study was

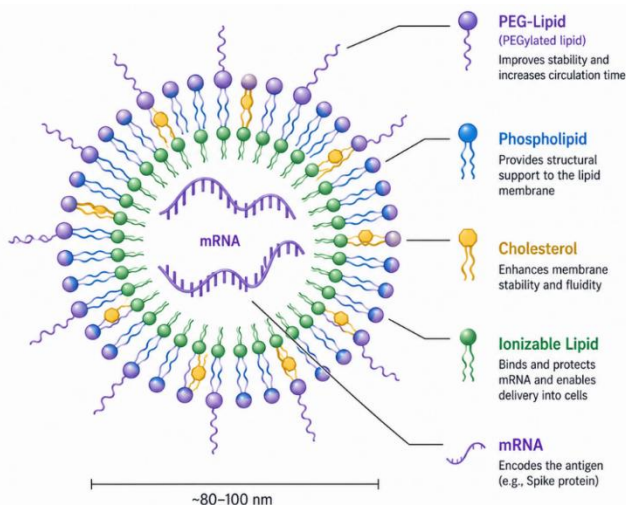
planned to include three cohorts, with the third cohort intended to expand the dataset and further characterize treatment effects, including liver-related outcomes. The trial was terminated early after enrolling only 16 participants, and the third cohort was never opened (ClinicalTrials.gov, 2026; Health Research Authority, 2026). The ClinicalTrials.gov record states that the study was terminated because the “development project was discontinued” (ClinicalTrials.gov, 2026).

Belcesiran does not appear to have failed because of an absence of pharmacodynamic activity. Repeated dosing continued to reduce blood AAT levels, suggesting that the RNAi mechanism remained active in the patient population (Health Research Authority, 2026). The more substantive issue was that AAT knockdown alone was insufficient to establish a compelling clinical path forward. For AATLD, the key development question is not simply whether circulating AAT can be reduced, but whether reducing hepatic Z-AAT production translates into meaningful liver benefit, such as reduced toxic protein accumulation, improvement or stabilization of fibrosis, or prevention of progression to cirrhosis and liver-related complications. Because ESTRELLA was stopped early, enrolled only a small number of participants and generated limited follow-up, it was unable to provide robust evidence on these disease-modifying outcomes.

Safety uncertainty may also have contributed to the cancellation of the program. Although the Health Research Authority summary described Belcesiran as generally safe and well tolerated overall, it also reported four serious adverse events considered possibly related to Belcesiran: pneumonia, atrial fibrillation, deterioration in lung function tests and worsening liver function, the latter explained by underlying cirrhosis due to AATLD. All affected participants recovered clinically (Health Research Authority, 2026). In a very small Phase II dataset, such events do not necessarily establish a causal safety signal, but they increase uncertainty, particularly for a therapy intended for patients who may already have clinically significant liver disease and who may also be vulnerable to pulmonary complications due to AAT deficiency.

Overall, the discontinuation of Belcesiran showed evidence of target engagement and had encouraging early tolerability data, but the curtailed Phase II study did not provide enough evidence that AAT lowering would translate into clinically meaningful liver benefit. (ClinicalTrials.gov, 2026; Health Research Authority, 2026).

	DRUG	COMPANY	TARGET	INDICATION
Rare Diseases	GIVOSIRAN	Alnylam	ALAS1	Acute hepatic porphyria
	LUMASIRAN	Alnylam	Glycolate oxidase	Primary hyperoxaluria type 1
	VUTRISIRAN	Alnylam	Transthyretin (TTR)	ATTR amyloidosis
	NEDOSIRAN	Novo Nordisk	LDHA	Primary hyperoxaluria type 1
	FAZIRSIRAN	Arrowhead	AAT (SERPINA1)	AAT deficiency liver disease
	FITUSIRAN	Alnylam / Sanofi	Antithrombin	Hemophilia
	SLN-124	Silence Therapeutics	TMPRSS6	Thalassemia
Cardiometabolic	INCLISIRAN	Alnylam / Novartis	PCSK9	Hypercholesterolemia
	OLPASIRAN	Amgen	Lp(a)	Cardiovascular disease
	ZILEBESIRAN	Alnylam	Angiotensinogen	Hypertension
	ZERLASIRAN	—	Lp(a)	Cardiovascular disease
	SOLBINSIRAN	Eli Lilly	ANGPTL3	Dyslipidemia / CV disease
Liver Diseases	CEMDISIRAN	Alnylam	Complement C5	Complement-mediated diseases
	ALN-HSD (ALN-HSD17B13)	Alnylam	HSD17B13	NASH
	BELCESIRAN	Alnylam / Dicerna	AAT	AAT liver disease
Infectious Diseases	AB-729	Arbutus	HBV proteins	Hepatitis B
	JNJ-3989	Arrowhead / J&J	HBV proteins	Hepatitis B
	ALN-HBV02	Alnylam	HBV proteins	Chronic HBV



Lipid nanoparticles (LNPs) are a clinically validated non-viral platform for delivering nucleic acid therapeutics, including mRNA and siRNA. Their clinical development has been driven by the success of COVID-19 mRNA vaccines and liver-directed RNA therapies, where LNPs have shown strong *in vivo* activity (Belaidi et al., 2025). Most LNPs contain four core components: an ionizable lipid, a helper phospholipid, cholesterol and a PEG-lipid. The ionizable lipid is especially important because it binds RNA during formulation, remains largely neutral in circulation and becomes protonated in the acidic endosome to support endosomal escape after cellular uptake (Belaidi et al., 2025). However, endosomal escape remains one of the main limits on LNP efficacy (Jozić et al., 2026; Belaidi et al., 2025).

Route of administration is a major determinant of LNP efficacy and safety. Intravenous administration is useful for systemic delivery, but it strongly directs LNPs toward the liver because circulating particles adsorb apolipoproteins, especially ApoE, and are taken up efficiently by hepatocytes and other liver-associated cells (Belaidi et al., 2025). This makes IV delivery well suited to liver diseases but less effective for extrahepatic targets such as the brain. For CNS therapies, the blood–brain barrier (BBB) adds another obstacle by limiting the entry of complex therapeutics, including nucleic acids, antibodies, immunotherapeutics and gene therapies (Iqbal, Hwang and Costain, 2026). As a result, CNS-directed LNPs often require additional engineering, such as optimized particle size, charge, lipid chemistry, surface ligands or receptor-mediated transcytosis

strategies targeting receptors such as transferrin receptor, insulin receptor, IGF1R or LRP1 (Belaidi et al., 2025; Iqbal, Hwang and Costain, 2026).

Different routes therefore create different safety concerns.

Intramuscular administration, used mainly for vaccines, supports local antigen expression and immune activation rather than broad organ targeting. IV administration produces systemic exposure, making infusion-related reactions, complement activation, hepatic uptake and off-target biodistribution more relevant (Belaidi et al., 2025). Intrathecal, intracerebroventricular and intracerebral routes can improve CNS exposure by bypassing or reducing reliance on vascular transport barriers, but they introduce local inflammation, neurotoxicity, procedural burden and patient acceptability concerns (Belaidi et al., 2025; Iqbal, Hwang and Costain, 2026). Focused ultrasound with microbubbles offers a less invasive way to transiently open the BBB after IV dosing, but its safety may vary in diseased, inflamed or aged brain tissue (Belaidi et al., 2025; Iqbal, Hwang and Costain, 2026).

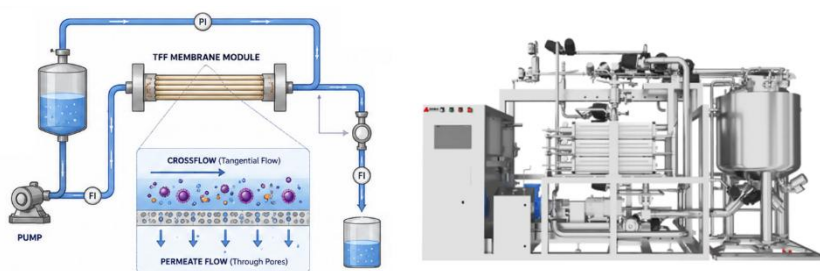
The toxicology profile of LNPs also depends on dose schedule. Acute toxicity is relatively well characterized and often includes transient flu-like symptoms, cytokine elevations and occasional infusion-related reactions. These effects depend on dose, infusion rate, lipid composition and patient context (Belaidi et al., 2025). Complement activation-related pseudo-allergy has been reported in high-infusion-rate animal models, and PEG-lipids remain under scrutiny because they may contribute to rare hypersensitivity-like responses (Belaidi et al., 2025).

Long-term toxicity is less well defined, especially for products that require repeated dosing. Chronic administration can promote anti-PEG antibody formation, accelerated blood clearance, altered biodistribution and reduced efficacy over time (Belaidi et al., 2025). Slowly metabolized lipids may also accumulate in the liver or other tissues, increasing the risk of chronic on-target and off-target toxicity. Biodegradable lipids are intended to reduce this risk by breaking down more readily after delivery, improving clearance and tolerability during repeat dosing (Belaidi et al., 2025). Overall, long-term safety must be evaluated in relation to route, target tissue, dosing frequency and disease context rather than generalized across all LNP products.

Manufacturing remains another major barrier to wider commercialization. LNP production requires reproducible large-scale manufacturing while maintaining critical quality attributes such as particle size, polydispersity, RNA encapsulation efficiency, sterility, stability and batch-to-batch consistency. This is difficult because LNPs self-assemble within milliseconds, and small changes in flow rate, mixing geometry, solvent dilution, shear stress, salt concentration, temperature or downstream

purification can alter particle properties. As a result, commercial manufacturing must reproduce laboratory-scale performance across much larger volumes while preserving potency, biodistribution, stability and safety.

To manage these challenges, commercial platforms increasingly use technologies such as parallel impinging jet mixers, continuous-flow reactors and automated GMP systems. These require precise fluid-dynamic control, sterile closed-system designs, high-accuracy pumps and sensors, in-line monitoring and validated process controls. The cost burden is substantial, including GMP facilities, cleanrooms, continuous manufacturing systems, automated fill-finish lines, analytical instruments, sterile single-use consumables, expensive ionizable lipids, RNA payload losses and failed batches. For these reasons, manufacturing scalability remains one of the most important practical barriers to broad RNA-LNP commercialization.



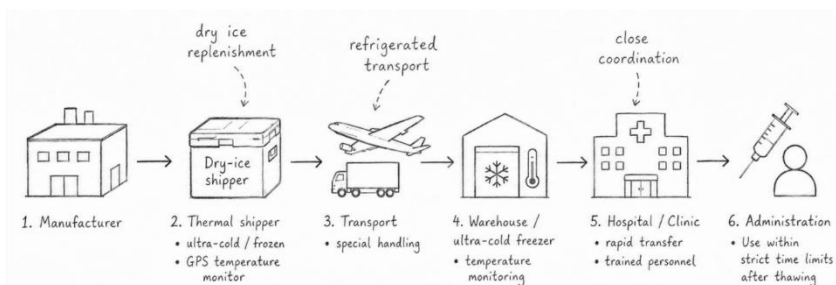
For example, Tangential flow filtration (TFF) is commonly used to remove ethanol, exchange buffers, concentrate nanoparticles and prepare sterile formulations, but LNPs are fragile nanoscale structures. TFF can therefore cause shear-related particle damage, RNA leakage, membrane fouling, aggregation, flux decline and inconsistent product recovery. These failures can reduce encapsulation efficiency or compromise entire batches.

Overall, LNP manufacturing is not only a formulation challenge but also a fluid-dynamics and process-control problem. Successful production depends on millisecond-scale mixing, controlled solvent exchange, diffusion kinetics, shear management and precise regulation of self-assembly. Even minor deviations can affect product quality. For this reason, the field is increasingly investing in advanced mixing platforms, real-time monitoring, AI-driven process optimization and lyophilization to improve scalability, stability and reduce dependence on cold-chain logistics.

The complexity and cost of LNP therapeutics extend beyond manufacturing into storage, distribution and real-world deployment. Thermal instability and cold-chain dependence remain major limitations for RNA-LNP products, as the COVID-19 vaccine rollout demonstrated.

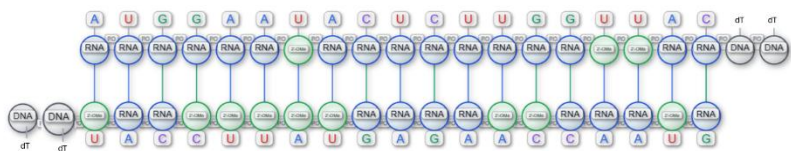
mRNA vaccines are highly temperature-sensitive because they are vulnerable to degradation through mRNA hydrolysis, lipid oxidation, RNA–lipid chemical interactions and nanoparticle aggregation, each of which can reduce potency during storage and distribution. During the pandemic, Pfizer-BioNTech’s mRNA vaccine initially required ultra-cold storage at approximately -70°C , whereas Moderna’s vaccine required frozen storage at around -20°C . These requirements created substantial logistical burdens across packaging, transportation, warehousing and administration (Fahrni et al., 2022).

For example, Pfizer-BioNTech’s vaccine had to be shipped in dry-ice containers with GPS-enabled temperature monitors, transferred rapidly to ultra-cold freezers after arrival and used within strict time limits after thawing. If storage was prolonged or disrupted, dry ice had to be replenished to maintain appropriate conditions (Fahrni et al., 2022). Such constraints required specialized freezers, validated thermal shippers, temperature-monitoring systems, trained personnel, refrigerated transport and close coordination across manufacturers, freight providers, airports, warehouses and healthcare facilities (Fahrni et al., 2022).



The COVID-19 vaccine experience therefore highlighted cold-chain dependence as both a formulation problem and an access problem. Improving RNA-LNP stability through formulation optimization, lyophilization, improved packaging and reduced temperature sensitivity is essential not only for preserving therapeutic efficacy, but also for lowering distribution costs, reducing wastage and improving equitable global access.

Example siRNA/LNP



Patisiran (Onpattro), an LNP-formulated RNAi therapeutic, established that chronic systemic delivery of siRNA is clinically feasible. However, its safety and administration burden reflect the inherent liabilities of lipid nanoparticles. In the APOLLO trial, patients treated with patisiran experienced higher rates of infusion-related reactions (19% vs 9% with placebo) and peripheral edema (30% vs 22%), along with systemic adverse events such as fatigue, diarrhea, and dizziness. These effects are consistent with known LNP-associated mechanisms, including complement activation, innate immune stimulation, and vascular responses. As a result, patisiran requires intravenous infusion and routine premedication (e.g., corticosteroids, antihistamines) to mitigate hypersensitivity and infusion reactions.

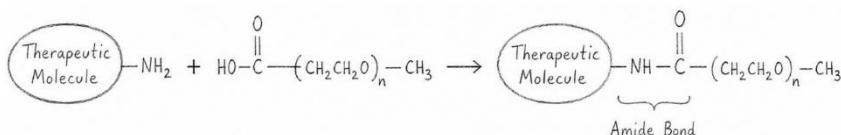
In contrast, vutrisiran is a next-generation RNAi therapeutic targeting the same gene (TTR) but uses a GalNAc conjugate rather than LNPs. This enables direct hepatocyte uptake via the asialoglycoprotein receptor, avoiding many of the systemic immune interactions associated with nanoparticle carriers. Consequently, vutrisiran can be administered as a subcutaneous injection once every three months, substantially reducing treatment burden.

Patisiran (Onpattro)	Category	Vutrisiran
TTR RNAi therapeutic	Target	TTR RNAi therapeutic
LNP-formulated siRNA	Delivery platform	GalNAc-conjugated siRNA
Systemic nanoparticle delivery	Hepatocyte uptake	Direct ASGPR-mediated uptake
Intravenous infusion	Administration	Subcutaneous injection
Repeat IV treatment visits	Dosing schedule	Once every 3 months
Required	Premedication	Not required
Infusion-related reactions observed Peripheral edema more common	Immune / infusion signal	No infusion-related reaction signal
Infusion reactions, peripheral edema, fatigue, diarrhea, dizziness	Common tolerability profile	Pain in extremity, arthralgia, dyspnea, decreased vitamin A
Higher: infusion center + monitoring + premedication	Treatment burden	Lower: simple repeat dosing paradigm

Vutrisiran demonstrates a distinct and generally improved tolerability profile. In the HELIOS-A study, the most common adverse events in patients with hATTR-PN were pain in extremity (15%), arthralgia (11%), dyspnea (7%), and decreased vitamin A levels (7%), with no signal of infusion-related reactions. Similarly, in ATTR-CM, no new safety concerns were identified. Notably, unlike patisiran, vutrisiran does not require premedication, underscoring the advantage of eliminating LNP-associated immune activation and enabling a more convenient, repeat dosing paradigm.

ONPATTRO requires weight-based IV infusion every 3 weeks, along with premedication (corticosteroids, acetaminophen, and antihistamines) and monitoring due to infusion-related reactions and broader systemic exposure. In contrast, GalNAc-conjugated siRNAs achieve targeted hepatocyte uptake via the asialoglycoprotein receptor, enabling subcutaneous administration, fixed dosing, and much less frequent schedules (monthly to every 3–6 months or longer) without routine pre-treatment. Overall, GalNAc conjugation has enabled more precise liver targeting and more convenient, less intrusive dosing compared with earlier non-targeted LNP approaches like ONPATTRO.

PEGylation



PEGylation, the covalent attachment of polyethylene glycol (PEG) to therapeutic molecules. The technology was discovered in the 1960s by Frank Davis who proposed its use to reduce the immunogenicity of protein drugs by shielding them from immune recognition and prolonging circulation time (Veronese & Pasut, 2005; Pasut & Veronese, 2012). This concept was experimentally validated in a landmark study by Abuchowski et al. (1977), which demonstrated that PEGylation of bovine liver catalase significantly decreased immunogenicity while extending plasma half-life.

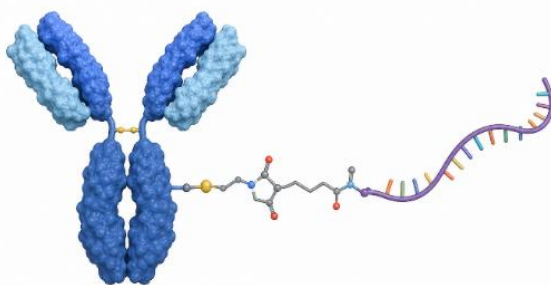
PEGylation transitioned from academic concept to commercial technology (Harris & Chess, 2003) with the FDA approval of Adagen in 1990, followed by Oncaspar in 1994. (Alconcel et al., 2011; Harris & Chess, 2003). From the mid-1990s onward, PEGylation rapidly expanded across diverse drug classes and architectures, including the development of PEGylated liposomes such as Doxil (Allen & Cullis, 2013), site-specific

mono-PEGylated proteins like Pegintron, and branched PEG systems exemplified by Pegasys (Veronese & Mero, 2008).

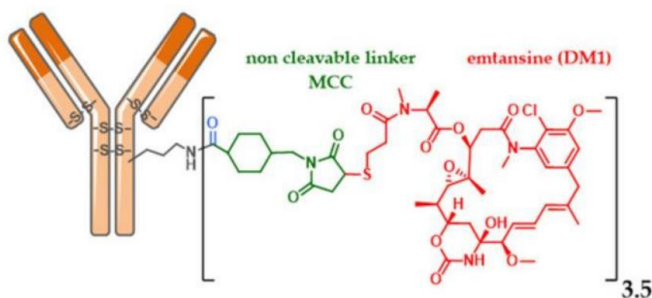
The technology further diversified to include PEGylated aptamers (Macugen™), antibody fragments (Cimzia™), peptides (Omontys), and small molecules (Movantik), reflecting its broad applicability across therapeutic modalities (Keefe et al., 2010; Fishbane et al., 2013). More recently, PEGylation has played a critical role in nanomedicine, particularly in lipid nanoparticle (LNP) systems used for nucleic acid delivery; a critical component of the Onpattro LNP and the widespread deployment of PEGylated LNP-based mRNA COVID-19 vaccines (Cullis & Hope, 2017; Hou et al., 2021).

While there are many more ligand conjugate technologies used including palmitate and antibodies most pro-drug technologies for tissue specificity is for liver with various forms of GalNAc.

Antibody conjugation



Antibody–oligonucleotide conjugates (AOCs) are an emerging delivery platform that couples the tissue specificity of monoclonal antibodies with the gene-modulating activity of oligonucleotide therapeutics, including siRNA, antisense oligonucleotides (ASOs), and related chemistries. In these constructs, oligonucleotides are covalently attached to antibodies or antibody fragments through linkers and bioconjugation strategies such as maleimide, disulfide, lysine–amine, or cysteine–thiol coupling, with the goal of promoting receptor-mediated uptake and intracellular delivery to otherwise difficult-to-access tissues (Junutula et al., 2008; Cuellar et al., 2015). Early antibody–siRNA conjugates demonstrated that chemically defined antibody conjugates could retain both antibody binding and RNA-silencing activity, although efficient



anti-HER2 Kadcyla® (ado-trastuzumab emtansine)

Kadcyla®

intracellular trafficking and endosomal escape remained important barriers to broader therapeutic translation (Cuellar et al., 2015).

Antibody-Drug Conjugates (ADC) began with first-generation constructs that used murine, chimeric, or early humanized antibodies, relatively unstable linkers, and low-potency cytotoxic payloads such as doxorubicin or duocarmycins. These agents established the feasibility of antigen-directed delivery but were limited by heterogeneous lysine conjugation, broad drug-to-antibody ratio distributions, premature payload release, immunogenicity, and narrow therapeutic indices, as seen with early approved agents such as Mylotarg® and Besponsa® (Beck et al., 2017; Tsuchikama and An, 2018; Shastry et al., 2023).

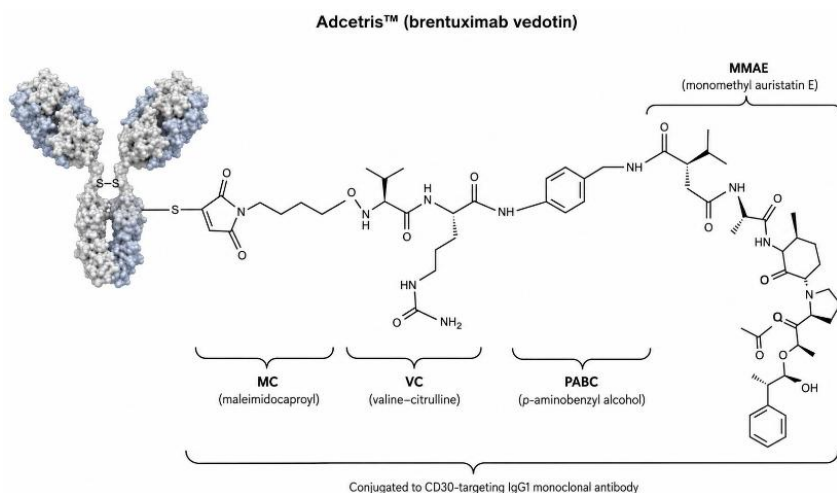
Second-generation ADCs improved this framework through more humanized antibodies, more stable linkers, and higher-potency payloads, particularly auristatins and maytansinoids. Compounds such as Adcetris®, Kadcyla®, Padcev®, and Elahere® showed that ADCs could produce meaningful clinical benefit across hematologic malignancies and solid tumors, but random lysine or reduced interchain cysteine conjugation continued to generate heterogeneous products, and high-DAR species could be associated with rapid clearance, off-target toxicity, or acquired resistance (Beck et al., 2017; Drago et al., 2021; Shastry et al., 2023).

Third-generation ADCs increasingly emphasize molecular precision: fully human antibodies or Fab-based formats, stable circulation-compatible linkers, site-specific conjugation, and defined DAR values. This engineering logic underlies newer agents such as Enhertu® and Trodelvy®, which use optimized linker–payload systems and topoisomerase I inhibitor payloads to expand activity in tumors with low or heterogeneous antigen expression, in part through bystander killing (Drago et al., 2021; Shastry et al., 2023).

Like modern ADCs, successful AOCs will likely depend on precise control of antibody format, conjugation site, linker stability, payload

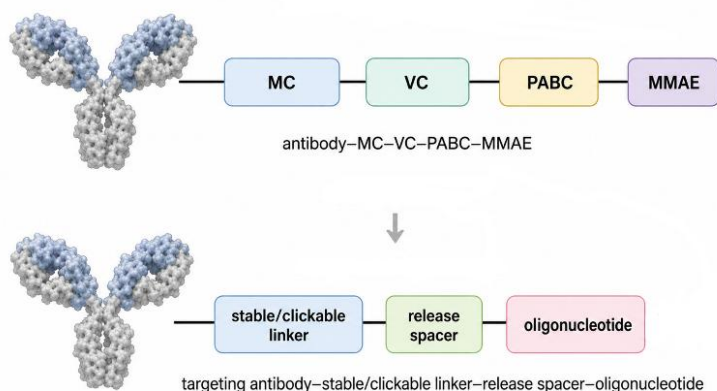
loading, and intracellular trafficking. However, AOCs face additional challenges imposed by oligonucleotide size, charge, nuclease sensitivity, endosomal escape, and the need to deliver the payload to the cytosol or nucleus rather than simply to the lysosome. Maturation of AOCs will similarly require iterative improvements in linker chemistry, site-specific conjugation, payload design, and a deeper understanding of tissue-specific uptake and intracellular processing.

Site-specific platforms, including Genentech's THIOMAB technology, improve conjugate homogeneity and pharmacokinetic performance by introducing defined cysteine attachment sites on antibodies (Nanna et al., 2020). AOCs have shown strong potential for extrahepatic delivery, especially to muscle and central nervous system tissues, where conventional oligonucleotide delivery remains difficult. For example, anti-CD71 antibody–siRNA conjugates have achieved efficient gene silencing in skeletal muscle and cardiac tissue, and companies such as Denali Therapeutics and Avidity Biosciences have advanced transferrin receptor–targeting antibody–oligonucleotide systems linked through noncleavable maleimide chemistry (Sugo et al., 2016). By exploiting receptor-mediated internalization, these approaches can improve tissue targeting, intracellular delivery, and the overall therapeutic index, positioning AOCs as a promising next-generation platform for precision RNA therapeutics (Winkler, 2013; Belgrad, Fakih and Khvorova, 2024).



Adcetris™ (brentuximab vedotin) provides a useful template for developing a new antibody–oligonucleotide conjugate (ADO/AOC) with a cleavable linker. In Adcetris, a CD30-targeting IgG1 antibody is joined to the cytotoxic payload MMAE through the MC–VC–PABC linker. The MC portion, maleimidocaproyl, serves as the antibody-attachment handle; its

maleimide group reacts with reduced cysteine thiols on the antibody to form the covalent ADC linkage. The VC segment, valine–citrulline, is a protease-cleavable dipeptide designed to remain stable in circulation but be cleaved after internalization into target cells, particularly in the lysosome. PABC, p-aminobenzyl alcohol/carbamate, acts as a self-immolative spacer that breaks down after VC cleavage to release the free payload. MMAE, monomethyl auristatin E, is the potent antitubulin drug that disrupts microtubule dynamics and kills the CD30-positive cancer cell after release. The uploaded review describes BV/Adcetris as MMAE conjugated to a CD30-targeting monoclonal antibody through a maleimidocaproyl–valine–citrulline–PABC linker.



To develop a new ADO, the same modular logic could be retained, but the small-molecule payload MMAE would be replaced with an oligonucleotide payload, such as an siRNA, ASO, splice-switching oligo or PMO. A practical strategy would be to keep an antibody-binding chemistry, such as cysteine–maleimide conjugation or a more stable next-generation cysteine linker, while replacing the MMAE attachment point with a click-compatible handle. For example, the antibody could be functionalized with an azide, alkyne, DBCO, or tetrazine-bearing linker, and the oligonucleotide could be synthesized with the complementary click handle at its 5' or 3' end.

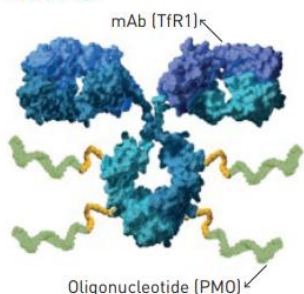
Site-specific attachment

A key technology leap from early ADC/ADO's is the cysteine conjugation where defined cysteine residues are engineered into selected

sites on the antibody heavy or light chain, creating reactive thiol handles that can be selectively coupled to thiol-reactive linker–payloads, thereby enabling site-specific attachment of oligonucleotides with controlled loading, improved conjugate homogeneity, and reduced variability compared with random lysine or native cysteine conjugation (Genentech, 2016).

While conventional antibody conjugation often relies on native lysine residues or reduced interchain cysteines, producing heterogeneous mixtures with variable payload loading and attachment sites, THIOMAB engineering introduces reactive cysteine residues at defined positions within the antibody heavy or light chain; thiol handles for site-specific linker–payload attachment. These engineered cysteines can react with thiol-selective chemistries such as maleimide, iodoacetamide, or pyridyl disulfide linkers, yielding more homogeneous conjugates with tighter control over drug-to-antibody or oligonucleotide-to-antibody ratios, reduced batch variability, and potentially improved plasma stability and therapeutic index. Although the platform was originally developed primarily for antibody–drug conjugates, the same design principles apply directly to antibody–oligonucleotide conjugates, where the placement of ASO, siRNA, or PMO payloads can strongly influence biodistribution, receptor-mediated uptake, intracellular trafficking, and manufacturability (Genentech, 2016).

Figure 1: AOC 1044, an Antibody Oligonucleotide Conjugate Targeting Exon 44 Skipping

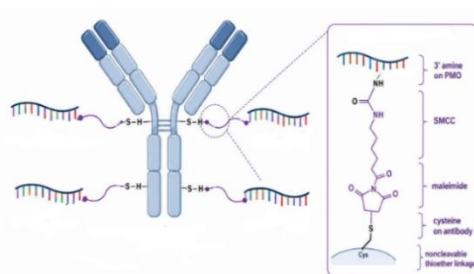


AOC 1044 as a Novel Therapeutic Approach for DMD Patients Amenable to Exon 44 Skipping: EXPLORE44™ Phase 1/2 Healthy Volunteer Data

Mark Stahl, Yiming Zhu, Varun Goel, Laura Leung, Tara Carmack, Yvonne Tami, Trista Hardin, Usue Etxaniz, Philip Kovach, Josiah Herzog, Steve Hughes, Elizabeth J. Ackermann

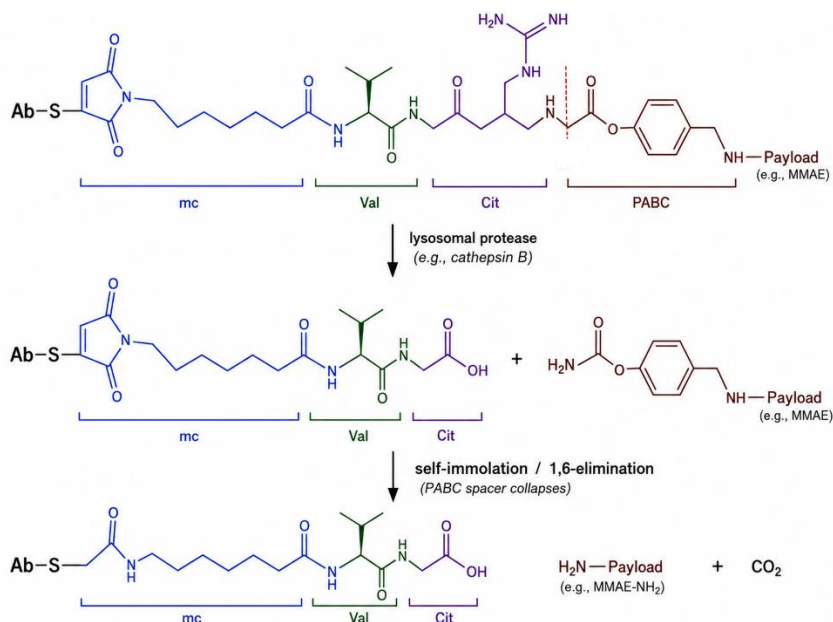
Avidity Biosciences, San Diego, CA, USA

Avidity’s AOC 1044 is a transferrin receptor 1 (TfR1)-targeted antibody–oligonucleotide conjugate developed to deliver an exon 44–skipping phosphorodiamidate morpholino oligomer (PMO) to skeletal muscle. The molecule comprises a humanized anti-TfR1 antibody conjugated to multiple copies of a PMO designed to promote exclusion of exon 44 from dystrophin pre-mRNA. In healthy volunteers, AOC 1044 showed dose-dependent delivery of PMO to skeletal muscle and statistically significant exon 44 skipping after a single dose (Stahl *et al.*, 2024).



Although the precise conjugation chemistry has not been unambiguously disclosed, the most strongly supported publicly available model for AOC 1044 is a noncleavable SMCC/maleimide–thiol architecture. In this scheme, SMCC is coupled to a primary amine at the 3' end of the PMO, generating a maleimide-functionalized PMO that is subsequently reacted with thiols exposed on the anti-human TfR antibody following TCEP-mediated reduction of cysteines. This assignment is consistent with broader AOC linker-design literature, which classifies SMCC as a noncleavable linker and specifically identifies Avidity's industrial AOC linker as SMCC (Li et al., 2025).

Linker technology



The most common clinically approved linker class is the protease-cleavable (maleimidocaproyl)-Val-Cit-PABC linker, usually written as mc-VC-PABC. This linker is used in several MMAE-based “vedotin” ADCs, including brentuximab vedotin, polatuzumab vedotin, enfortumab vedotin, tisotumab vedotin and telisotuzumab vedotin (Jang and Chang, 2025). Its common architecture contains a maleimide-based antibody attachment group, a caproyl spacer, a valine-citrulline dipeptide designed for lysosomal protease cleavage, and a self-immolative PABC spacer that facilitates release of free MMAE. Its clinical success reflects a useful combination of systemic stability and intracellular enzymatic release, although the maleimide-thiol attachment and MMAE payload are also associated with class-specific liabilities such as deconjugation concerns and peripheral neuropathy (Lin et al., 2025; Su et al., 2021).

Approved antibody-drug conjugate (ADC) linkers share a common design objective: they must keep a highly potent payload attached to the antibody during systemic circulation, while enabling payload release after antigen binding, internalization and lysosomal processing in target cells. In this sense, all clinically approved linkers are pharmacological “control elements”, not passive tethers: they influence plasma stability, exposure to free payload, intracellular release, bystander killing and toxicity (Su et al., 2021; Jang and Chang, 2025). Current approved ADCs use both cleavable

and non-cleavable linker strategies, reflecting different solutions to the same central problem of balancing stability with efficient payload delivery (Lin et al., 2025).

Approved non-cleavable linkers include Trastuzumab emtansine which uses the SMCC linker with DM1, while belantamab mafodotin uses a maleimidocaproyl linker with MMAF (Jang and Chang, 2025). In these systems, payload release depends on intracellular antibody degradation rather than a defined enzymatic linker cleavage event. This strategy can improve plasma stability and reduce premature release (Su et al., 2021; Lin et al., 2025).

Several approved ADCs use chemically cleavable linkers that respond to intracellular conditions. Gemtuzumab ozogamicin and inotuzumab ozogamicin use AcButDMH linkers with calicheamicin, historically described as acid-labile hydrazone-type systems designed to release payload in acidic endosomal/lysosomal compartments. Mirvetuximab soravtansine uses sulfo-SPDB, a disulfide-containing linker paired with DM4, making release sensitive to the intracellular reducing environment. Loncastuximab tesirine uses a PEG–Val–Ala–PABC linker with a pyrrolobenzodiazepine dimer payload, combining hydrophilic PEGylation with protease-cleavable peptide release (Jang and Chang, 2025).

Overall, the approved linker landscape differs mainly along four axes: cleavable versus non-cleavable release, trigger mechanism (protease, acid, reduction, hydrolysis) or antibody degradation payload compatibility, and bystander potential. Protease-cleavable peptide linkers dominate many modern solid-tumor ADCs because they can combine stable circulation with efficient lysosomal release. Non-cleavable linkers favor systemic stability. Hydrazone, disulfide and hydrolysable linkers provide alternative release mechanisms but can raise different stability and toxicity trade-offs.

Cell penetrating peptides

The cell membrane protects the interior of every living cell with remarkable precision. It permits small nutrients, ions, and selected signals to pass, yet it excludes many molecules that researchers would like to deliver for diagnosis, imaging, and therapy. For decades, this barrier limited the therapeutic use of large or fragile cargoes, including proteins, peptides, nucleic acids, siRNA, nanoparticles, and imaging agents. Cell-penetrating peptides changed that story by offering a way to carry biological cargo across the plasma membrane and into the cell.

Researchers first recognized this possibility in 1988 while studying the HIV-1 transactivator of transcription protein, Tat. They observed that Tat could enter mammalian cells and influence viral gene expression,

suggesting that part of the protein possessed an unexpected membrane-crossing ability (Guo et al., 2016; Gori et al., 2023). This finding challenged the assumption that proteins required classical receptors, channels, or disruptive delivery methods to gain intracellular access. It also raised an important question: could short protein regions themselves behave as transport modules?

In 1991, Alain Joliot and colleagues, working at the Centre National de la Recherche Scientifique in Paris, showed that a 60-amino-acid peptide from the *Drosophila melanogaster* Antennapedia homeodomain could enter mammalian nerve cells. This peptide retained the biological activity expected of an Antennapedia homeodomain because it bound a homeobox consensus DNA sequence from the Hox-1.3 promoter in a gel-shift assay.

This work then led researchers to ask which part of the 60-amino-acid homeodomain was responsible for cell entry. Later studies showed that uptake was lost when researchers mutated the C-terminal region of the homeodomain, directing attention to the third α -helix in that region. In 1994, Derossi, Joliot and colleagues chemically synthesized shorter peptides from this third helix and tested whether they could still cross cell membranes. They found that 20-amino-acid and 16-amino-acid fragments retained membrane-crossing ability, whereas shorter 15-amino-acid fragments did not. The active 16-amino-acid sequence, RQIKIWFQNRRMKWKK, later became known as penetratin or Antp.

Antennapedia homeodomain:

RKRGRQTYTRYQTLELEKEFHFNRYLTTRRRRIEIAHALCLTERQIKI
WFQNRRMKWKKEN

Antp/penetratin, derived from the third helix of the Antennapedia homeodomain:

RQIKIWFQNRRMKWKK

Cell-penetrating peptides, or CPPs, now describe a diverse family of short peptides, commonly around 5–40 amino acids long, that can move across cellular membranes and transport attached or associated cargo. Many CPPs carry a positive charge at physiological pH because they contain arginine and lysine residues. These amino acids help them interact with negatively charged membrane components, including phospholipids and

heparan sulphate proteoglycans; positively charged CPPs bind to negatively charged heparan sulphate proteoglycans on the cell surface, which helps start cellular uptake. Other CPPs use amphipathic or hydrophobic structures to associate with lipid bilayers and promote internalization. (Gori et al., 2023; Guo et al., 2016).

Scientists classify CPPs in several overlapping ways. Some arise from natural proteins, such as Tat, penetratin, VP22, lactoferrin-derived peptides, and toxin-derived sequences. Others combine natural and synthetic elements to form chimeric peptides, while fully synthetic CPPs allow researchers to design charge, stability, amphipathicity, and targeting properties from the start. Researchers also group CPPs by physicochemical behavior, including cationic, amphipathic, hydrophobic, and even anionic peptides (Gori et al., 2023).

CPPs enter cells through more than one route. Some cross membranes directly through energy-independent mechanisms such as pore formation, membrane thinning, carpet-like disruption, or inverted micelle formation. Others enter through energy-dependent endocytosis, including macropinocytosis, clathrin-mediated endocytosis, caveolae-mediated endocytosis, and related pathways. The exact route depends on the CPP sequence, cargo size, concentration, incubation time, temperature, cell type, and membrane composition. Because many CPP-cargo complexes enter endosomes first, endosomal escape remains one of the central challenges in CPP-mediated delivery (Gori et al., 2023; Yang, Liu and Sha, 2025).

Their promise lies in their ability to move difficult cargoes into cells. Researchers have used CPPs to deliver anticancer drugs, proteins, oligonucleotides, siRNA, plasmid DNA, imaging probes, quantum dots, nanoparticles, and vaccine antigens. In cancer research, CPPs have gained attention because they can improve intracellular delivery and tumor penetration. Recent work highlights their role in tumor immunotherapy, where CPPs may support cytokine therapy, adoptive cell therapy, cancer vaccines, immune checkpoint blockade, and tumor-targeted nanocarriers. Peptides such as iRGD show promise because they can improve tumor tissue penetration and help therapeutic agents reach poorly accessible tumor regions (Yang, Liu and Sha, 2025).

Despite this promise, CPPs still face important barriers before broad clinical translation. Many CPPs lack tissue specificity, and uncontrolled uptake may increase off-target effects. Some peptides show limited stability in biological fluids, while others struggle with poor endosomal release after cellular uptake. Researchers therefore continue to improve CPP design through cyclisation, stapling, retro-inverso modification, dendrimeric structures, tumor-targeting motifs and nanoparticle conjugation. These strategies aim to increase stability, reduce toxicity, enhance selectivity and

improve cargo release inside target cells (Gori et al., 2023; Yang, Liu and Sha, 2025).

DaxibotulinumtoxinA-lanm, marketed as DAXXIFY®. DAXXIFY is a botulinum toxin type A product that acts as an acetylcholine-release inhibitor and neuromuscular-blocking agent. It is approved for the temporary improvement of moderate-to-severe glabellar lines and for the treatment of cervical dystonia in adults. Unlike many botulinum toxin formulations that use human serum albumin as a stabilizing component, DAXXIFY is formulated with RTP004, a proprietary 35-amino-acid peptide excipient.

RTP004 contains highly basic, Tat-like protein-transduction domains. Its reported sequence, RKKRRQRRRGKKKKKKKKKKKKKKKKKKKKGRKKRRQRRR, includes repeated RKKRRQRRR motifs, which closely resemble the classic HIV-Tat cell-penetrating sequence. These positively charged domains flank a polylysine core, giving RTP004 strong cationic character like many CPPs.

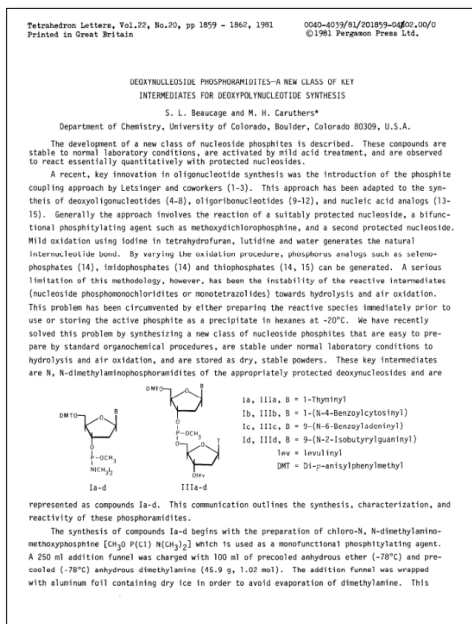
Ionis Pharmaceuticals and Bicycle Therapeutics entered a collaboration in 2021 to test whether Bicycle's constrained peptide technology could improve the targeted delivery of Ionis' oligonucleotide therapeutics. Drug delivery is especially difficult for skeletal and cardiac muscle, where systemically administered antisense oligonucleotides and siRNAs must leave the bloodstream, cross tissue barriers and enter cells before they can engage intracellular targets. Conventional oligonucleotides show limited potency in these tissues.

A 2025 study from Ionis and Bicycle identified BCY17901, a small bicyclic peptide of approximately 2 kDa that binds human transferrin receptor 1 (TfR1), a recycling receptor expressed on skeletal muscle cells and cardiomyocytes. BCY17901 was conjugated to ASO and siRNA to improve receptor-mediated uptake into muscle and heart. In human TfR1 knock-in mice, BCY17901 conjugation improved ASO potency by 5.2–8.4-fold in quadriceps, 6.9–7.6-fold in gastrocnemius and at least 5.8-fold in heart compared with unconjugated ASOs. For siRNA, BCY17901 conjugation improved potency by 5.5-fold in quadriceps, 4.7-fold in gastrocnemius and 2.7-fold in heart compared with a lipid-conjugated siRNA control.

The approach translated into nonhuman primates, despite BCY17901 having about 10-fold lower affinity for cynomolgus monkey TfR1 than for human TfR1. BCY17901-conjugated MALAT1 ASO reduced target RNA by 71% in quadriceps and 63% in heart. BCY17901-conjugated HPRT siRNA reduced target RNA by 86% in quadriceps and 75% in heart. The study estimated that delivering a hypothetical 3 mg/kg oligonucleotide-equivalent dose would require roughly 250–300 mg of a Bicycle–ASO

conjugate, compared with about 6000 mg for a monoclonal antibody–ASO conjugate.

ASO Synthesis

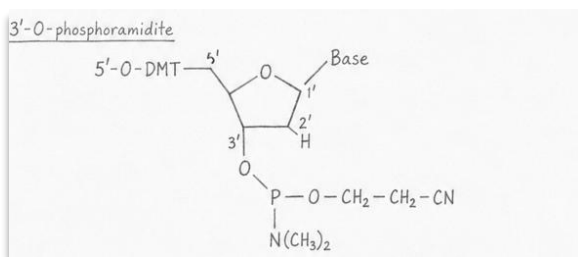


The development of antisense oligonucleotides depended on major advances in chemical synthesis that made production practical, reproducible, and ultimately automated. Modern ASO synthesis emerged from a series of innovations in solid-phase oligonucleotide chemistry, particularly work carried out by Robert L. Letsinger at Northwestern University and by Marvin H. Caruthers and colleagues at the University of Colorado Boulder (Egli, 2023; Beaucage and Caruthers, 1981).

Robert L. Letsinger, a chemist at Northwestern University from 1946 until his retirement in 1991, made foundational contributions to synthetic DNA chemistry (Egli, 2023). His work helped establish several central steps in oligonucleotide synthesis, including the solid-phase approach, the phosphotriester method and the P(III) phosphite–triester method. These developments were later refined into the phosphoramidite technology that became central to automated DNA and RNA synthesis (Egli, 2023).

A major technical challenge in early oligonucleotide synthesis was the efficient formation of internucleotide phosphodiester linkages. Letsinger

and co-workers introduced the phosphite coupling approach, in which protected nucleosides were reacted through phosphorus(III) intermediates and subsequently oxidized to form natural phosphate internucleotide bonds (Beaucage and Caruthers, 1981). This strategy represented an important advance because it allowed more efficient coupling than earlier approaches; however, the reactive intermediates, such as nucleoside phosphomonochloridites and monotetrazolides, were unstable toward hydrolysis and air oxidation (Beaucage and Caruthers, 1981). As a result, the reactive species often had to be prepared immediately before use or stored under restrictive conditions, limiting the practicality of the method for routine or automated synthesis (Beaucage and Caruthers, 1981).



In 1981, Serge L. Beaucage and Marvin H. Caruthers, working in the Department of Chemistry at the University of Colorado, Boulder, reported a new class of deoxynucleoside phosphoramidite intermediates (Beaucage and Caruthers, 1981). Their paper described N,N-dimethylaminophosphoramidites of protected deoxynucleosides as stable compounds that could be prepared by standard organic chemistry procedures and stored as dry powders under normal laboratory conditions (Beaucage and Caruthers, 1981). This solved a central limitation of earlier phosphite chemistry: the instability of the activated phosphorus intermediates. Caruthers demonstrated that these phosphoramidites could be activated under mild acidic conditions, particularly using 1H-tetrazole, and could condense with protected nucleosides to form internucleotide phosphite linkages in essentially quantitative yield before oxidation to the natural phosphate form (Beaucage and Caruthers, 1981).

Beaucage and Caruthers also showed that the new reagents were compatible with polymer-supported synthesis, an essential requirement for automated oligonucleotide production. In trial reactions using nucleoside-functionalized silica gel, the phosphoramidite reagents produced dinucleotides including d(TpG), d(CpG), d(ApG) and d(GpG) in high yields of 100%, 98%, 94% and 93%, respectively, after oxidation and deprotection

(Beaucage and Caruthers, 1981). These results demonstrated that phosphoramidite chemistry could support efficient stepwise chain extension on a solid support, providing the chemical foundation for automated oligonucleotide synthesis and, later, for the scalable production of ASOs (Beaucage and Caruthers, 1981; Egli, 2023).

By the late twentieth century, oligonucleotide chemistry had moved from a specialized synthetic process to an industry standard. Letsinger's early work at Northwestern established key principles of solid-phase and phosphite-based synthesis, while Beaucage and Caruthers' work at Colorado provided stable phosphoramidite building blocks that made high-yield, routine and automated synthesis feasible (Beaucage and Caruthers, 1981; Egli, 2023). These developments made it possible to prepare cost-effective ASOs with defined sequence, length and chemical composition, enabling their later application as research tools and therapeutic agents.

Automation of ASO synthesis

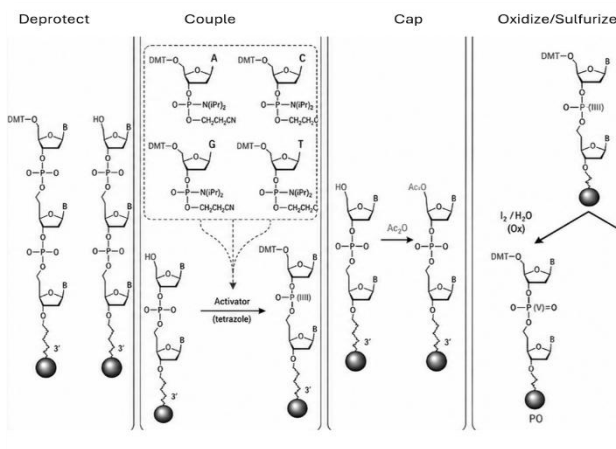
The automation of antisense oligonucleotide synthesis developed from the earlier transition between manual phosphoramidite chemistry and computer-controlled solid-phase oligonucleotide synthesis. Once Beaucage and Caruthers introduced stable deoxynucleoside phosphoramidite intermediates in 1981, oligonucleotide synthesis became suitable for repeated, programmed chemical cycles because each nucleotide addition could be carried out through a predictable sequence of reactions on a solid support (Beaucage and Caruthers, 1981). This chemical reliability was essential for automation; a synthesizer depends on the reproducibility of each cycle of 1) deprotection, 2) coupling, 3) capping and 4) oxidation. This four-step cycle is the basis of modern column-based oligonucleotide synthesis, in which nucleotide monomers are progressively added to solid-phase supports such as controlled pore glass or polystyrene beads.

In automated column-based synthesis, the growing oligonucleotide chain is attached to a solid support inside a synthesis column. A computer-controlled system then delivers reagents in a fixed order. First, the protecting group on the terminal nucleotide is removed during deprotection; second, the next phosphoramidite monomer is coupled to the growing chain; third, unreacted chains are capped to prevent them from participating in later cycles; and fourth, the newly formed phosphite linkage is oxidized to the more stable phosphate linkage (Ma et al., 2024). After the desired sequence has been assembled, the oligonucleotide is cleaved from the support under alkaline conditions and purified by methods such as oligonucleotide purification columns, polyacrylamide gel electrophoresis or high-performance liquid chromatography (Ma et al., 2024).

The emergence of automated DNA synthesizers in the 1980s therefore represented a practical turning point in ASO production. Ma et al. (2024) states that Applied Biosystems developed one of the first automated DNA synthesizers using the column-based method, improving the accessibility of synthetic oligonucleotides. These instruments used synthesis columns as support and computer-programmed solenoid valves to control the flow of reagents into the column, enabling the programmed synthesis of single-stranded DNA (Ma et al., 2024). For ASO synthesis, this automation was especially important because antisense molecules must be produced with a defined sequence, length and chemical composition. Manual synthesis made this possible only on a limited scale, whereas automated synthesis made routine and reproducible production feasible.

Automation also improved the scale and throughput of oligonucleotide production. Commercial column-based instruments, including the MerMade series from BioAutomation and the Dr. Oligo series from Biolytic, are identified by Ma et al. (2024) as widely used examples of automated oligonucleotide synthesis platforms. The same review notes that modern automated column-based synthesis can support throughput ranging from 96 to 1536 oligonucleotides, while reducing the cost of sequences under 100 bases to approximately US\$0.05–0.15 per nucleotide and lowering error rates to approximately one mutation per 200 base pairs or less (Ma et al., 2024). These improvements were significant for ASOs because therapeutic and research-grade antisense molecules require consistent quality, controlled impurities and scalable production.

The automation of ASO synthesis can therefore be understood as both a chemical and engineering achievement. Chemically, phosphoramidite synthesis supplied stable monomers and efficient coupling reactions. Technologically, automated synthesizers supplied timed reagent delivery, controlled washing steps and repeatable reaction cycles. Together, these advances allowed synthetic oligonucleotides to move from specialist laboratory products to widely available research, diagnostic and pharmaceutical materials. Ma et al. (2024) explicitly links the automation and commercialization of phosphoramidite chemistry in the 1980s with the breakthrough development of oligonucleotide synthesis, noting that oligonucleotides are now used across scientific research, clinical research, pharmaceutical development and molecular diagnostic probe synthesis.



Automated phosphoramidite synthesis also has technical limitations. Although it is mature and reliable, the synthesis of oligonucleotides longer than approximately 200 bases remain difficult; even very high per-cycle efficiencies produce accumulated errors across many cycles. The practical synthesis limit is often closer to 100 bases, depending on cost and quality requirements (Ma et al., 2024). ASO manufacturing still faces challenges associated with reagent cost, organic solvent use, toxic chemical waste and the need for purification after synthesis (Ma et al., 2024).

More recent forms of automation have expanded beyond conventional single-column synthesis. Microarray-based oligonucleotide synthesis allows large numbers of distinct oligonucleotides to be produced in parallel on silicon chips or glass slides. According to Ma et al. (2024), these platforms use methods such as photolithography, electrochemical synthesis and ink-jet printing to generate large numbers of oligonucleotide sequences while reducing reagent use and cost. For example, the review describes photolithographic platforms capable of synthesizing up to 500,000 oligonucleotides on a single chip, ink-jet systems that can produce hundreds of thousands to over one million oligonucleotides in parallel, and electrochemical systems capable of synthesizing millions of unique oligonucleotides (Ma et al., 2024). Although these methods are more commonly associated with oligonucleotide libraries and gene synthesis rather than individual therapeutic ASO manufacture, they show how automation has shifted the field toward high-throughput, miniaturizing and digitally controlled synthesis.

Enzymatic Synthesis

Instead of relying on chemical coupling, enzymatic approaches use polymerases, ligases or enzyme cascades to form phosphodiester bonds under milder, usually aqueous, conditions (Pichon and Hollenstein, 2024; Zhou et al., 2025). Polymerase-mediated synthesis can, in principle, generate much longer sequences than chemical synthesis, as shown by *in vitro* transcription methods used to produce long modified mRNA molecules (Pichon and Hollenstein, 2024). However, natural polymerases are usually template-dependent, meaning they copy an existing nucleic acid template rather than freely writing an arbitrary user-defined sequence. This creates a central challenge: enzymatic synthesis must be made programmable if it is to compete with the sequence control of phosphoramidite chemistry.

While controlled enzymatic DNA synthesis is relatively advanced, RNA synthesis remains more difficult. The key structural complication is the presence of both 2'-OH and 3'-OH groups in ribonucleotides. Blocking only the 3'-OH can allow later incorporation of therapeutically relevant 2' modifications, but selective 3' protection is synthetically challenging. Blocking the 2',3'-cis-diol is easier and can reduce unwanted 2'-5' branching, but it limits access to 2'-modified therapeutic RNA structures (Pichon and Hollenstein, 2024).

Recent work has shown that controlled enzymatic RNA synthesis is possible using suitable combinations of reversible terminator nucleotides and template-independent polymerases such as poly(U) polymerase (PUP) and poly(A) polymerase (PAP). Pichon and Hollenstein (2024) describe the use of 3'-O-allyl and 3'-O-propargyl ribonucleotides with engineered PUP variants, achieving high coupling yields across multiple synthetic cycles. Similarly, Zhou et al. (2025) reports that CID1 poly(U) polymerase mutants can support aqueous, stepwise RNA synthesis using reversible terminator nucleotides bearing 3'-O-allyl ether blocking groups. Together, these studies suggest that enzymatic RNA synthesis is moving beyond proof-of-principle toward a credible platform technology. Despite this progress, the approach remains less mature than enzymatic DNA synthesis and far less established than conventional phosphoramidite chemistry.

XNA and chemically modified oligonucleotides

A major future goal is the enzymatic synthesis of chemically modified oligonucleotides, including xeno nucleic acids (XNAs), locked nucleic acids (LNAs), threose nucleic acids (TNAs), hexitol nucleic acids (HNAs), phosphorothioates and 2'-modified RNAs. These modifications are especially important in therapeutic oligonucleotides because they can

improve nuclease resistance, binding affinity, pharmacokinetics and biological activity (Pichon and Hollenstein, 2024). However, modified nucleotides are often poor substrates for natural polymerases, and suitable reversible protecting groups are harder to design for non-natural sugar or backbone structures.

Enzyme cascades and ligase-based strategies

The future of enzymatic oligonucleotide synthesis is unlikely to rely on polymerases alone. Enzyme cascades and ligase-based strategies offer complementary routes, particularly for modified therapeutic oligonucleotides. Pichon and Hollenstein (2024) discuss an isothermal biocatalytic method in which a modified oligonucleotide is generated from a self-priming hairpin template using a polymerase and an endonuclease. This cascade can produce clinically relevant oligonucleotides at up to 100 mg scale and has a reported process mass intensity of 530 kg raw material per kg product, compared with more than 4000 kg for phosphoramidite synthesis (Pichon and Hollenstein, 2024).

Ligases also provide a promising alternative because they join shorter oligonucleotide fragments rather than adding one nucleotide at a time. DNA and RNA ligases can tolerate some nucleobase, sugar and backbone modifications, making them useful for assembling longer or heavily modified sequences (Pichon and Hollenstein, 2024). This is particularly relevant for oligonucleotide therapeutics, where the target molecule may contain many precisely positioned modifications. In the longer term, hybrid systems combining polymerases, ligases, endonucleases, microarrays and automated reagent delivery may offer the most flexible approach to manufacturing diverse oligonucleotide products.

The evolution of enzymatic oligonucleotide synthesis reflects a broader shift from purely chemical manufacturing toward controlled biocatalysis. Phosphoramidite chemistry remains indispensable because of its reliability, modification compatibility and industrial maturity. However, enzymatic synthesis offers attractive advantages, including milder aqueous conditions, potential reduction in hazardous waste, compatibility with automation and the ability to access longer sequences (Pichon and Hollenstein, 2024; Zhou et al., 2025).

At present, controlled enzymatic DNA synthesis is the most advanced area, especially through TdT-based systems using reversible terminators or polymerase–nucleotide conjugates. RNA synthesis is emerging rapidly but still faces challenges in protecting-group design and polymerase compatibility. XNA and heavily modified therapeutic oligonucleotides

remain the most difficult frontier. Future progress will likely depend on three converging advances: better reversible blocking groups, more highly engineered polymerases and integration with high-throughput synthesis technologies such as inkjet printing, chips, photochemistry and electrochemistry. As Zhou et al. (2025) suggest, combining enzymatic methods, microarrays and digital technologies may create the next generation of DNA synthesis platforms, while Pichon and Hollenstein (2024) argue that combining polymerase- and ligase-based strategies could ultimately enable the production of oligonucleotides of many types, lengths and modification patterns.

Discovery

High-throughput screening is central to modern therapeutic discovery because it allows large numbers of molecular candidates to be tested rapidly for binding, activity, specificity, delivery, and safety. In RNA therapeutics, screening strategies vary widely depending on the modality: aptamers are enriched from vast randomized nucleic-acid libraries, oligonucleotide drugs are selected from sequence-designed panels, mRNA therapeutics require optimization of both construct and delivery system, and antibody discovery relies on large immune or display libraries. Although these platforms differ in chemistry and workflow, they share a common goal: converting enormous molecular diversity into a small set of candidates with the potency, selectivity, manufacturability, and biological performance needed for therapeutic development.

Aptamers

SELEX is a high-throughput screening method for identifying DNA or RNA aptamers from very large randomized nucleic-acid libraries. In a typical SELEX screen, the starting library contains approximately 10^{13} – 10^{15} unique oligonucleotide sequences, meaning that trillions of candidate binders can be screened in parallel against a target molecule (Li et al., 2023). The target may be a purified protein, small molecule, whole cell, virus, tissue sample, or immobilized ligand. Binding sequences are retained, while non-binding sequences are removed by washing, filtration, magnetic-bead capture, affinity chromatography, nitrocellulose partitioning, or capillary electrophoresis (Li et al., 2023).

After each binding round, retained sequences are amplified by PCR for DNA aptamers or by reverse transcription, PCR and transcription for RNA aptamers. Screening stringency is increased over successive rounds by reducing target concentration, shortening incubation time, increasing wash intensity, or adding competitor molecules. Counter-screening can also be used to remove aptamers that bind related but unwanted targets. Modern high-throughput SELEX often combines the enrichment process with next-generation sequencing, allowing researchers to track millions of sequence reads and identify enriched aptamer families rather than relying only on endpoint cloning (Darmostuk et al., 2015; Li et al., 2023).

For synthesis, SELEX libraries are generally produced as randomised DNA oligonucleotide pools by commercial oligo vendors. Examples include IDT oPools, which support pooled DNA oligos of 40–350 bases and up to 20,000 oligos per pool, and Twist Bioscience Oligo Pools, which use silicon-based DNA synthesis to generate hundreds of thousands of oligos per run for complex screening libraries (IDT, 2019; Twist Bioscience, 2026). Custom SELEX services are also offered by specialist CROs such as Creative Biolabs, CD Bioparticles and Novaptech, which provide aptamer screening against proteins, cells and other target formats.

RNA-targeting therapeutic screening

High-throughput screening of RNA-targeting therapeutics depends on whether the candidates are oligonucleotide drugs or small molecules that bind RNA. For oligonucleotide drugs such as siRNAs, antisense oligonucleotides, splice-switching oligonucleotides and microRNA mimics or inhibitors, screening usually begins with computational design of large sequence panels complementary to the target RNA; La Jolla Labs PTx software is a common platform for design. These panels may range from dozens to hundreds of candidates for a single transcript to genome-wide RNAi libraries covering tens of thousands of genes.

These candidates are usually screened in multi-well plate formats after transfection into cells. The readouts include qPCR, RT-qPCR, RNA sequencing, reporter-gene activity, western blotting, ELISA, fluorescence imaging, high-content microscopy and cell-viability assays. Hits are selected based on knockdown efficiency, splice correction, protein reduction, potency, durability and low cytotoxicity. Secondary screens assess off-target effects, immune stimulation, transcriptome-wide specificity and activity in disease-relevant cell models (Roberts, Langer and Wood, 2020; Hofman and Corey, 2023). Commercial RNAi screening libraries and synthesis platforms are available from Thermo Fisher Scientific, Horizon Discovery/Revvity, MilliporeSigma MISSION, Dharmacon/Horizon, Qiagen and other

oligonucleotide vendors; these vendors offer predesigned, custom, arrayed or pooled siRNA, shRNA and microRNA libraries for high-throughput functional genomics screens (Thermo Fisher Scientific, 2026; Horizon Discovery, 2026; MilliporeSigma, 2026).

For small molecules that target RNA, the screen size depends heavily on the platform. Conventional biochemical or cell-based high-throughput screening often tests thousands to tens of thousands of compounds. For example, Life Chemicals markets RNA-focused screening libraries containing more than 29,600 drug-like compounds, while other HTS collections such as ChemBridge or Maybridge commonly contain around 50,000 compounds or more (Life Chemicals, 2025; Thermo Fisher Scientific, 2026; Northwestern HTAL, 2026). Alternative methods such as small-molecule microarrays can screen immobilised compound libraries against labelled RNA targets; Arrayjet's SMM-10 platform contains 10,000 pre-selected compounds (Arrayjet, 2026). At the ultra-large scale, DNA-encoded libraries can screen millions to billions of small molecules in pooled affinity-selection formats, and DEL approaches have been applied to RNA targets such as HIV-1 TAR RNA (Peterson and Liu, 2023; Chen et al., 2022).

mRNA therapeutic screening

High-throughput screening for mRNA therapeutics usually focuses on two linked screening problems: screening the mRNA construct and screening the delivery formulation. Construct screening compares mRNA designs that vary in 5' cap structure, 5' and 3' untranslated regions, coding-sequence codon optimization, modified nucleosides, open reading frame design, poly(A) tail length and purification method. These variants are tested in cell-based assays to measure protein expression, duration of expression, mRNA stability, innate immune activation and cytotoxicity (Sahin, Karikó and Türeci, 2014; Hou et al., 2021).

The largest screening bottleneck is often the delivery system, especially lipid nanoparticles. Automated formulation platforms can screen hundreds to thousands of lipid nanoparticle formulations. One published high-throughput mRNA-LNP platform reports screening up to 384 formulations per plate and states that several plates per run can support comparison of hundreds or thousands of novel LNPs (Lokugamage et al., 2022). Another in vivo barcoded LNP study screened over 400 nanoparticle formulations containing 384 unique ionisable lipids in only nine mice, using peptide-encoding mRNA barcodes and LC-MS/MS readout (Rhym et al., 2023).

Common technologies include automated liquid handling, microfluidic LNP mixing, reporter mRNA expression assays, luciferase or fluorescent-protein readouts, flow cytometry, high-content imaging, cytokine assays, LC-MS/MS barcode quantification and in vivo biodistribution studies. Vendors and platforms used in this space include TriLink BioTechnologies and GENEWIZ/Azenta for custom mRNA synthesis, Cytiva/Precision NanoSystems NanoAssemblr systems for microfluidic nanoparticle formulation, PreciGenome NanoGenerator Flex-S Plus, which advertises 96 samples per hour, and Unchained Labs Sunscreen, which automates screening of up to 96 formulations (TriLink BioTechnologies, 2026; Azenta, 2026; PreciGenome, 2026; Unchained Labs, 2026).

Antibody screening

High-throughput antibody screening is used to identify antibodies with the desired binding, specificity and biological function. Screening can begin with hybridomas, immunised-animal B cells, human B-cell repertoires, single-cell B-cell sequencing, phage display, yeast display, mammalian display or synthetic antibody libraries. In display-based antibody discovery, the number of antibody variants screened can be extremely large. Modern phage-display libraries commonly contain around 10^{10} – 10^{11} antibody variants, with published therapeutic antibody libraries ranging from 1.5×10^{10} to 3.6×10^{11} variants (Almagro et al., 2019).

In phage display, antibody fragments such as scFv or Fab molecules are displayed on bacteriophage particles and panned against immobilized or solution-phase antigen. Non-binding phage are removed, while antigen-binding phage are recovered, amplified in bacteria and subjected to additional rounds of selection. Yeast display and mammalian display allow antibody binding to be screened by fluorescence-activated cell sorting, often enabling selection by antigen-binding strength, expression level and cross-reactivity. After primary screening, individual hits are tested by ELISA, flow cytometry, surface plasmon resonance, biolayer interferometry, homogeneous fluorescence assays or cell-based functional assays (Hoogenboom, 2005; Almagro et al., 2019).

Secondary antibody screens measure affinity, epitope binning, receptor blocking, ligand neutralisation, immune-cell recruitment, antibody-dependent cellular cytotoxicity, complement activation, internalisation, species cross-reactivity and developability. Developability screening often includes expression titre, aggregation risk, thermal stability, hydrophobicity, polyspecificity and manufacturability. Vendors and technologies commonly used include GenScript for high-throughput gene-to-antibody production,

Sino Biological and Creative Biolabs for phage-display and antibody-discovery services, Carterra LSA, Sartorius Octet BLI, Cytiva Biacore SPR, Berkeley Lights/Bruker Beacon and Sphere Fluidics Cyto-Mine for high-throughput antibody characterisation or single-cell screening. GenScript describes high-throughput gene-to-antibody production using high-throughput gene synthesis and transient antibody expression, with standard projects completed in as few as 18 business days (GenScript, 2026).

Chemistry, Manufacturing, and Controls (CMC)

The chemistry, manufacturing, and controls (CMC) framework for RNA therapeutics must accommodate two related but distinct development paradigms. For large indications, the CMC program must ultimately support reproducible, scalable, commercial manufacturing, validated analytical methods, justified specifications, long-term stability, process robustness, and lifecycle management. For ultra-rare and N-of-1 therapies, the CMC program may be more compressed and risk-based, particularly where the ASO belongs to a well-characterized chemical class, is manufactured using conventional methods, and is formulated as a simple aqueous solution or lyophilized powder for reconstitution (FDA, 2021). In both settings, however, the central CMC objective is the same: to ensure that the ASO drug substance and drug product are consistently identifiable, pure, potent, stable, and suitable for clinical administration.

Simple ASOs without conjugation

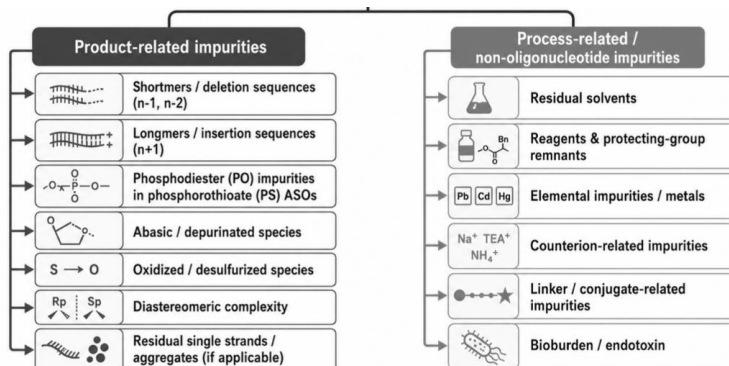
ASO CMC begins with precise definition of the drug substance. Unlike many small molecules, an ASO cannot be fully described by molecular formulas alone. The sequence, chain length, backbone chemistry, sugar modifications, nucleobase composition, terminal modifications and salt form must be defined and controlled. For phosphorothioate-containing ASOs, stereochemistry is also an important consideration because each phosphorothioate linkage introduces a chiral phosphorus center. Conventional phosphorothioate ASOs are often manufactured as reproducible mixtures of diastereomers rather than as single stereochemically pure entities. Consequently, the CMC control strategy must demonstrate consistency of the resulting product profile rather than necessarily resolving every stereoisomer (EMA, 2024).

Most ASOs are manufactured by solid-phase oligonucleotide synthesis, commonly using phosphoramidite chemistry. The standard synthetic cycle includes deprotection or detritylation, coupling of a protected nucleotide building block, oxidation or sulfurization to generate the intended internucleotide linkage, and capping of unreacted chains. Following chain assembly, the oligonucleotide is cleaved from the solid support and deprotected to generate the crude product. The core drug substance workflow then typically proceeds through purification, isolation, and, where applicable, lyophilization or preparation of the active substance in solution (Chen et al., 2026; EMA, 2024). Common purification approaches include anion-exchange chromatography, ion-pair reversed-phase chromatography, hydrophobic interaction chromatography,

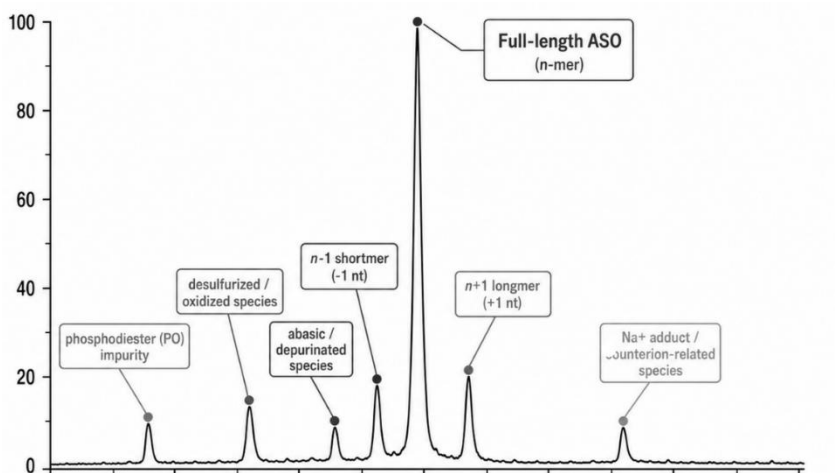
ultrafiltration/diafiltration, desalting, and combinations of orthogonal methods.

A major feature of ASO CMC is the need to understand and control both product-related and process-related impurities. Product-related impurities may include shortmers, longmers, deletion sequences, insertion sequences, phosphodiester impurities in phosphorothioate ASOs, depurinated or abasic species, oxidized or desulfurized products, cross-linked species, and degradation products formed during manufacture or storage. Non-oligonucleotide impurities may include residual solvents, reagents, protecting groups, elemental impurities, counterion-related impurities, and impurities derived from linkers, solid supports, or conjugation chemistry. Because many ASO-related impurities are structurally similar to the parent molecule, and because some may co-elute or differ only subtly in mass or sequence position, a single analytical method is rarely sufficient to support a complete control strategy (EMA, 2024).

Analytical characterization is central to ASO development. Mass spectrometry, including LC-MS and tandem MS, is commonly used for molecular weight determination, sequence confirmation, and impurity identification. Chromatographic methods such as ion-pair reversed-phase HPLC, anion-exchange HPLC, and other orthogonal separation methods are used for purity, related substances, assay, and in-process controls. Additional techniques may include UV spectroscopy, melting temperature analysis, NMR, capillary electrophoresis, elemental analysis, Karl Fischer water determination, GC for residual solvents, and compendial microbiological and endotoxin methods. For drug product, especially injectable presentations, CMC controls extend to pH, osmolality, visible and subvisible particulates, sterility assurance, container-closure integrity, extractables and leachables, and compatibility with the final dosage form (FDA, 2021; EMA, 2024).



For large-population ASO therapeutics, CMC development typically matures progressively from early clinical supply to a robust commercial control strategy. Early-stage development may rely on platform knowledge, qualified methods, provisional specifications, and limited stability data sufficient to support clinical investigation. As development advances toward registration, the process must become more product-specific and better characterized. This includes identification of critical quality attributes, understanding of critical process parameters, establishment of in-process controls, validation of analytical procedures, justification of specifications, generation of representative batch data, stability studies supporting shelf life, and demonstration that the intended commercial manufacturing process can consistently deliver material of appropriate quality (Chen et al., 2026; EMA, 2024).



For ultra-rare and N-of-1 ASO therapies, the CMC paradigm is more flexible but still quality-driven. FDA’s draft guidance for individualized ASO drug products focuses on therapies for severely debilitating or life-threatening diseases caused by unique genetic variants where only one or two individuals are prospectively identified. Within this context, the guidance emphasizes well-characterized ASO chemical classes, conventional manufacturing methods, and relatively simple formulations, while recommending CMC information sufficient to support first-in-human exposure (FDA, 2021). Such information includes the description of the drug substance, manufacturing process, characterization, drug substance and drug product controls, reference standards or materials, container closure systems, and stability. Thus, even when development timelines are

compressed, the CMC package must still provide confidence in identity, quality, purity, strength, and patient safety.

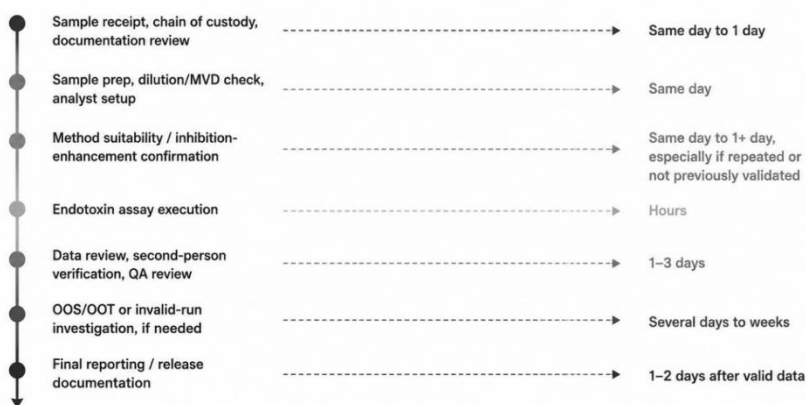
The practical difference between commercial ASO CMC and individualized ASO CMC is therefore primarily one of scale, timing, and evidentiary depth. A large-indication ASO program must support reproducible manufacturing across many batches, potentially multiple sites, and a long commercial lifecycle. An individualized ASO program must prioritize speed, use of prior knowledge, focused risk assessment, and sufficient analytical confidence to justify administration to one or a very small number of patients. In both cases, ASO CMC requires integration of sequence-defined molecular design, controlled chemical synthesis, impurity understanding, orthogonal analytical methods, appropriate specifications, formulation and container-closure controls, and stability assessment. The goal is to ensure that every ASO, whether developed for a broad population or a single patient, is manufactured and controlled to a standard appropriate for its clinical context.

Bioburden control for antisense oligonucleotides (ASOs) manufactured under GMP is an important element of the overall contamination control strategy. Because ASOs may be produced through multi-step synthesis, purification, concentration, and formulation processes, microbial control should be maintained through qualified raw materials, controlled water systems, validated cleaning procedures, environmental monitoring, closed or appropriately protected processing, and defined in-process hold conditions. Bioburden testing should be performed at appropriate process points based on risk, particularly before sterile filtration or final aseptic processing, to confirm that microbial levels remain within established alert, action, and acceptance limits. Results should be trended and investigated when excursions occur, with corrective actions documented under the quality system. For sterile ASO drug products, bioburden control supports—but does not replace—validated sterilization, aseptic processing controls, endotoxin control, and final sterility assurance.

Endotoxin

Endotoxin control is a critical quality attribute for ASOs intended for parenteral or intrathecal administration. Endotoxin levels should be controlled through qualified raw materials, controlled water systems, appropriate equipment cleaning, validated depyrogenation or endotoxin-reduction steps where applicable, and routine in-process and release testing. Testing should be performed using a compendial bacterial endotoxins test method, with method suitability demonstrated for the ASO matrix to address

potential assay interference. Endotoxin acceptance criteria should be justified based on the intended route of administration, maximum dose, patient population, and applicable regulatory or pharmacopeial expectations.



Endotoxin testing should be performed using a validated bacterial endotoxins test method suitable for the ASO drug substance, intermediate, or drug product being evaluated. The assay should be conducted using qualified reagents, calibrated equipment, endotoxin-free consumables, and controlled test conditions to prevent false-positive or false-negative results. Samples should be handled aseptically and diluted, where necessary, using endotoxin-free water to bring the test article within the validated assay range and to reduce any product-related interference. The selected method may be gel-clot, chromogenic, or turbidimetric, provided that method suitability has been demonstrated for the specific ASO matrix.

Prior to routine sample testing, method suitability should be confirmed by demonstrating that the ASO sample does not inhibit or enhance endotoxin detection at the proposed dilution. This is typically performed by testing unspiked sample preparations alongside sample preparations spiked with a known amount of control standard endotoxin. Recovery of the endotoxin spike should fall within the predefined acceptance range specified by the applicable pharmacopeial method and site procedure. The maximum valid dilution should be calculated based on the endotoxin limit, product concentration, and assay sensitivity, and routine testing should be performed at a dilution that is justified, validated, and documented.

For each assay run, appropriate controls should be included, such as reagent negative controls, positive product controls, endotoxin standard controls, and, where applicable, standard curve controls. The test is considered valid only if all system suitability and control criteria are met.

Sample results should be reported in endotoxin units per millilitre, endotoxin units per milligram, or another justified unit appropriate to the material and specification. Results exceeding the established acceptance criterion should be investigated under the quality system, including assessment of sample handling, reagent performance, assay interference, laboratory contamination, and potential process-related sources of endotoxin. Confirmatory or repeat testing should be scientifically justified and documented according to approved GMP procedures.

Stability

Stability studies should be designed to support the proposed storage conditions, shelf life, and in-use handling of the ASO drug substance and drug product. GMP stability programs typically include long-term, accelerated, and, where appropriate, stress or forced-degradation studies to evaluate chemical, physical, and microbiological stability. Stability-indicating analytical methods should monitor relevant ASO quality attributes, such as identity, purity, assay, degradation products, water content, appearance, pH, concentration, particulate matter, sterility or microbial quality, and endotoxin as applicable. Data should be trended over time, and any out-of-specification or significant changes should be investigated under the quality system. The stability program should also assess container-closure compatibility, freeze-thaw sensitivity, light exposure, and hold times for intermediates, drug substance, and formulated drug product where relevant.

Antisense oligonucleotide (ASO) stability testing should evaluate both the intrinsic chemical stability of the molecule and its susceptibility to enzymatic degradation. ASOs are short, single-stranded synthetic nucleic acid therapeutics that rely on sequence-specific binding to RNA targets; however, unmodified oligonucleotides are vulnerable to degradation by endonucleases and exonucleases. As a result, most therapeutic ASOs incorporate chemical modifications, such as phosphorothioate backbones and modified sugar chemistries, to improve nuclease resistance, binding affinity and pharmacological performance (Crooke et al., 2018; Collotta et al., 2023). Stability testing should therefore include stability-indicating analytical methods capable of detecting degradation products, truncations, depurination, desulfurisation, oxidation, aggregation or other relevant impurities, as well as changes in assay, purity, appearance, pH, water content and concentration where applicable.

For GMP development, ASO stability studies should support the proposed storage conditions, retest period or shelf life for both drug

substance and drug product. Programs commonly include long-term and accelerated studies, together with forced-degradation or stress studies to demonstrate method specificity and identify likely degradation pathways. Because the primary degradation pathway for ASOs is often nuclease-mediated cleavage, in vitro stability studies using biological matrices or defined nuclease systems can provide useful supporting information on how sequence and chemical modifications influence metabolic stability (Sutton et al., 2025). For formulated ASO drug products, stability protocols should also consider container-closure compatibility, adsorption or leachables risk, freeze-thaw exposure, light sensitivity, in-use hold times, particulate matter, sterility or bioburden, and endotoxin where relevant to the route of administration.

Document No.:	SPEC-DP-ASO1-001	Effective Date:	TBD
Version:	0.1 Draft	Specification Type:	GMP release and stability specification
Product Name:	ASO1 Sterile Injection	Status:	Draft template – Not approved for GMP release until product-specific validation, stability, toxicology and regulatory review are complete
Drug Substance:	ASO1 sodium salt, single-stranded antisense oligonucleotide		
Dosage Form:	Sterile aqueous solution for injection		
Strength:	TBD mg/mL ASO1	Container Closure:	USP Type I glass vial with sterile elastomeric stopper and aluminum seal, or equivalent
Fill Volume:	TBD mL per vial		
Route of Administration:	TBD (e.g., intrathecal, subcutaneous, intravitreal)		
Storage Condition:	Store frozen or refrigerated as justified by stability data; protect from light if applicable		

No.	TEST / QUALITY ATTRIBUTE	METHOD	ACCEPTANCE CRITERION	RELEASE	STABILITY
1	Appearance	Visual inspection	Clear, colorless to slightly opalescent solution; essentially free from visible particulates; no discoloration	Yes	Yes
2	Identity: Molecular mass	ESI-MS (LC-MS)	Observed MW conforms to 7154.5 Da $\pm$ 2.0 Da	Yes	As needed
3	Identity: Chromatographic retention	IE-HPLC or IP-RP-HPLC	Retention time of principal peak conforms to reference standard	Yes	Yes
4	Identity: Sequence confirmation	LC-MS/MS or other orthogonal method	Sequence confirmed or bridged to reference standard / drug substance lot	Yes	As needed
5	Assay / Strength	UV at 260 nm or HPLC-UV	90.0 – 110.0% of label claim	Yes	Yes
6	Full-length ASO content	Stability-indicating HPLC/UPLC	Report full-length content: NLT 90.0% (or product-specific)	Yes	Yes
7	Overall purity	IE-HPLC or IP-RP-HPLC	NLT 90.0% total ASO purity	Yes	Yes
8	Individual specified impurities	Stability-indicating HPLC/UPLC	Each specified impurity: NMT 1.5% (or product-specific)	Yes	Yes
9	Unspecified individual impurities	Stability-indicating HPLC/UPLC	Each unspecified impurity: NMT 1.0%	Yes	Yes
10	Total impurities / degradants	Stability-indicating HPLC/UPLC	NMT 10.0% (or product-specific)	Yes	Yes
11	Phosphodiester impurity	LC-MS or HPLC	NMT 1.0% (or product-specific)	Yes	Yes
12	Aggregates / higher order species	SEC-HPLC or non-denaturing HPLC	Report: NMT 2.0% (or product-specific)	Conditional	Conditional
13	pH	USP <791>	6.5 – 8.0 (or product-specific)	Yes	Yes
14	Osmolality	USP <785>	250 – 400 mOsm/kg	Yes	Yes
15	Particulate matter, subvisible	USP <788>	Meets USP <788> for small-volume injections	Yes	Yes
16	Visible particulate matter	USP <790>	Meets compendial requirements	Yes	Yes
17	Sterility	USP <71>	No growth	Yes	Yes*
18	Bacterial endotoxins	USP <85>	NMT route- and dose-based limit: • Intrathecal: NMT 0.2 EU/kg/hour • Subcutaneous: NMT 5 EU/kg/hour • Intravitreal: NMT 2.0 EU/dose/eye	Yes	Yes

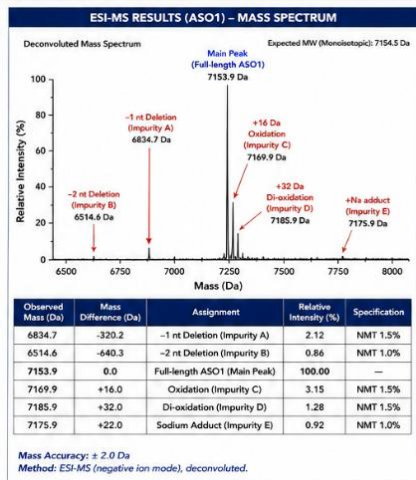
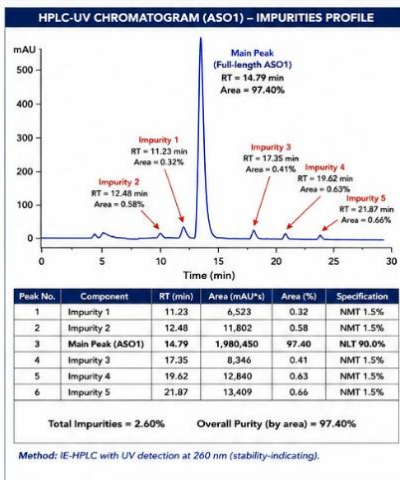
*Sterility is a release test. For stability, sterility or container closure integrity (CCI) may be used as stability endpoints.

ADDITIONAL NOTES:

- Specifications are based on current knowledge and may be tightened as more data become available.
- Methods are stability-indicating and qualified/validated per ICH Q2(R2).
- Endotoxin limits are established based on maximum proposed dose and route of administration.
- This is a sample template and must be product-specifically justified and approved.

BATCH RELEASE DECISION:

A batch may be released only if all release tests meet acceptance criteria, all deviations are resolved, sterility assurance and CCI are met (if applicable), stability data support the proposed use period, and QA has approved the batch record and CoA.



CMC for Lipid Nanoparticle Drug Products

Lipid nanoparticles (LNPs) are an increasingly important non-viral delivery platform for nucleic acid therapeutics, including mRNA, siRNA, gene-editing components, and potentially other oligonucleotide-based drug substances. Their role in drug development is not simply as passive excipients, but as functional delivery systems that protect nucleic acid cargo, support cellular uptake, influence biodistribution, and affect pharmacology,

safety, and stability. As a result, the chemistry, manufacturing, and controls (CMC) strategy for an LNP product must address both the nucleic acid cargo and the nanoparticle delivery system as an integrated drug product (Cameau et al., 2022; Johnson, Ho and Kapatrala, 2024).

A typical LNP CMC process begins with definition and control of raw materials. These include the nucleic acid drug substance, ionizable lipid, helper phospholipid, cholesterol, PEG-lipid, ethanol or other organic solvents, aqueous buffers, water for injection, filters, tubing, single-use assemblies, and container-closure components. Lipid excipients require particular attention because their purity, impurity profile, safety, and consistency can significantly affect the quality of the finished nanoparticle. Novel or non-compendial lipids may require substantial CMC and safety justification, including impurity evaluation, residual solvent control, elemental impurity risk assessment, genotoxic impurity assessment, and documentation of manufacturing controls. (MilliporeSigma, 2026).

LNP formation is commonly achieved by rapid mixing of a lipid solution in ethanol with an aqueous nucleic acid solution under controlled conditions. During this process, the lipids self-assemble around the nucleic acid cargo to form nanoparticles. Critical process parameters may include lipid concentration, nucleic acid concentration, lipid-to-cargo ratio, N/P ratio, aqueous-to-organic flow ratio, total flow rate, mixing geometry, temperature, buffer composition, pH, ionic strength, and hold times. Because particle size, polydispersity, morphology, surface charge, encapsulation efficiency, and potency are sensitive to the mixing environment, LNP manufacturing requires tight process control and scalable equipment that can maintain comparable mixing conditions from small-scale formulation screening through clinical and commercial production (Cameau et al., 2022).

After nanoparticle formation, the process typically proceeds through dilution, concentration, diafiltration, solvent removal, buffer exchange, sterile filtration, and fill-finish. Tangential flow filtration is commonly used to remove ethanol and unencapsulated small molecules while concentrating the LNPs into the final formulation buffer. These downstream steps can themselves influence product quality because LNPs may be sensitive to shear, temperature, filter type, membrane chemistry, processing time, and freeze-thaw exposure. Sterile filtration and filling must also be carefully evaluated because nanoparticle products can be affected by filter compatibility, adsorption, particle-size shifts, visible or subvisible particles, and container-closure interactions. For some products, frozen storage, refrigerated storage, or lyophilization may be considered, but each approach requires product-specific stability justification (Cameau et al., 2022; Jeffs, Ho and Kapatrala, 2024).

The analytical control strategy for LNPs is necessarily multidimensional. In addition to conventional drug product tests such as appearance, identity, assay, pH, osmolality, sterility, endotoxin, subvisible particles, residual solvents, and container-closure integrity, LNP-specific characterization is required. Common critical quality attributes include particle size, polydispersity index, encapsulation efficiency, nucleic acid integrity, lipid identity and content, lipid impurity profile, residual ethanol, surface charge or zeta potential, potency, in vitro expression or functional activity, and stability-indicating degradation products. Because standardized nanomaterial methods are still evolving, complementary analytical methods and carefully justified sample preparation, sampling, and method-suitability strategies are especially important (Stephens, 2022; Cameau et al., 2022).

CMC development for LNPs must also address manufacturing scale, automation, facility design, and data integrity. Scale-up from milliliter-scale formulation development to clinical or commercial-scale manufacturing can introduce variability in mixing, temperature control, residence time, solvent handling, TFF performance, and sterile filtration. For this reason, scale-down models, automated platform technologies, single-use systems, electronic batch records, real-time monitoring, and robust process characterization are frequently emphasized as tools for reducing variability and improving GMP readiness. (Johnson, Ho and Kapatrala, 2024; Jeffs, Ho and Kapatrala, 2024).

The nanomaterial nature of the product does not create an entirely separate quality standard, but it does increase the need to characterize how the nanoparticle's physical and chemical attributes relate to safety, efficacy, and quality. Developers are expected to identify critical quality attributes, understand process risks, justify analytical methods, bridge formulation or process changes, retain samples for comparability, and maintain lifecycle control. Early interaction with regulators is particularly important because the regulatory framework for LNPs continues to evolve, and each formulation may have unique features depending on lipid composition, cargo type, route of administration, dose, therapeutic area, and intended patient population (Stephens, 2022; Johnson, Ho and Kapatrala, 2024).

Compared with naked ASO drug products, LNP-based products introduce a major additional layer of CMC complexity. A naked ASO formulation may often be developed as a relatively simple aqueous or lyophilized oligonucleotide product, with the primary CMC burden focused on ASO identity, purity, impurity profile, assay, microbiological quality, endotoxin, stability, and container compatibility. By contrast, an LNP product must control not only the nucleic acid cargo, but also multiple lipid components, lipid impurities, nanoparticle assembly, particle size distribution, encapsulation efficiency, residual solvent, surface properties,

filterability, physical stability, and interactions among cargo, lipids, process materials, and container closure. This greatly increases analytical burden, process-development work, raw-material qualification, facility requirements, regulatory justification, and comparability testing. In practical terms, LNPs can improve delivery and enable therapeutic applications that naked ASOs may not achieve, but they usually add substantial cost, time, technical risk, and operational complexity to CMC development and GMP manufacturing.

Conclusion

RNA therapeutics began with a simple but transformative idea: that nucleic acids are not only carriers of biological information, but programmable molecules that can be designed, synthesized, modified, delivered, and translated into clinical action.

The history of this field is one of iteration. Each generation of RNA therapeutic technology solved one problem and exposed the next. Early antisense oligonucleotides showed that sequence-specific recognition could alter gene expression, but they also revealed the limitations of unmodified nucleic acids: instability, rapid clearance, poor tissue access, and nonspecific toxicity. Phosphorothioate backbones improved nuclease resistance and pharmacokinetics, but introduced new questions around protein binding, inflammation, and tolerability. Sugar modifications such as 2'-O-methyl, 2'-fluoro, 2'-MOE, locked nucleic acids, constrained ethyl chemistries, and newer analogues expanded the medicinal chemistry toolkit, allowing potency, stability, affinity, and safety to be tuned with increasing precision. Splice-switching oligonucleotides showed that RNA could be remodeled rather than simply destroyed. siRNA therapeutics harnessed endogenous RNA interference to make gene silencing catalytic and durable. mRNA therapeutics reversed the logic entirely, using synthetic RNA not to suppress a message, but to introduce one.

These advances did not arise from a single breakthrough. They emerged through disciplined cycles of design, synthesis, testing, failure, redesign, and translation. The first molecules were crude by today's standards, but they established the central principle that still defines the field: RNA therapeutics are information medicines made possible by chemistry. The sequence provides specificity, but chemistry determines whether that sequence can survive, distribute, enter cells, engage the correct machinery, avoid the wrong biology, and produce a therapeutic effect in patients.

This iterative progress is enabling a rational form of drug design. In traditional small-molecule discovery, screening often begins with large chemical libraries and uncertain target engagement. In RNA therapeutics, the starting point can be genetic information itself: a pathogenic mutation, a toxic transcript, a splice defect, a disease-associated isoform, a regulatory RNA, or a protein-coding message that needs to be supplied. Once the biological hypothesis is defined, therapeutic candidates can be designed

directly against the relevant sequence or regulatory element. Screening is still essential, but it becomes more targeted. Instead of searching broadly for activity, the field can focus on optimizing chemistry, position, modification pattern, delivery format, potency, selectivity, durability, and safety.

This changes the economics and tempo of discovery. RNA therapeutics can compress the distance between genetic diagnosis and therapeutic hypothesis. Candidate panels can be designed in silico, synthesized rapidly, and screened in disease-relevant systems, including patient-derived cells. Iterative chemistry allows weak leads to be improved systematically rather than abandoned blindly. Off-target analysis, transcriptome profiling, allele selectivity, splice correction, pharmacodynamic biomarkers, and dose-response behavior can be incorporated early. The result is not that RNA drug discovery becomes easy. It does not. Rather, it becomes more rational, more modular, and more directly connected to the biology of disease.

The same logic is reshaping the path to the clinic. RNA therapeutics are increasingly developed through platform knowledge: shared chemistries, shared manufacturing processes, shared analytical methods, shared toxicology principles, and shared delivery strategies. A GalNAc-conjugated siRNA, a 2'-MOE gapmer ASO, a splice-switching oligonucleotide, or an LNP-formulated mRNA each carries lessons from previous molecules in the same class. Those lessons inform dose selection, tissue targeting, safety monitoring, pharmacodynamic endpoints, and chemistry, manufacturing, and controls. Each approved drug makes the next one less mysterious.

Yet the field's greatest lesson is also its warning: programmability is not enough. A sequence is not a drug. A potent molecule in a dish is not a medicine. A beautiful mechanism is not a clinical benefit. RNA therapeutics must pass through every practical gate of drug development: delivery, biodistribution, endosomal escape, intracellular trafficking, durability, tolerability, analytical characterization, scale-up, purity, stability, regulatory review, and patient access. The failures in the field are as instructive as the successes. They show that potency without delivery is insufficient, affinity without specificity can be dangerous, and target knockdown without clinical benefit is not enough.

This is why the next era of RNA therapeutics will belong not only to those who can design sequences, but to those who can engineer systems. Better conjugates will expand tissue reach beyond the liver. Improved lipid nanoparticles and alternative delivery vehicles will make mRNA and RNA editing more precise. New backbone and sugar chemistries will tune stability, protein binding, immune recognition, and intracellular behavior. Computational design will improve target selection and reduce off-target risk. Patient-derived models will make screening more disease-relevant.

Manufacturing innovation will determine whether personalized and nano-rare therapies can move from heroic exceptions to repeatable clinical practice.

The final measure of the field will not be its elegance. It will be execution. We can design medicines with unprecedented precision, but can we deliver them reliably, manufacture them reproducibly and bring them to patients quickly enough to matter?

The logistics are the hard part: chemistry, delivery, toxicology, manufacturing, clinical operations, regulatory alignment, and access. The future of RNA medicine will be built by those who drive this.

“Amateurs talk strategy. Professionals talk logistics.”

General Omar Bradley

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